# Battery Storage and Natural Gas Generator Market Power

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**Abstract**

Utility-scale batteries are often valued for their ability to integrate renewables. They also compete directly with the fossil generators that exercise market power during peak. I study this competitive effect in Texas exploiting the staggered entry of battery capacity across counties. I find that batteries make the residual demand of nearby gas generators more elastic, displace their evening output, and lower their evening markups. An additional 100 MWh of storage reduces gas operating profits by 6 percent, and between 73 percent and all of this loss is market power rent. Gas generators keep setting prices, but at smaller margins. The resulting $47 million annual transfer flows mainly to consumers, while the production cost savings from shifting energy across hours are small. Batteries earn their arbitrage revenue from the markups they erode, so competition limits the private return to storage.

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# 1 Introduction

Battery storage is viewed as an important technology for the energy transition because it complements intermittent renewable generation. By shifting energy across time, storage smooths the variability of wind and solar output and allows renewables to supply a larger share of generation (Karaduman, 2023; Andres-Cerezo and Fabra, 2025). The welfare gains from this complementarity are typically framed as reductions in environmental externalities when storage-backed renewables displace fossil-fuel generation. An underappreciated role of storage, however, is that it competes directly with the conventional generators. Fossil-fuel plants, specifically the marginal generators during peak demand hours, earn markups that reflect their market power (Borenstein, Bushnell and Wolak, 2002; Woerman, 2023). Because battery dispatch also targets the same peak hours, their entry makes the residual demand facing the incumbents more elastic and mitigates their market power. This competitive effect is distinct from the reduction in external costs, and it has not been quantified. Whether it matters in aggregate, and who captures the surplus it creates, are important empirical questions.

In this paper, I quantify how battery storage entry affects the market power of incumbent natural gas generators. I study the Texas electricity market (ERCOT), an energy-only market in which frequent transmission congestion allows local generators to extract scarcity rents. ERCOT provides an ideal setting for this analysis. More than 7 GW of battery capacity has been deployed in ERCOT since 2021, a scale of market penetration substantially larger than those evaluated in prior studies.[1] To identify the competitive effects of this entry, I construct a unit-level panel that links each gas generator's output, offers, and marginal cost to storage entry in their local market. To account for congestion, I define local markets using price divergence and high-resolution grid topology (Arkolakis and Walsh, 2024; Mercadal, 2022; Woerman, 2023), and measure the slope of each generator's residual demand from the offers of its local rivals.

To guide the empirical analysis, I develop a simple two-period model where a strategic natural gas generator competes with a price-taking storage fleet. Storage affects the generator's markup through two channels. Through the *level channel*, storage shifts the generator's residual demand, lowering it when batteries discharge and raising it when they charge. Through the *slope channel*, storage makes residual demand flatter, which raises the cost of withholding output. The model's prediction maps each channel to an estimable parameter. The generator's optimal output is linear in its marginal cost, with an intercept set by the level of residual demand and a slope set by its elasticity. Storage therefore enters output through two terms: a direct effect that captures displacement, and an interaction with cost that captures the slope channel, because a flatter residual demand displaces higher-cost generators more. The markup rule then shows that storage lowers markups through both channels in discharge hours, while its effect in charge hours is ambiguous.

My empirical strategy follows from the model prediction. Because the level channel reverses sign between charge and discharge hours, I estimate the marginal effect for each hour of the day rather

[1] This capacity is roughly ten to one hundred times the storage analyzed in Butters, Dorsey and Gowrisankaran (2025), Karaduman (2023), and Kirkpatrick (2026).

than pooling across the day. Because the first-order condition holds only when a generator runs between its operating limits, I estimate the output rule on those hours and instrument marginal cost with the product of the pre-entry heat rate and the natural gas price. Because the cost interaction identifies the slope channel only in sign, I also measure the slope of each generator's residual demand directly from the offers of its local rivals. Finally, because both channels lower operating profit, a fall in profit alone does not show that storage reduces market power. I therefore decompose profit into a competitive rent and a market power rent using a benchmark competitive price.

I estimate these model parameters leveraging the staggered entry of storage capacity across ERCOT. As storage enters, incumbent natural gas generators are exposed to different levels of new competition across counties. My research design compares generators facing local battery entry with generators in counties that storage has not yet reached, before and after entry, and it uses the growing installed capacity in each county as the intensity of exposure. I carry the model's output rule into this design by interacting local storage with each generator's predetermined cost, so the estimates recover the same displacement and cost gradient as the model. Exploiting this spatial and temporal variation, I find four main results.

First, storage entry erodes the market power of incumbent natural gas generators during the evening peak hours. An additional 100 MWh of local storage reduces the markup of local gas generators by \$2.71 per MWh at 7 PM and by \$1.50 per MWh on average over the evening peak, about 5 percent of the pre-entry evening markup. Markups do not change overnight or in the morning, when batteries charge. Operating profit falls by \$5.45 per MW-day, about 6 percent of the pre-entry average. The reduction in market power grows with the installed dose, and the event study estimates provide no evidence of pre-entry trends.

Second, using the computed residual demand slope from the bid data, my estimates imply that residual demand of nearby gas generators becomes more elastic after storage enters. The effect is most pronounced for combined cycle generators in the evening. Storage also displaces gas output in the afternoon and evening, and it displaces higher-cost generators more, as the model predicts. My instrumental variables estimate of the generator's optimal output rule confirms that output responds to cost as the model requires. These results hold when I control for the solar and wind capacity that often enters alongside storage.

Third, the lost profit is mainly market power rent rather than a lower competitive price. I compare the realized price with the competitive price that a cost-based dispatch of the same gas output would produce within the local market, following the approach of Borenstein, Bushnell and Wolak (2002). I use a more conservative lower bound of competitive price by using the cost of the generator that ERCOT identifies as the price-setter. Market power rent accounts for between 73 percent and all of the profit loss under the two benchmarks. Gas generators also continue to set the price as often as before storage entry. If storage lowered gas profits only by adding supply, it would push cheaper units or storage itself to the margin. Instead, gas generators keep setting the price at a smaller margin above cost, so storage disciplines their pricing rather than only adding

supply.

Finally, I quantify the welfare implications of these competitive effects. The competitive effect of storage is mainly a transfer. Local storage lowered gas operating profits by about $47 million per year, of which $34 million to $55 million was market power rent. Storage owners kept about $6 million of this transfer through energy arbitrage, and consumers and the other generators gained the remaining $41 million. My estimates suggest that most of it went to consumers, although the split between the two is imprecise. The efficiency gain from shifting energy is small. Valued at marginal cost, the energy arbitrage across hours is negligible, because the cost spread between charging and discharging hours barely covers round-trip losses. Valued at market prices, energy arbitrage earns storage owners $5.8 million per year. The difference is the markup of gas generators, so storage earns its arbitrage revenue from the markups it erodes, and competition limits its private return from arbitrage. Energy arbitrage is also a small part of what storage earns. Storage owners earned another $35.6 million per year from ancillary services, 86 percent of their total revenue of $41.4 million, or $54 per kW-year.[2] Even this total covers only about a quarter of the annualized capital cost of storage, and the measured social benefits, including avoided emissions at their upper bound, also fall short of it. At current penetration, the case for storage rests on its reliability services and on who gains from its competitive effect, rather than on energy arbitrage.

This paper contributes to three strands of literature. First, it contributes to the broad industrial organization literature on the effect of entry on incumbent firm behavior. A body of empirical work shows that incumbents respond to the threat or realization of entry across industries, from retail (Arcidiacono et al., 2020; Busso and Galiani, 2019; Taylor and Muehlegger, 2026) to airlines (Goolsbee and Syverson, 2008), and pharmaceuticals (Ellison and Ellison, 2011). Arcidiacono et al. (2020) study the entry of Walmart supercenters and find that nearby retailers lose revenue with no significant change in prices. Taylor and Muehlegger (2026) uses high-frequency microdata to evaluate how incumbents adjust to entry. I extend this framework to wholesale electricity markets, where I observe the incumbent's output, pricing, and cost which allow me to measure realized margin at a more granular level. Moreover, the entrant in my setting is not a symmetric rival because storage buys when incumbents face little competition and sells when they face the most, so its competitive effect varies across hours of the day, and its own profit depends on the markups it erodes.

Second, this paper contributes to the literature on competition and market power in wholesale electricity markets. A large body of work documents how generators exercise market power under transmission constraints and localized congestion (Borenstein, Bushnell and Wolak, 2002; Hortaçsu and Puller, 2008; Ito and Reguant, 2016; Woerman, 2023), and recent studies show that transmission expansion (Ryan, 2021; Doshi, 2024), financial arbitrageurs (Mercadal, 2022), and the price exposure of renewables (Fabra and Imelda, 2023) can reduce it. Closer to this paper, two recent

[2]These figures are in line with industry estimates. Modo Energy reports that the average ERCOT battery earned $141 per kW in 2022, $192 per kW in 2023, and $55 per kW in 2024, and that energy arbitrage provided about 15 percent of battery revenue in 2023 (Vermillion, 2024, 2025). My average spans 2018 to 2024, excludes Winter Storms Uri and Elliott, and is in 2020 dollars, which explains its lower level.

studies examine how large-scale solar entry affects conventional generators. Jha and Leslie (2025) show that it softens evening competition through start-up costs, while Bushnell and Novan (2021) show that it shifts the marginal technology toward less efficient units. I study a different large-scale entry, storage. I show that storage disciplines market power through a similar residual demand channel that this literature identifies as central to transmission-driven discipline.

Finally, this paper speaks to the growing literature on the economics of utility-scale storage. This work has largely quantified the aggregate, market-level impact of storage, documenting how it substitutes for or complements renewable capacity (Karaduman, 2023; Andres-Cerezo and Fabra, 2025), lowers intraday price spreads (Lamp and Samano, 2022), shifts long-run investment equilibria (Butters, Dorsey and Gowrisankaran, 2025), and relieves local congestion (Kirkpatrick, 2026). Less is known about how entry changes the behavior of incumbents. Closest to this paper, Reaser and Taylor (2025) uses the entry of storage in ERCOT to estimate its effect on nodal prices, and builds a structural dispatch model that computes counterfactual prices without storage offers. I leverage the same source of variation focusing on the marginal generators and measure how storage changes the output, residual demand, and markups. I further separate the market power component of their lost profit from the effect of a lower competitive price. Other recent papers study different margins of storage entry. Dudek (2026) studies how storage reduces renewable curtailment in California, and Visaggio (2025) studies how ownership structure shapes the market incentives of batteries paired with renewables in Texas. I complement these papers by focusing on its competition with gas generators. To my knowledge, this paper is the first to use unit-level storage dispatch to trace the channel from an individual storage unit's entry to the behavior of incumbent gas generators. While prior work characterizes the theoretical price impact of storage (Andrés-Cerezo and Fabra, 2023), I provide causal evidence that observed storage entry erodes realized markups. I also use a welfare framework that separates transfers from net social gains, and show that the private return to storage arbitrage depends on the market power it disciplines.

The remainder of this paper is organized as follows. Section 2 provides an overview of the Texas electricity market. Section 3 provides the data and descriptive statistics on natural gas and storage units. Section 4 develops the theoretical model. Section 5 presents the empirical strategy. Section 6 presents the effects on output, residual demand, markups, and operating profit. Section 7 quantifies the welfare and distributional effects, and Section 8 concludes.

## 2 Storage and The Texas Electricity Market

In this section, I provide an overview of the Texas electricity market, highlighting the unique features of ERCOT that make it an ideal setting to study the impact of storage entry on natural gas generator market power.

The Texas electricity market provides an ideal setting to study the competitive interactions between storage and natural gas generators. The Electric Reliability Council of Texas (ERCOT) manages this grid, delivering power to more than 26 million customers, about 90% of the state's

electric load, and operating a wholesale market that schedules the dispatch of over 1,500 generators in real time. Because it has limited interconnection with other regions, the market is relatively insulated from external shocks, creating a highly contained environment for empirical analysis. This system is particularly useful to study because of its generation mix; natural gas accounts for a dominant 44.3% of the generation capacity, followed by wind (25.2%) and solar (13.2%) (ERCOT, 2025).[3] At the same time, massive recent growth in battery storage capacity is actively altering these established competition dynamics. Furthermore, because ERCOT operates as an energy-only market with no capacity payments, it creates strong financial incentives for generators to respond strategically to real-time conditions.

The massive growth of utility-scale battery storage in ERCOT introduces new competitors to incumbent natural gas generators. Storage units operate as fully dispatchable assets that offer operational flexibility, functioning as generators when exporting and as load when charging. They participate in both the day-ahead and real-time markets providing both energy and ancillary services.[4] Figure 1 documents this expansion, with ERCOT storage capacity growing to over 7.6 GW since 2021.[5]

Because battery storage can respond quickly to price signals, these units can precisely target their dispatch during high-revenue periods where incumbent generators exercised market power. Panel B documents this pattern where storage discharges predominantly during the evening peak (5 PM to 9 PM). Panels C and D provide further descriptive evidence of competition during these hours: the share of storage units clearing at peak prices has increased, while the corresponding share of marginal natural gas generators has declined as more storage capacity enters. Before formally estimating the magnitude of this competitive effect, I first outline the features of the ERCOT market design that facilitate these interactions.

ERCOT shares a similar market feature with other deregulated electricity markets in the U.S. Like PJM, MISO, and CAISO, it runs a day-ahead and a real-time market and sets a separate wholesale electricity price at each node of the grid. This nodal price, also called the locational marginal price, reflects the cost of generation, transmission constraints, and line losses at that location.[6] Generators submit offers for each hour of the next day in the day-ahead market. The real-time market then runs every five minutes, and the Security Constrained Economic Dispatch (SCED) clears it and sets the real-time nodal prices.

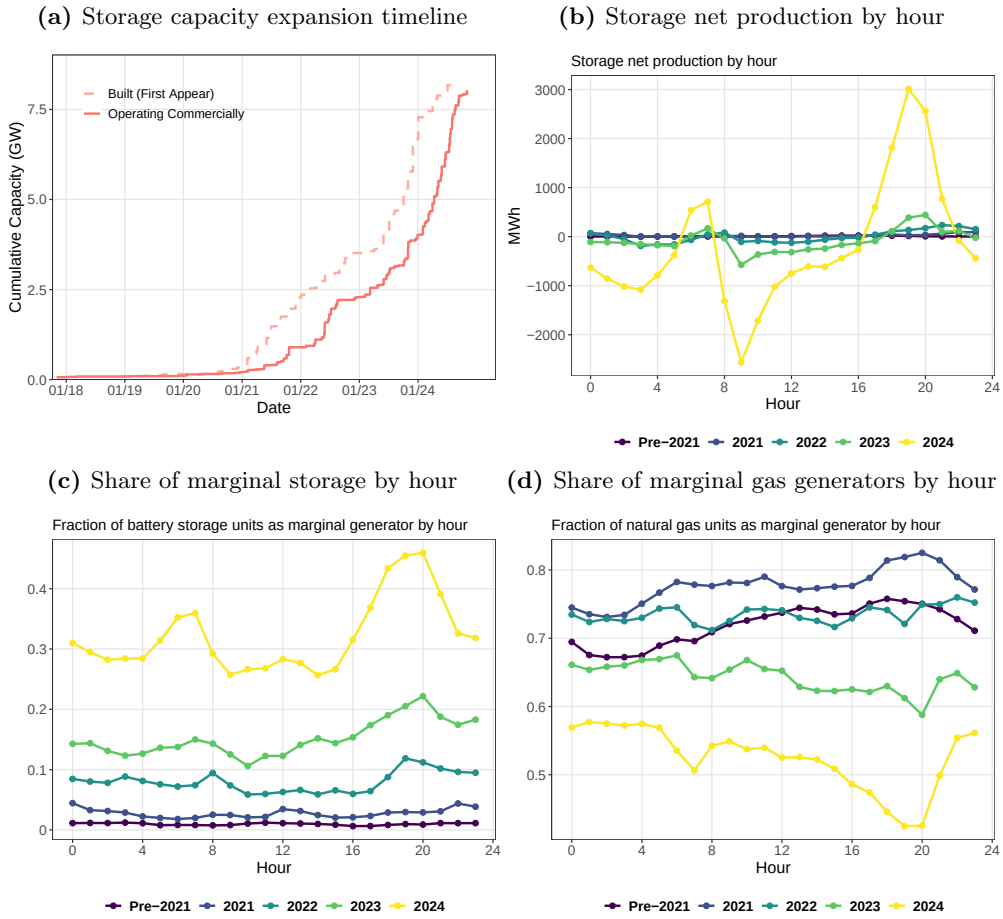
[3] Figure E.21 shows the spatial distribution of each unit based on their fuel type in ERCOT. Table D.10 summarizes the generator units in ERCOT territory as of the end of 2024, while Figure E.23 details the installed capacity over my sample period: Natural gas units dominate the installed capacity (38.9% or 65.23 GW), followed by wind and solar PV units, respectively. Natural gas and coal capacity remain roughly the same over my sample, while solar PV and wind capacity grow significantly by about 25 GW and 20 GW.

[4] Because they serve as both generation and load, batteries can provide a wide range of regulation and reserve services. During summer 2023, for example, ERCOT Contingency Reserve Service (ECRS) accounted for more than 75% of total battery revenues, whereas real-time energy settlement contributed just under 15% (Miller and Nakajima, 2024). Figure E.22 illustrates how a single storage resource can simultaneously fulfill multiple AS responsibilities.

[5] In Q2 2025, ERCOT's installed storage capacity surpassed California's. A fundamental difference between the two markets is that ERCOT's expansion is driven by price signals and efficient interconnection queues, rather than the state-level procurement mandates or subsidies seen in California (Mathrani, 2025; Ahmed et al., 2026).

[6] Each generator is assigned to a settlement point and receives the nodal price at that location.

**Figure 1:** Storage expansion and its competition trends with natural gas generators in ERCOT




*Notes:* Panel A shows the cumulative storage capacity built and commercially operating over time. Panel B shows the aggregate storage net production pattern by hour. Net production is defined as aggregate storage discharge less aggregate storage charge. Panel C shows the share of storage units that receive the highest offered price in each hour. Panel D shows the share of natural gas generators that receive the highest offered price in each hour. Section B.1 provides the same yearly trend in hourly demand, net demand, aggregate solar, wind, and natural gas generation.

Transmission congestion creates local market power in every nodal market. When a constraint binds, it splits the grid into local markets and shields the generators within the local market from rivals on the other side of the constraint (LaRiviere and Lyu, 2022; Fell, Kaffine and Novan, 2021; Woerman, 2023; Doshi, 2024). Even small generators can then become pivotal and exercise market power by raising their offers above marginal cost. Storage competes with gas generators inside these local markets. The mechanism I study therefore requires only nodal pricing, congestion, and storage that responds to prices, which every deregulated electricity market has.

ERCOT's market design differs from other markets in three ways that affect generators' incentives to raise their offers. First, it is an energy-only market. PJM, MISO, ISO-NE, and NYISO pay generators for available capacity through centralized auctions, and CAISO does so through bilateral resource adequacy contracts. ERCOT made no capacity payments during my sample, so generators recover their fixed costs from energy and ancillary service revenue, including scarcity prices in tight

hours.[7] Second, its offer cap was $9,000/MWh until 2022 and $5,000/MWh after, compared with $1,000/MWh to $2,000/MWh in markets subject to FERC Order 831.[8] Third, ERCOT mitigates local market power more narrowly. When a transmission constraint is classified as non-competitive, SCED caps each offer at the greater of a cost-based mitigated offer cap and the price that would prevail without that constraint (ERCOT, 2024; Potomac Economics, 2025). Offers that affect only constraints classified as competitive are not mitigated. Other markets apply structural tests for pivotal suppliers to constraints as they bind, such as PJM's three-pivotal-supplier test (Joskow, 2022). Together, these features leave gas generators in ERCOT more scope to exercise local market power than in other markets.

# 3 Data and Descriptive Evidence

This section describes the data sources and the construction of the panel dataset used to analyze the impact of storage entry on natural gas generator market power. I first outline the sample construction and the empirical definitions of local markets, marginal costs, and markups. I then provide descriptive statistics for natural gas generators and storage units.

## 3.1 Data Sources and Sample Construction

I combine multiple datasets to construct a panel at the generator-by-hour level covering January 1, 2018 to December 1, 2024. The main components are the generator and storage unit characteristics, supply offers and bids, the market outcomes, and marginal costs. The novel component of this dataset is that I observe each storage unit's power capacity, storage duration, bid/offer curves, and unit-level charge and discharge quantity.

**Generator and battery storage data.** I identify unique generators from ERCOT's 60-day Security Constrained Economic Dispatch (SCED) Report and the Monthly Outlook for Resource Adequacy (MORA) Report.[9] For all generators, I define effective capacity as the maximum production observed in the SCED Report, and I use the first date a generator produces electricity outside of testing as its initial commercial operating date (COD).

Because ERCOT reports power capacity but not energy capacity for battery storage, I complement this data using EIA Form 860M, the MODO Energy Battery Buildout Report, and other public sources to determine each unit's storage duration. Furthermore, because storage acts as both a generator and a load, it operates under different names when submitting supply offers versus load bids; I manually link these identifiers for each storage unit. Finally, I use EIA Form 860

[7] Ancillary services in ERCOT include regulation (REGUP, REGDN) and operating reserves (Responsive Reserve [RRS], Non-Spin [NSPIN], and ERCOT Contingency Reserve Service [ECRS]).

[8] Under FERC Order 831, non-ERCOT markets allow offer caps to increase to $2,000/MWh to set the market clearing price, but strictly on the condition that these higher offers are justified by verified short-run marginal costs.

[9] Two or more generators may be associated with a single storage or generation unit. A generator's name might also change before and after testing. For combined-cycle units, I use the maximum production from the generator combination with the highest possible output as the effective capacity.

to assign the load zone and county for each generator, and the Decision Making Entity from the SCED MORA Report to identify the parent company.[10]

**Market outcome data.** I collect 15-minute interval data on offer and bid curves, dispatch instructions, operating status, and the identity of marginal generators directly from the SCED reports. I also extract 5-minute nodal prices at each settlement point, mapping them to individual generators and aggregating them to 15-minute intervals. I then aggregate these market outcomes to the hourly level, constructing variables for total hourly generation, average nodal price, and an indicator for whether a generator was marginal at any point during the hour.

To define the local markets created by transmission congestion, I group generators based on nodal price divergence and network connectivity at each interval, following Mercadal (2022) and Woerman (2023). I first group generators whose nodal prices are within \$1/MWh or 1% of the larger price. I then verify transmission connectivity within these groups using transmission lines and substation shapefiles collected by Arkolakis and Walsh (2024). Generators sharing both similar nodal prices and grid connections constitute a local market. Figure B.3 shows examples of market definition for two intervals, one with no congestion and one with congestion. The number of local markets varies from 1 to 137 across intervals, with a median of 8.[11] Within these local markets, I estimate residual demand for each generator by stacking the bids of all other generators, computing the slope at the average nodal price using an Epanechnikov kernel for bids within \$5/MWh of the nodal price. The resulting slope represents the density of rival supply, measured in MW per (\$/MWh). This captures how many additional MW of rival capacity would enter the local market if the nodal price were to increase by one dollar.

**Marginal cost.** I construct the marginal cost of natural gas generators as the sum of variable fuel cost plus emission permit costs following Hausman (2025) and Doshi (2024):

$$c_{it} = \underbrace{hr_{it} \cdot p_{it}^{Fuel}}_{\text{Fuel Cost}} + \underbrace{ER_{it}^{\text{SO}_2} p_t^{\text{SO}_2} + ER_{it}^{\text{NO}_x} p_t^{\text{NO}_x}}_{\text{Emission Cost}}.$$

I use hourly generation data from SCED and collect information on heat content and emissions to compute heat rates and emission rates at the plant-by-type-by-week level from the Environmental Protection Agency's (EPA) Continuous Emission Monitoring System (CEMS) from 2018 to 2024.[12] To compute fuel cost, I collect daily natural gas price data from S&P Capital IQ. Natural gas units

[10] I manually crosscheck this ownership data with EIA Form 860 and employ AI tools to determine the parent company for each generator. Generators owned by multiple companies are coded under a single parent company.

[11] Figure B.2 presents the distribution within my sample and across hours by year of sample. The number of local markets has increased over the years as more generators are added to the ERCOT grid.

[12] Heat rate and emission rate vary depending on the operating condition of a generator, as discussed by Demirer and Karaduman (2026). I compute the heat rate by dividing the weekly total heat content and emissions for that week by the weekly total net generation for each plant-type combination. For units with no observation in CEMS, I assign the average heat rate for the plant with the same Type (Combined Cycle, Simple Cycle, or Gas Steam Turbine). I winsorize the extreme values to be within the reported heat rate range for the same generator type from EPA eGrid, to account for unreasonably high and low values from CEMS.

source their fuel from different hubs. I use the natural gas price location information for each unit from SNL Energy.[13] To compute emission cost, I collect permit prices for $SO_2$ and $NO_x$ from S&P Capital IQ and apply the emission cost to units subject to the Acid Rain $SO_2$ Program and CSAPR $NO_x$ Program.

**Markup.** I use realized markup as the primary outcome variable to measure market power, defined as:

$$\mu_{iht} = \mathbb{1}[q_{iht} > 0] (p_{iht} - c_{it}) .$$

This markup is zero when the generator is not producing electricity, and equal to the difference between its nodal price $p_{iht}$ and marginal cost $c_{it}$ when it is producing. This construction captures both the extensive margin (whether the generator is producing) and the intensive margin (how much markup it earns when producing). This is consistent with the electricity market literature that measures markup over all unit-time rather than only when a unit is dispatched (Bushnell, Mansur and Saravia, 2008; Borenstein, Bushnell and Wolak, 2002; Reguant, 2014).

**Other data.** To control for local weather conditions, I construct a balanced hourly panel of local temperature using station data from the NOAA Integrated Surface Database (ISD) (NOAA, 2024), interpolating minor gaps and assigning local temperatures to each power plant based on its nearest weather station. I also construct hourly renewable energy potential at the county level from the global horizontal irradiance data from National Solar Radiation Database (NSRDB) (NREL, 2024) and wind speed from ERA5 global reanalysis data (C3S, 2018).

**Sample construction.** I limit my estimation sample to marginal natural gas generators, defined at the generator-by-hour level. To prevent endogenous selection from storage shifting the merit order, I restrict the sample to generator-hour pairs that set the nodal price prior to local storage entry, requiring at least 100 pre-treatment observations to ensure I capture established operational patterns. I exclude the period of Winter Storms Uri and Elliott from my estimation sample. I assign generators, storage units, and settlement points to counties with the same imputed county field, so that treatment and outcomes refer to the same county. I drop Howard County, where a battery operated before my sample begins, because it has no untreated period. Next, I describe the gas generator's characteristics.

### 3.2 Descriptive Statistics on Natural Gas Generators

Table 1 presents the descriptive statistics on natural gas generators. My sample consists of 553 natural gas generators (210 Simple Cycle, 294 Combined Cycle, and 49 Steam Turbine generators),

[13] For units missing this information in the S&P Capital IQ database, I assign Waha Hub for units in the West load zone, NGPL TX-OK for units in the North load zone, TETCO STX for units in the South load zone, and Katy for units in the Houston load zone.

**Table 1:** Summary statistics of natural gas generators

| | All | | | Treated | | | Never-Treated | | |
|---|---|---|---|---|---|---|---|---|---|
| | All | 12AM –12PM | 12PM –12AM | All | 12AM –12PM | 12PM –12AM | All | 12AM –12PM | 12PM –12AM |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| **Panel A: Market Outcome and Characteristics (2018–2021)** | | | | | | | | | |
| Nodal Price ($/MWh) | 28.7 | 21.7 | 35.7 | 29.0 | 21.9 | 36.1 | 27.7 | 21.1 | 34.5 |
| | (79.2) | (38.4) | (104.8) | (80.8) | (39.8) | (106.7) | (73.2) | (32.9) | (97.7) |
| Capacity Factor | 0.11 | 0.08 | 0.13 | 0.10 | 0.08 | 0.11 | 0.14 | 0.11 | 0.17 |
| | (0.18) | (0.16) | (0.20) | (0.17) | (0.15) | (0.19) | (0.20) | (0.18) | (0.21) |
| Probability Operating | 0.34 | 0.31 | 0.38 | 0.33 | 0.30 | 0.36 | 0.39 | 0.34 | 0.44 |
| | (0.47) | (0.46) | (0.48) | (0.47) | (0.46) | (0.48) | (0.49) | (0.47) | (0.50) |
| Probability Marginal | 0.10 | 0.10 | 0.09 | 0.10 | 0.10 | 0.09 | 0.09 | 0.09 | 0.09 |
| | (0.29) | (0.30) | (0.29) | (0.30) | (0.30) | (0.29) | (0.29) | (0.29) | (0.29) |
| Realized Markup ($/MWh) | -2.78 | -9.76 | 4.20 | -4.33 | -11.38 | 2.74 | 2.74 | -3.95 | 9.45 |
| | (79.88) | (39.97) | (105.23) | (81.49) | (41.38) | (107.13) | (73.53) | (33.84) | (97.96) |
| Resid. Demand Slope (MW per $/MWh) | 1234 | 1443 | 1064 | 1227 | 1435 | 1054 | 1259 | 1470 | 1094 |
| | (703) | (663) | (689) | (705) | (668) | (689) | (695) | (646) | (687) |
| Market Size (GW) | 86.6 | 88.1 | 85.1 | 85.6 | 87.1 | 84.0 | 90.4 | 91.7 | 89.0 |
| | (29.5) | (28.5) | (30.3) | (30.5) | (29.6) | (31.4) | (24.8) | (23.6) | (25.8) |
| **Panel B: Generator Characteristics** | | | | | | | | | |
| Heat Rate (MMBtu/MWh) | | 10.7 | | | 11.0 | | | 9.8 | |
| | | (3.0) | | | (3.0) | | | (3.1) | |
| $CO_2$ Rate (tons/MWh) | | 0.6 | | | 0.6 | | | 0.5 | |
| | | (0.2) | | | (0.2) | | | (0.2) | |
| $NO_x$ Rate (lbs/MWh) | | 1.145 | | | 1.163 | | | 1.079 | |
| | | (1.974) | | | (1.752) | | | (2.625) | |
| $SO_2$ Rate (lbs/MWh) | | 0.010 | | | 0.011 | | | 0.007 | |
| | | (0.044) | | | (0.050) | | | (0.006) | |
| Marginal Cost ($/MWh) | | 31.79 | | | 32.68 | | | 28.63 | |
| | | (11.28) | | | (10.29) | | | (13.84) | |
| Avg. Capacity (MW) | | 368 | | | 334 | | | 488 | |
| | | (447) | | | (451) | | | (413) | |
| **Panel C: Sample Composition** | | | | | | | | | |
| # of Unique Generators | | 553 | | | 432 | | | 121 | |
| # Simple Cycle | | 210 | | | 184 | | | 26 | |
| # Combined Cycle | | 294 | | | 209 | | | 85 | |
| # Steam Turbine | | 49 | | | 39 | | | 10 | |
| # of Counties | | 49 | | | 30 | | | 19 | |
| # in Houston Zone | | 220 | | | 208 | | | 12 | |
| # in North Zone | | 136 | | | 56 | | | 80 | |
| # in South Zone | | 152 | | | 140 | | | 12 | |
| # in West Zone | | 45 | | | 28 | | | 17 | |

*Notes.* This table presents summary statistics of natural gas generators in ERCOT. Each entry reports the mean and counts of natural gas generators, with standard deviations in parentheses. Panels A and B report statistics computed over 2018–2021, the period before significant storage build-out in ERCOT. Panel C reports sample composition over the full estimation period (2018–2024). Columns (1)–(3) report statistics for all natural gas generators; Columns (4)–(6) for generators sharing the same county with storage; Columns (7)–(9) for generators located not in the same county with any storage. Panel A describes the variation in market outcome and characteristics within a day. Panel B describes the generator characteristics and marginal costs. Panel C describes sample composition.

with an average capacity of 366 MW and an average marginal cost of $31/MWh. Geographically, these generators cluster heavily in the Houston, South, and North load zones.

Columns (1)–(3) detail within day variation in market outcomes: later hours (12 PM–12 AM) which include the evening peak hours consistently feature higher nodal prices, capacity factors, and markups compared to early hours. Figure B.4 presents a detailed daily pattern of these outcomes over the years. These elevated afternoon and evening markups link directly to tighter market conditions, specifically, smaller local market sizes and steeper residual demand that enhance generator market power. Columns (4)–(6) present identical statistics for generators that share the same county with storage units, while Columns (7)–(9) present statistics for generators located in different counties from any storage units. Because the operational patterns and market conditions of these groups closely mirror the overall sample, the treatment group in my difference-in-differences design is highly representative of ERCOT natural gas generators. The nodal price spread between early and late periods does not differ significantly between the treatment and control groups. Section 5.2 and Appendix C.1 test for selection in storage siting directly.

### 3.3 Descriptive Statistics of Battery Storage Units

This section describes the characteristics, ownership, siting, and operation of battery storage units in ERCOT.

Storage in ERCOT consists mostly of short-duration batteries. Table 2, Column (1), shows that 13.1 GWh of storage (218 units) is built by the end of my sample, with 12.3 GWh (168 units) operating commercially. The average unit has 49.8 MW of capacity and a 1.36-hour duration. Units are spread across the state, with the largest shares in the West and South load zones.

Storage that shares a county with a natural gas generator provides my identifying variation. Column (2) shows 6.3 GWh of such capacity (106 units), with 6 GWh (75 units) operating commercially. Capacity and duration resemble the full sample. Because gas generators cluster near population centers, these units sit more often in the Houston and South zones and less often in the West. Column (3) shows that the first entrant into each county varies widely in capacity.

Ownership and collocation differ across storage units. Panel B shows that about half of the units belong to storage-only firms, while the rest belong to firms that also own renewables. Storage often shares a node with solar PV, mainly in the North zone, and rarely with wind, which is entirely in the West. Storage and gas units rarely share a node.

Storage operation appears consistent with price arbitrage. Figure 2 shows that units tend to charge when nodal prices are low, demand is low, and renewable output is high. Units also tend to charge during less congested periods (larger local markets) and discharge during congested hours. Woerman (2023) documents that firms exercise more market power in smaller markets. This pattern suggests that storage entry may affect competition with gas generators where market power is greatest.

**Table 2:** Summary statistics of storage units

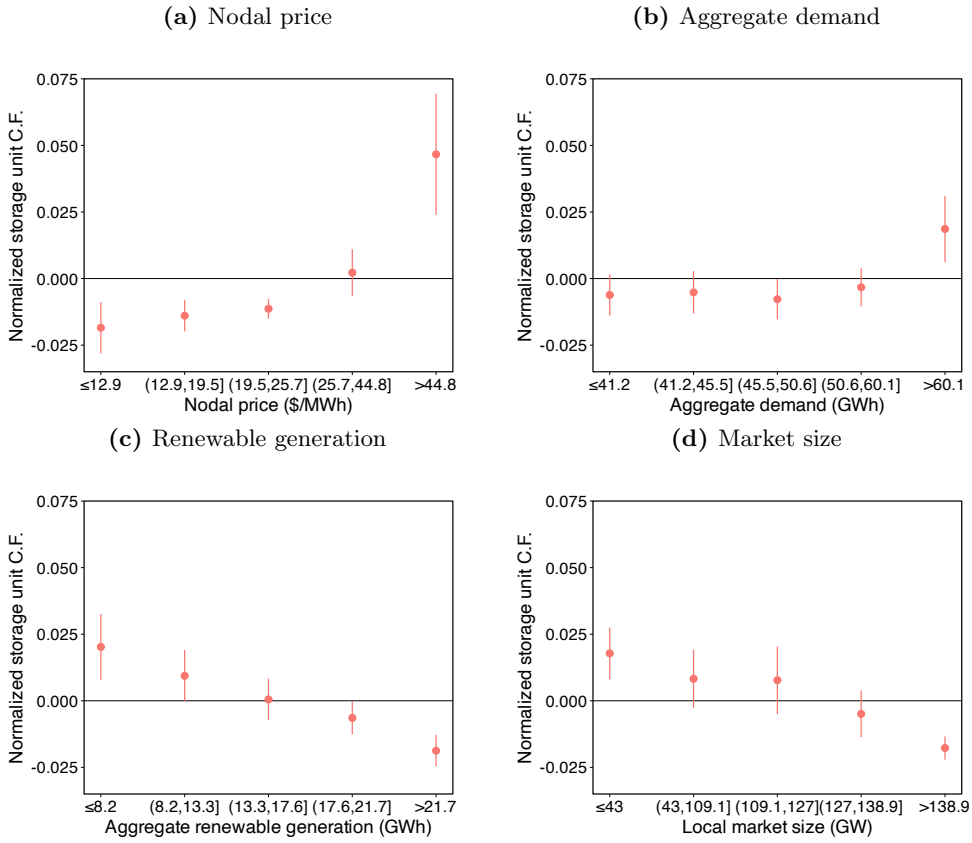
| | All Storage (1) | In County w/ Gas (2) | First in County w/ Gas (3) | Collocated w/ Solar (4) | Collocated w/ Wind (5) |
|---|---|---|---|---|---|
| **Panel A: Storage Characteristics** | | | | | |
| Avg. Capacity (MW) | 48.8 | 51.0 | 61.4 | 61.7 | 48.3 |
| | (51.0) | (53.3) | (61.7) | (38.9) | (46.9) |
| Avg. Energy Capacity (MWh) | 73.5 | 79.6 | 89.1 | 84.4 | 60.8 |
| | (87.4) | (94.5) | (96.5) | (43.4) | (65.8) |
| Avg. Duration (hr) | 1.36 | 1.36 | 1.34 | 1.46 | 1.33 |
| | (0.61) | (0.64) | (0.47) | (0.72) | (0.91) |
| Year of First Appearance | 2022.8 | 2023.0 | 2022.4 | 2022.7 | 2021.5 |
| | (1.7) | (1.6) | (1.7) | (1.5) | (2.3) |
| Year of First Operation | 2023.0 | 2023.0 | 2022.9 | 2023.4 | 2022.5 |
| | (1.6) | (1.4) | (1.6) | (1.5) | (2.7) |
| Cumulative Capacity Built (GWh) | 13.1 | 6.3 | 2.7 | 2.4 | 0.5 |
| Cumulative Capacity Operational (GWh) | 12.3 | 6.0 | 2.7 | 2.0 | 0.5 |
| **Panel B: Ownership Structure** | | | | | |
| By Parent Company | | | | | |
| Share w/ NG Portfolio | 0.07 | 0.07 | 0.07 | 0.04 | 0.22 |
| Share w/ Wind Portfolio | 0.44 | 0.35 | 0.43 | 0.67 | 1.00 |
| Share w/ Solar Portfolio | 0.50 | 0.47 | 0.57 | 1.00 | 1.00 |
| Share Independent | 0.49 | 0.53 | 0.43 | 0.00 | 0.00 |
| By Site Collocation (ERCOT Designation) | | | | | |
| Share Collocated w/ Natural Gas | 0.02 | 0.05 | 0.03 | 0.00 | 0.00 |
| Share Collocated w/ Solar | 0.14 | 0.13 | 0.17 | 1.00 | 0.00 |
| Share Collocated w/ Wind | 0.05 | 0.00 | 0.00 | 0.00 | 1.00 |
| **Panel C: Sample Composition** | | | | | |
| # of Battery Units (Operational) | 168 | 75 | 30 | 24 | 9 |
| # of Battery Units (Built) | 218 | 106 | 32 | 27 | 9 |
| # of Counties | 73 | 30 | 30 | 16 | 6 |
| # in Houston Zone | 19 | 19 | 4 | 2 | 0 |
| # in North Zone | 30 | 16 | 10 | 14 | 0 |
| # in South Zone | 53 | 30 | 12 | 6 | 0 |
| # in West Zone | 66 | 10 | 4 | 2 | 9 |

*Notes.* Summary statistics of ERCOT storage units as of December 1, 2024. I report means and counts, with standard deviations in parentheses. Column (1) includes all units. Column (2) includes units in a county with a natural gas generator. Column (3) includes the first storage unit to enter each of those counties. Columns (4) and (5) include units collocated with solar PV and wind generators.

# 4 Theoretical Framework

I build a simple two-period model to show how storage entry changes the competitive behavior of natural gas generators (gas generators, hereafter). Storage affects a gas generator's markup through two channels. It shifts the *level* of the residual demand the generator faces, which changes the quantity it serves. It also changes the *slope* of that residual demand, which changes how easily the rest of the market substitutes away from it. The appendix extends the model to transmission congestion, strategic storage, and portfolio ownership.

**Figure 2:** Battery storage capacity factor by market conditions




*Notes:* Each panel shows the conditional mean and 95% confidence interval of storage capacity factor by quantile of the explanatory variable. Capacity factor is energy charged or discharged divided by the maximum possible in each interval; positive values indicate charging and negative values indicate discharging. Market size is total installed capacity in the local market. Following Lamp and Samano (2022), I estimate $\text{StorageCF}_{it} = \sum_q \beta_q \mathbb{1}(X_{it} \in q\text{-th quantile of } \mathbf{X}) + \varepsilon_{it}$ for generator $i$ in interval $t$. Nodal price and market size vary by generator and interval; demand and renewable generation vary by interval. Standard errors are two-way clustered by generator and day.

## 4.1 Setup

The model has two periods, $t \in \{L, H\}$, with inelastic load $D_L < D_H$, and each period clears at a single price $p_t$. A competitive fringe of price-taking units has supply $S_t^c(p) = \alpha_t + \gamma p$ with $\gamma > 0$ (Fabra and Imelda, 2023; Ito and Reguant, 2016). A storage fleet charges in the low-demand period ($s_L \leq 0$) and discharges in the high-demand period ($s_H \geq 0$). A strategic gas generator $i$ with marginal cost $c_{it}$ chooses output $q_{it}$ to maximize $\pi_{it} = (p_t - c_{it})\, q_{it}$, taking residual demand as given. The market clears in each period,

$$S_t^c(p_t) + q_{it} + s_t(p_t) = D_t. \tag{1}$$

**Storage supply.** Each battery maximizes arbitrage profit. To be able to sell one MWh in the discharge period, a battery with round-trip efficiency $\eta$ must buy $1/\eta$ MWh in the charge period, so its marginal cost is $m_s = p_L/\eta$. Because the objective is linear in the arbitrage quantity, it

discharges at full capacity when $p_H > m_s$ and stays idle otherwise. A single battery's offer is a step, consistent with awarded storage quantities clustering on the vertical segment of the offer curve observed in the storage bid data (Figure E.22). Batteries differ in efficiency, state of charge, and price expectations, so a higher price pulls successively more of them past their threshold, and the aggregate supply is smooth (Karaduman, 2023),

$$s_t(p_t) = \bar{s}_t + \sigma_t \, p_t, \qquad \sigma_t \geq 0. \tag{2}$$

The level $\bar{s}_t$ is storage that clears at any relevant price. It is positive when storage discharges and negative when it charges. The slope $\sigma_t$ is storage near its threshold, which responds to the price in both periods.

### 4.2 The Generator's Problem

The gas firm's optimal markup is proportional to the quantity it serves and inversely proportional to the slope of its residual demand. Substituting fringe supply and storage into Equation (1), the firm faces residual demand

$$q_{it} = A_t - E_t \, p_t, \qquad A_t = D_t - \alpha_t - \bar{s}_t, \qquad E_t = \gamma + \sigma_t. \tag{3}$$

The first-order condition gives the markup rule

$$\mu_{it} \equiv p_t - c_{it} = \frac{q_{it}}{E_t}, \tag{4}$$

standard in the electricity market literature (Jha and Leslie, 2025; Woerman, 2023; Ryan, 2021; Doshi, 2024; Fabra and Imelda, 2023). Solving jointly with residual demand gives the equilibrium

$$p_t^\star = \tfrac{1}{2}\left(\frac{A_t}{E_t} + c_{it}\right), \qquad q_{it}^\star = \tfrac{1}{2}(A_t - E_t c_{it}), \qquad \mu_t^\star = \frac{A_t}{2E_t} - \frac{c_{it}}{2}. \tag{5}$$

The markup rises with the level of residual demand and falls with its slope. Figure 3 illustrates both channels. I abstract from any effect of storage on the generator's own marginal cost.

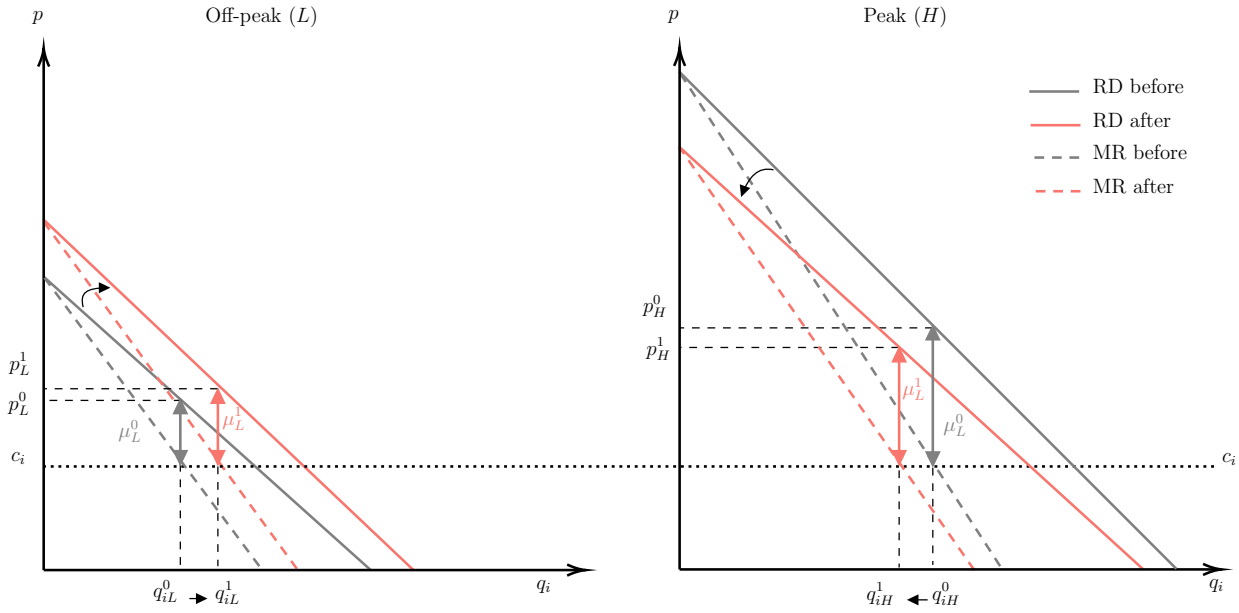
### 4.3 Storage Entry and the Generator's Outcomes

Local storage shifts both the level and slope of residual demand. Let storage capacity, $B$, be the installed battery energy capacity in the generator's local market, thus,

$$\frac{\partial A_H}{\partial B} \leq 0, \qquad \frac{\partial A_L}{\partial B} \geq 0, \qquad \frac{\partial E_t}{\partial B} = \frac{\partial \sigma_t}{\partial B} \geq 0. \tag{6}$$

In this simple model, storage lowers the level of residual demand during the discharge period, raises it when it charges, and flattens it in both periods. The following comparative statics describe how storage affects the generator's output and markup.

**Figure 3:** Illustration of market power effect of storage entry




*Notes:* Residual demand of a natural gas generator before (gray) and after (red) storage entry, in the charge (left) and discharge (right) periods. Setting MR $= c$ yields the markup $\mu$. Storage entry shifts residual demand outward in the charge period and inward in the discharge period, and makes it flatter in both.

**Generation.** Differentiating the equilibrium output with respect to $B$,

$$\frac{\partial q_{it}}{\partial B} = \tfrac{1}{2}\left(\frac{\partial A_t}{\partial B} - c_{it}\frac{\partial E_t}{\partial B}\right), \qquad \frac{\partial^2 q_{it}}{\partial B\,\partial c_{it}} = -\tfrac{1}{2}\frac{\partial E_t}{\partial B} \leq 0. \tag{7}$$

In the discharge period, both terms lower the gas generator's output. In the charge period, the added charging load raises output while the flatter residual demand lowers it, so the sign is an empirical question. The cross-derivative implies that storage displaces higher-cost generators more, because a flatter residual demand leaves less room for units whose cost is close to the price.

**Markup.** Differentiating the markup rule (4) separates the two channels,

$$\frac{\partial \mu}{\partial B} = \underbrace{\frac{1}{E}\frac{\partial q}{\partial B}}_{\text{displacement}} - \underbrace{\frac{q}{E^2}\frac{\partial E}{\partial B}}_{\text{slope}}. \tag{8}$$

The slope term is always negative because a more elastic residual demand raises the cost of withholding. In the discharge period, both terms are negative, so the markup falls. In the charge period, the two terms oppose, and the markup rises only if storage raises the quantity served proportionally more than it flattens residual demand, $\frac{1}{q}\,\partial q/\partial B > \frac{1}{E}\,\partial E/\partial B$.

### 4.4 Summary of Theoretical Results

The model produces three theoretical predictions that motivate the empirical analysis. First, storage flattens the residual demand of gas generators in both periods. Second, in the discharge period, storage lowers both the output and the markup of gas generators, with higher-cost generators losing more output. Third, in the charge period, storage increases load but still flattens residual demand, so the net effects on output and markup are ambiguous, and the markup can rise only if output rises.

## 5 Empirical Strategy

I take the model to the data using a panel of marginal gas generators in ERCOT at the hourly level. I estimate the marginal effect of storage separately by hour of the day, with $q_{iht}$, $\mu_{iht}$, and $c_{it}$ the output, markup, and marginal cost of generator $i$ on date $t$ at hour $h$. Since storage production patterns exhibit a daily cycle, I group hours into charging periods: overnight (midnight to 6 AM), midday (10 AM to 4 PM); and discharging periods: the evening peak (5 PM to 9 PM). I normalize output as a capacity factor, $y_{iht} = q_{iht}/K_i$, where $K_i$ is the capacity of generator $i$ to account for large variation in gas generators capacity in ERCOT. The capacity factor puts every generator on a common scale, and multiplying by $K_i$ recovers the output in MWh.

I employ two research designs. I begin with estimating the optimal output rule from the theoretical model by the instrumental variable approach. The parameter estimates describe how generator output responds to its own cost as the model requires, and motivate the interaction between storage and cost that informs the second design. I then employ a continuous two-stage difference-in-differences design as the main design for all outcomes: output, operating states, the residual demand slope, markups, and operating profit.

### 5.1 Displacement of Gas Generation

I bring the theoretical prediction from the previous section to the data. The generator's optimal output in Equation (5) is linear in its marginal cost, $q^\star = \frac{1}{2}(A - E\,c)$. Dividing by capacity keeps the rule linear, $y^\star = \frac{1}{2}(\tilde{A} - \tilde{E}c)$ with $\tilde{A} = A/K$ and $\tilde{E} = E/K$. I assume local storage shifts the intercept and slope linearly at hour $h$, $\tilde{A}_h(B) = a_h + a'_h B$ and $\tilde{E}_h(B) = e_h + e'_h B$. Substituting and writing $c = \bar{c}_k + \tilde{c}$, with $\bar{c}_k$ the mean marginal cost of technology $k$, gives

$$y^\star_h = \tfrac{1}{2}\left(a_h - e_h \bar{c}_k\right) + \tfrac{1}{2}\left(a'_h - e'_h \bar{c}_k\right) B - \tfrac{1}{2} e_h\, \tilde{c} - \tfrac{1}{2} e'_h\, B\, \tilde{c}. \tag{9}$$

The coefficient on $B$ is the displacement at the typical cost of the technology, and the coefficient on $B\tilde{c}$ shows how displacement varies with cost. Because $e'_h \geq 0$ in the model, higher-cost generators are displaced more.

I estimate the empirical analog of Equation (9) on interior hours,

$$y_{iht} = \delta^{(1)} B_{m(i),t} + \delta^{(2)} \tilde{c}_{it} + \delta^{(3)} \left( B_{m(i),t} \tilde{c}_{it} \right) + \mathbf{X}_{iht}\theta + \alpha_i + \nu_{\ell(i),y(t),h} + u_{iht}. \tag{10}$$

The outcome $y_{iht}$ is the capacity factor in percentage points. The treatment $B_{m(i),t}$ is the cumulative storage energy capacity in the generator's county $m(i)$, in units of 100 MWh. The cost term $\tilde{c}_{it} = c_{it} - \bar{c}_k$ is marginal cost in \$/MWh, centered at the mean of its technology on interior hours. The generator fixed effect $\alpha_i$ absorbs permanent differences across generators, and the load-zone-by-month-by-hour fixed effect $\nu_{\ell(i),y(t),h}$ absorbs regional shifts in load and fringe supply. The controls $\mathbf{X}_{iht}$ are system net load, a cubic in local temperature, and local renewable capacity and potential output: county solar and wind capacity, and their products with local irradiance and a normalized wind power curve. I cluster standard errors by county. I estimate the equation pooled, by technology $k \in \{\text{CC}, \text{SC}, \text{GS}\}$, and by technology and hour. Because some estimates rely on as few as 18 county clusters, I construct confidence intervals with critical values from a $t$ distribution with $G - 1$ degrees of freedom, where $G$ is the number of counties behind each estimate (Bester, Conley and Hansen, 2011; Colin Cameron and Miller, 2015).

Interior hours are hours in which the generator runs strictly between its low and high sustained limits. The first-order condition in Equation (4) holds with equality only in the interior. At the low sustained limit, a committed generator's output reflects the startup and minimum-load costs of staying online (Jha and Leslie, 2025), and at the high sustained limit, capacity binds.

Marginal cost is potentially endogenous. Realized heat rates depend on realized output, so the constructed cost is mechanically a function of output (Demirer and Karaduman, 2026). The constructed cost does not factor in maintenance, startup, and ramping costs, so it is a noisy proxy. The identity of the marginal unit might respond to the shocks that enter $u_{iht}$.[14] I instrument $\tilde{c}_{it}$ and $B_{m(i),t}\tilde{c}_{it}$ with $\tilde{z}_{it}$ and $\tilde{z}_{it}B_{m(i),t}$, where $z_{it} = \text{HR}_i^{\text{pre}} \times P_t^{\text{gas}}$ is the product of the generator's pre-entry heat rate and the Henry Hub gas price, centered within technology. The instrument shifts cost strongly, because marginal cost is dominated by fuel cost which by construction is heat rate times gas price. The pre-entry heat rate does not move with the current operating point, which breaks the mechanical link between cost and output.

Identification rests on four assumptions. Let $W_{iht}$ collect the fixed effects and controls. The instruments must satisfy relevance and exclusion restriction, $\mathbb{E}[\tilde{z}_{it}u_{iht} \mid W_{iht}] = 0$ and $\mathbb{E}[\tilde{z}_{it}B_{m(i),t}u_{iht} \mid W_{iht}] = 0$. Storage must be conditionally exogenous, $\mathbb{E}[B_{m(i),t}u_{iht} \mid W_{iht}] = 0$, and the generator must optimize in the interior.

There are several potential threats to these identification assumptions. First, the Henry Hub price shifts the costs of all gas generators at once and may therefore shift residual demand directly; replacing the month fixed effects with load-zone-by-date-by-hour fixed effects absorbs this common shock. Second, storage enters the estimating equation as an exogenous regressor, so its coefficient

[14] Ganapati, Shapiro and Walker (2020) instrument estimated marginal costs with energy-price shocks to address measurement error, and Fabra and Reguant (2014) treat hourly marginal cost as endogenous because the identity of the marginal unit is set by unobserved supply and demand shocks.

is exposed to comparisons with already-treated counties and to siting on local trends; the two-stage design below addresses both. Third, storage can also change which generators operate in the interior, so I report how sensitive the estimate is to excluding hours near the operating limits and estimate the effect on operating states directly. Finally, the coefficients in Equation (9) equal the residual demand slope only under constant marginal cost. With increasing marginal cost, $MC = c + kq$, the cost coefficient identifies $E/(2 + Ek)$, which approaches $1/k$ when residual demand is flat. I therefore use this specification to test for the sign of the cost interaction and measure the slope directly.

### 5.2 The Two-Stage Difference-in-Differences Design

The treatment is the entry of storage into a generator's county. I define the commercial operation date $\text{COD}_{m(i)}$ as the first date a same-county storage unit is awarded a positive quantity in the real-time market. The treatment is the continuous dose $B_{m(i),t}$, and $\mathbb{1}\{B_{m(i),t} > 0\}$ is the entry indicator. The county boundary approximates the competitive neighborhood that transmission constraints create (Figure E.24).

For an hourly outcome $Y_{iht}$, the estimating equation is

$$Y_{iht} = \sum_{h=0}^{23} \beta_h \, \mathbb{1}\{\text{hour} = h\} \, B_{m(i),t} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht}, \tag{11}$$

where $\eta_{\ell(i),t,h}$ is a load-zone-by-date-by-hour fixed effect and $\mathbf{X}_{iht}$ contains the same controls as in Equation (10). For the capacity factor, I carry the cost gradient from the structural model into this design by adding an interaction with predetermined fuel cost,

$$y_{iht} = \sum_{h=0}^{23} \mathbb{1}\{\text{hour} = h\}\big(\beta_h B_{m(i),t} + \gamma_h \, \tilde{z}_{it} B_{m(i),t}\big) + \lambda \tilde{z}_{it} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht}. \tag{12}$$

The coefficient $\beta_h$ is the displacement at mean predetermined cost, and $\gamma_h$ is the reduced-form analog of $\delta^{(3)}$. I use predetermined cost rather than realized cost because realized cost responds to storage. I cluster standard errors at the county level.

Counties adopt storage at different dates and the dose grows over time. Under this staggered timing, the two-way fixed-effects estimator uses already-treated counties as controls, which can attenuate or reverse the sign of the true effect when effects are heterogeneous (Goodman-Bacon, 2021; De Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak, Jaravel and Spiess, 2024). I therefore use the two-stage estimator of Gardner et al. (2024). The first stage estimates the fixed effects and control coefficients, including $\lambda$, on never-treated and not-yet-treated observations. The second stage regresses the first-stage residuals on the storage terms. The standard errors account for the first stage.

With a continuous and time-varying dose, $\beta_h$ is a positively weighted average of the local slopes

of the dose-response function (Yitzhaki, 1996; Gardner et al., 2024),

$$\hat{\beta}_h = \mathbb{E}\left[\sum_d w(d)\,\frac{\partial\,\mathbb{E}[Y_{iht}(d) - Y_{iht}(0) \mid B_{m(i),t} = d]}{\partial d}\right], \tag{13}$$

where the weights $w(d) \geq 0$ reflect the distribution of the dose among treated generators. The coefficient is therefore a marginal effect scaled to per 100 MWh rather than the effect of moving from no storage to 100 MWh. The identification results of Callaway, Goodman-Bacon and Sant'Anna (2024) assume a dose fixed at a single value per unit, which does not hold in this setting.

Identification rests on three assumptions. The first is parallel trends: absent local entry, the untreated outcome is additive in the fixed effects and controls, $\mathbb{E}[Y_{iht}(0) \mid \alpha_i, \eta_{\ell(i),t,h}, \mathbf{X}_{iht}] = \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta$. The second is no anticipation, $Y_{iht}(d) = Y_{iht}(0)$ for $t < \text{COD}_{m(i)}$. The third is no interference, so a generator's outcome depends only on its own county's dose.

I examine whether there is parallel trends in the pre-treatment period with an event study that replaces the dose with indicators for years relative to entry,

$$Y_{iht} = \sum_{\substack{k=-3 \\ k\neq -1}}^{3} \tau_k\,\mathbb{1}\{K_{m(i),t} = k\} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht}, \tag{14}$$

where $K_{m(i),t}$ counts years relative to entry, binned up to three years pre and post, and the year before entry is the reference. I estimate it by hour group and at the daily level for operating profit. Anticipation is unlikely, because storage appears in the SCED data long before commercial operation, with a median gap of 204 days (Figure E.25), and the data are published with a sixty-day lag. The remaining concern is that storage enters where local conditions are changing. The generator and load-zone-by-date-by-hour fixed effects absorb permanent and regional differences, and the event study then provides a falsification test for county-specific trends. Storage is also often sited alongside new solar capacity, whose output displaces gas generation directly. To address this I control for the local renewable capacity and potential.

I also test directly whether storage enters where prices or gas conduct make entry attractive (Appendix C.1). Storage siting is not random across Texas. Counties with more solar capacity and wider pre-period price spreads receive more storage, and receive it earlier (Table C.1). Solar capacity is the strongest predictor, consistent with the co-location pattern above. Among counties with gas generation, pre-period prices do not predict whether a county receives storage (joint $p = 0.30$), although they still predict how much and when. The county fixed effects absorb these level differences. The estimates would be biased only if storage selects on trends, and I find no evidence that it does. County price spreads show no trend before entry (joint $p = 0.43$, Figure C.5). Within counties, batteries also do not locate closer to gas units with higher pre-period markups or prices (joint $p = 0.33$, Table C.2). The markup coefficient is close to zero and precisely estimated, so siting does not select on the conduct I measure.

For output, the outcome is the capacity factor, including hours with zero output, estimated

by technology and weighted by capacity. Weighting makes $\beta_h$ a capacity-weighted average of generator-level effects, so multiplying it by capacity and summing recovers the change in MWh. This is the displacement I carry into the welfare analysis. I also estimate the effect on indicators for running, running at minimum load, running in the interior, and running at maximum output, which show how storage changes the way generators operate. For the residual demand slope, the outcome $E_{iht}$ is the slope of the local residual demand curve constructed from the offer stack.[15] For the markup, the outcome is $\mu_{iht} = p_{iht} - c_{it}$, set to zero when the generator does not run. For operating profit, I aggregate to the generator-day and use daily operating profit per MW of capacity, $\pi_{it} = \sum_h \mu_{iht} q_{iht} / K_i$, weighted by capacity, with load-zone-by-date fixed effects,

$$\pi_{it} = \beta^{\pi} B_{m(i),t} + \alpha_i + \eta_{\ell(i),t} + \mathbf{X}_{it}\theta + \varepsilon_{it}. \tag{15}$$

I also estimate (15) separately for the competitive rent and the market power rent defined in Section 6.2. Because the estimator is linear, the two effects sum to $\beta^{\pi}$.

# 6 Results

## 6.1 Displacement results

This section presents the estimates on output and the slope of the residual demand of gas generators. First, I compare the structural estimates of Equation (10) with the two-stage difference-in-differences (2SDD) estimates. I then present the impact on hourly output by technology, the operating margins, and the residual demand slope. I then estimate the aggregate displacement that informs the welfare analysis.

Table 3 presents the pooled estimates. Column (1) and Column (2) present the OLS and IV estimates of Equation (10) on interior hours. An additional 100 MWh of local storage lowers the capacity factor of a generator with an average marginal cost by 0.79 percentage points. The OLS and IV estimates are nearly identical, which suggests that simultaneity in the constructed marginal cost is modest. The instruments are highly relevant, yielding first-stage cluster-robust $F$-statistics of 327 for cost and 2,447 for the interaction, with partial $R^2$ of 0.40 and 0.81. The cost coefficient is negative consistent with the optimal output rule. The interaction term is also negative, so storage displaces higher-cost generators more.

Column (3) applies the two-stage estimator to the same interior sample. The storage coefficient is $-0.55$ (s.e. 0.17), about 70 percent of the IV estimate, and the confidence intervals overlap. The two-stage estimator uses only generators in not-yet-treated counties as controls, so it avoids the comparisons with already-treated generators that affect the storage coefficient in Equation (10). The interaction with the predetermined cost shifter is $-0.013$ (s.e. 0.003), close to the structural interaction of $-0.016$. This implies that higher-cost generators are displaced more under both designs. Column (4) is my preferred specification for the welfare analysis. It includes all hours,

[15] Section 3 details the construction of $E_{iht}$. A larger value denotes a flatter, more elastic residual demand.

**Table 3:** Local storage effect on capacity factor: structural and two-stage estimates

| | Capacity factor (pp) | | | |
|---|---|---|---|---|
| | OLS | IV | 2SDD | 2SDD |
| | (1) | (2) | (3) | (4) |
| Storage $B$ (100MWh) | -0.785*** | -0.790*** | -0.553*** | -0.904*** |
| | (0.291) | (0.292) | (0.165) | (0.266) |
| Cost $\tilde{c}$ ($/MWh) | -0.070*** | -0.070*** | | |
| | (0.019) | (0.017) | | |
| Cost $\tilde{c} \times B$ | -0.016*** | -0.016*** | | |
| | (0.004) | (0.003) | | |
| Predetermined cost $\tilde{z} \times B$ | | | -0.013*** | -0.004 |
| | | | (0.003) | (0.008) |
| Generator FE | ✓ | ✓ | ✓ | ✓ |
| Load zone-Month-Hour FE | ✓ | ✓ | | |
| Load zone-Date-Hour FE | | | ✓ | ✓ |
| Local renewable capacity and potential | ✓ | ✓ | ✓ | ✓ |
| Local temperature | ✓ | ✓ | ✓ | ✓ |
| Sample | Interior | Interior | Interior | All hours |
| Weights | None | None | None | Capacity |
| Observations | 3,770,307 | 3,770,307 | 3,705,777 | 13,472,250 |
| Counties | 49 | 49 | 49 | 49 |
| First-stage $F$, $\tilde{c}$ | | 327.3 | | |
| First-stage $F$, $\tilde{c} \times B$ | | 2447.3 | | |
| Partial $R^2$, $\tilde{c}$ | | 0.397 | | |
| Partial $R^2$, $\tilde{c} \times B$ | | 0.810 | | |

*Notes.* The outcome is the hourly capacity factor in percentage points. $B$ is cumulative storage energy capacity in the generator's county in 100 MWh. Columns (1) and (2) estimate Equation (10), with marginal cost centered within technology and instrumented in Column (2) by the pre-entry heat rate times the Henry Hub price. Columns (3) and (4) use the two-stage difference-in-differences estimator, with storage interacted with predetermined cost, the pre-entry heat rate times the Henry Hub price, centered within technology. Interior hours have output strictly between the low and high sustained limits. Standard errors clustered by county in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

including hours with zero output and hours at an operating limit, and weights by capacity. Storage lowers the capacity factor by 0.90 percentage points (s.e. 0.27).

The two-stage estimate is more robust to the sample of operating hours. Appendix Table C.3 shows that excluding hours within 2 percent of capacity of either operating limit shrinks the IV estimate to $-0.26$ and makes it insignificant. The same restriction leaves the two-stage estimate at $-0.44$ and significant (Appendix Table C.5). Much of the IV estimate is driven by generators operating close to their limits. The IV estimate is otherwise stable across choice of fixed effect and sample weights. It is $-0.81$ when I include hours at the operating limits, $-0.66$ with capacity weights, and $-0.81$ with load-zone-by-date-by-hour fixed effects that absorb the common natural gas price shock.

Table 4 presents the welfare specification by technology. Storage lowers the capacity factor of combined cycle generators by 0.37 percentage points (s.e. 0.17), or about 3.4 MWh at their mean capacity of 921 MW. The estimates for simple cycle ($-0.93$) and steam turbine ($-0.09$) generators are not statistically significant. The pooled estimate of $-0.90$ exceeds the combined cycle and steam

turbine estimates because the pooled first stage shares fixed effects across technologies and so also compares generators of different types. For aggregation, I therefore use the technology-specific estimates.

**Table 4:** Local storage effect on capacity factor by technology

| | Capacity factor (pp) | | | |
|---|---|---|---|---|
| | All gas | Combined Cycle | Simple Cycle | Steam Turbine |
| | (1) | (2) | (3) | (4) |
| Storage $B$ (100MWh) | -0.904*** | -0.367** | -0.931 | -0.090 |
| | (0.266) | (0.172) | (0.573) | (0.496) |
| Predetermined cost $\tilde{z} \times B$ | -0.004 | -0.013 | -0.007 | 0.011 |
| | (0.008) | (0.012) | (0.009) | (0.012) |
| Generator FE | ✓ | ✓ | ✓ | ✓ |
| Load zone-Date-Hour FE | ✓ | ✓ | ✓ | ✓ |
| Local renewable capacity and potential | ✓ | ✓ | ✓ | ✓ |
| Local temperature | ✓ | ✓ | ✓ | ✓ |
| Sample | All hours | All hours | All hours | All hours |
| Weights | Capacity | Capacity | Capacity | Capacity |
| Observations | 13,472,250 | 3,540,612 | 7,155,074 | 2,424,701 |
| Counties | 49 | 32 | 24 | 18 |

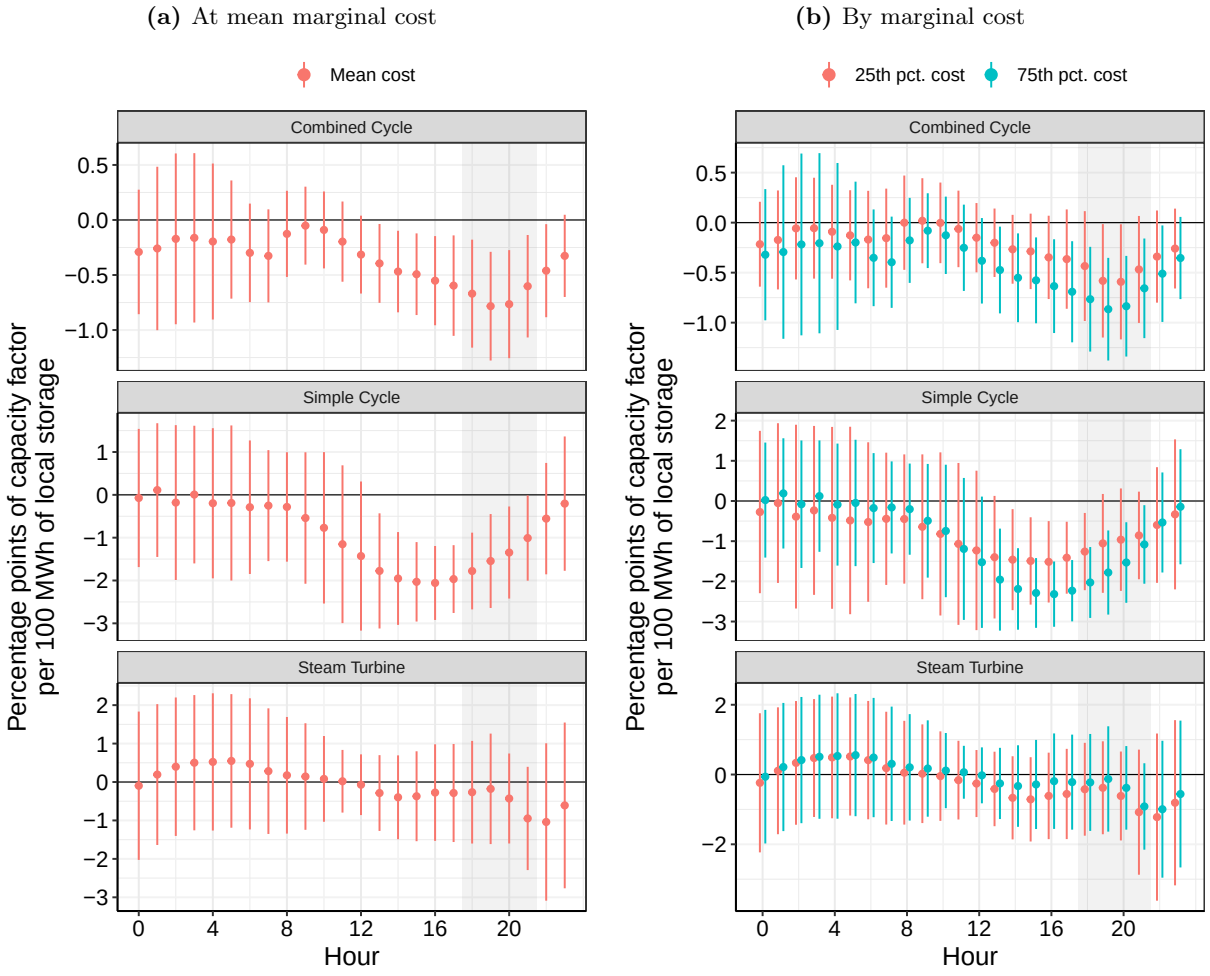
*Notes.* Two-stage difference-in-differences estimates on all hours, weighted by nameplate capacity. Column (1) pools all gas generators and repeats Column (4) of Table 3. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Hourly generation displacement** Storage displaces gas generation in the afternoon and evening. Figure 4 presents the hourly two-stage estimates by technology. Combined cycle output falls significantly from 1 PM to 10 PM, with the largest effect of 0.78 percentage points at 7 PM, or about 7.2 MWh at mean capacity. Simple cycle output falls significantly from 1 PM to 9 PM, with the largest effect of 2.06 percentage points at 4 PM, or about 2.8 MWh at mean capacity. None of the overnight estimates is significant, and none of the steam turbine estimates is significant in any hour. The confidence intervals use $t_{G-1}$ critical values, where $G$ is the number of counties behind each estimate.

The theoretical model predicts that storage displaces higher-cost generators more. Panel (b) of Figure 4 evaluates the displacement at the 25th and 75th percentiles of predetermined cost within each technology. Consistent with the prediction, the higher-cost simple cycle generator loses 0.6 to 0.8 percentage points more capacity factor than the lower-cost one from 1 PM to 8 PM, and the cost interaction is significant in these hours. The higher-cost combined cycle generator loses 0.1 to 0.3 percentage points more in every hour, although the interaction is not significant hour by hour. Steam turbines show the opposite gradient, which is significant only in a few overnight and morning hours. Event studies by technology show no pre-trends in output (Appendix Figure C.8).

I perform two robustness checks that support a causal reading of the afternoon displacement. First, the local renewable controls matter little for the afternoon estimates (Appendix Figure C.9). They lower the combined cycle estimates by about 0.1 percentage points in every hour and remove

**Figure 4:** Hourly displacement by technology




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, in percentage points, by hour and technology, on all hours and weighted by capacity. Panel (a) evaluates the effect at the mean predetermined cost of each technology. Panel (b) evaluates it at the 25th and 75th percentiles of predetermined cost. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level. The shaded band marks the evening peak.

most of the overnight simple cycle displacement. The afternoon simple cycle displacement changes little, from 2.3 to 2.0 percentage points at 4 PM. This implies that the afternoon displacement is unlikely to be driven by solar capacity that enters alongside storage. Second, the displacement is larger in the Houston and North load zones, where it reaches 1.5 percentage points in the evening, than in the wind-rich West and South zones. It is also larger on days with high solar or wind output, when arbitrage is more valuable (Appendix Figure C.10).

**Operating margins** Table 5 presents estimates on operating margin. For combined cycle, the evening displacement works through fewer hours at maximum output, which fall by 1.4 percentage points. For simple cycle, storage lowers the probability of running at minimum load during midday by 1.6 percentage points. Steam turbines behave differently. In the evening peak, storage increases

**Table 5:** Effect of local storage on operating margin

| | Running | At minimum load | Interior | At maximum output |
|---|---|---|---|---|
| *Combined Cycle* | | | | |
| Overnight (0–6) | -0.390 | -0.424 | -0.520 | 0.554 |
| | (0.296) | (0.481) | (0.653) | (0.492) |
| Midday (10–16) | -0.162 | -0.032 | 0.301 | -0.431 |
| | (0.274) | (0.323) | (0.520) | (0.457) |
| Evening (17–21) | -0.408 | 0.276 | 0.697 | -1.381** |
| | (0.278) | (0.311) | (0.632) | (0.640) |
| *Simple Cycle* | | | | |
| Overnight (0–6) | 2.585 | -0.901 | 3.510 | -0.024 |
| | (3.174) | (0.570) | (3.668) | (0.252) |
| Midday (10–16) | 1.899 | -1.593** | 3.250 | 0.242 |
| | (2.087) | (0.757) | (2.622) | (0.424) |
| Evening (17–21) | 1.549* | -1.516 | 2.433 | 0.632 |
| | (0.857) | (1.037) | (2.283) | (1.165) |
| *Steam Turbine* | | | | |
| Overnight (0–6) | 1.109* | -0.233 | 2.125*** | -0.784* |
| | (0.569) | (1.065) | (0.720) | (0.374) |
| Midday (10–16) | 1.651* | -0.047 | 1.706** | -0.008 |
| | (0.817) | (0.655) | (0.806) | (0.343) |
| Evening (17–21) | 1.712*** | 0.637 | 2.672*** | -1.598*** |
| | (0.490) | (0.625) | (0.744) | (0.338) |

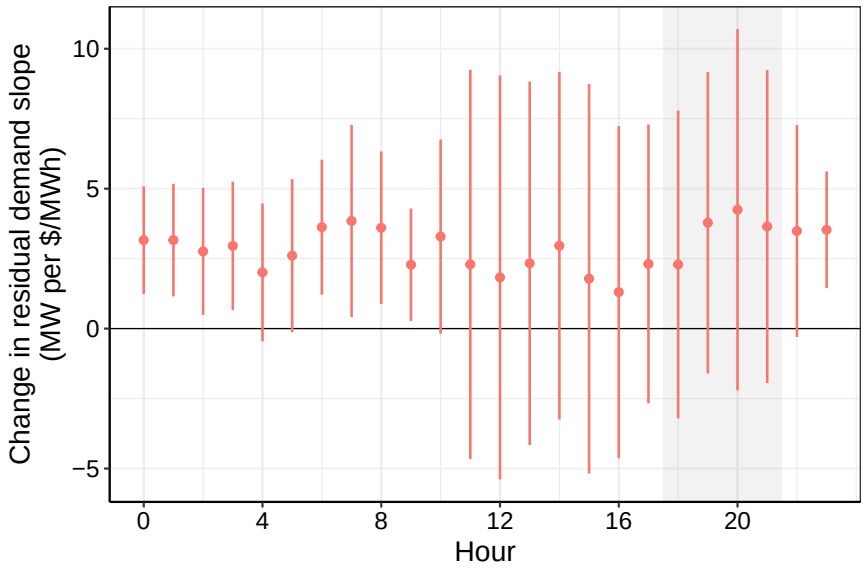
*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on indicators for running, running at minimum load, running in the interior, and running at maximum output, in percentage points, on hours with both operating limits observed. The last three columns sum to the first. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the probability that a steam turbine runs by 1.7 percentage points and runs in the interior by 2.7 percentage points, while lowering the probability of running at maximum output by 1.6 percentage points. Steam turbines run more but further from their maximum output.

**Residual demand slope** The model predicts that storage makes the residual demand of gas generators flatter. To test this prediction, I use the slope of the local residual demand constructed from the bid data as the outcome in the two-stage design. Before storage entry, the slope is about 1,300 MW per $/MWh overnight and about 750 to 900 MW per $/MWh in the afternoon and evening. Figure 5 shows that each 100 MWh of local storage increases the slope by 1.3 to 4.3 MW per $/MWh, or 0.2 to 0.5 percent of the baseline. The effect is statistically significant overnight and in the morning, and positive but imprecisely estimated in the evening. The event study estimates exhibit no pre-trend in any hour group (Appendix Figure C.12). Across technology, the change in slope is stronger for combined cycle generators in the evening, when the model predicts market power binds. Their residual demand becomes flatter by 5.5 to 8.6 MW per $/MWh per 100 MWh of storage between 6 PM and 11 PM (Appendix Figure C.11).

Both the inferred residual demand slope from IV estimates and two-stage estimates on the

**Figure 5:** Effect of local storage on the residual demand slope




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the slope of the local residual demand curve, in MW per $/MWh, by hour, for all gas generators. A larger value is a flatter, more elastic residual demand. The slope is winsorized at the 99th percentile within technology. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

measured slope agree in sign. The negative interaction implies a flatter residual demand for combined cycle and simple cycle generators. The measured slope agrees for combined cycle generators overnight and in the evening, where the effect of 7.5 MW per $/MWh is significant, and for simple cycle generators at midday and in the evening. In magnitude, the slope implied by the cost coefficient is about 1.7 MW per $/MWh for a combined cycle generator at mean capacity, roughly three orders of magnitude below the measured slope of 1,300 MW per $/MWh. The static model with constant marginal cost cannot explain this discrepancy. With increasing marginal cost, $MC = c + kq$, the optimal output becomes $q^\star = (A - Ec)/(2 + Ek)$, and when residual demand is flat the cost response approaches $-1/k$. The cost coefficient then reflects the curvature of the generator's own cost, factoring in heat rate curvature and ramping and minimum load constraints, rather than the residual demand alone. Therefore, I use the measured slope for the slope channel and interpret the structural interaction only for its sign.

**Counterfactual output** I construct counterfactual output without storage for each generator and hour as observed output minus the estimated effect of local storage. The counterfactual uses the unit-specific effect from the welfare specification, which lets displacement vary with each generator's predetermined cost. Output without storage cannot be negative or exceed capacity, so I bound the counterfactual at these limits. The lower bound binds when a generator is off and the estimate implies that storage raised its output, which would make output without storage negative. The upper bound almost never binds. The lower bound binds in 30 percent of generator-hours with

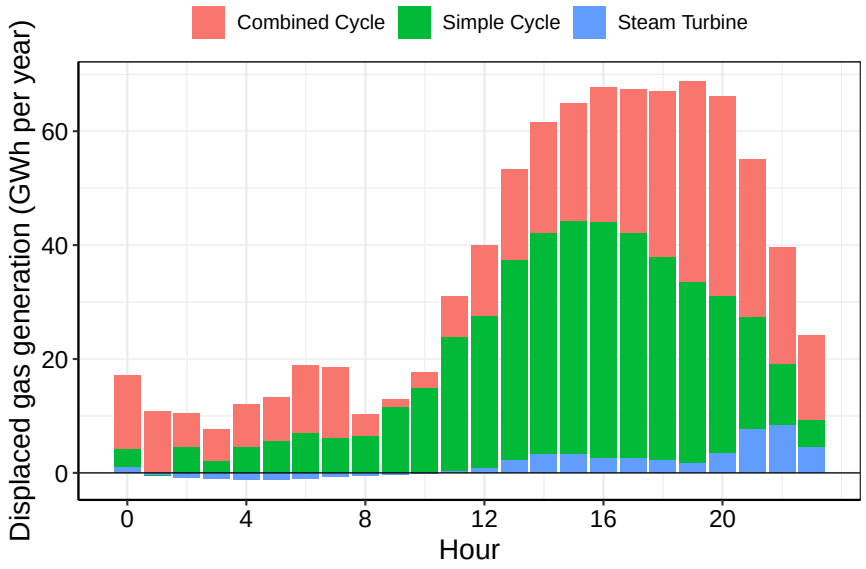
local storage for steam turbines, 7 percent for simple cycle units, and about 1 percent for combined cycle units. It raises displacement from 811 GWh to 849 GWh per year, mostly among steam turbines. Because the bound makes the totals nonlinear in the coefficients, I compute standard errors by simulation. I draw the hourly coefficients 500 times from their estimated joint normal distribution, recompute the bounded totals in each draw, and report the standard deviation across draws (Krinsky and Robb, 1986). Appendix Table C.7 reports the unbounded totals.

**Aggregate displacement** Local storage displaced about 849 GWh of gas generation per year over the sample (s.e. 305). Of the total, 371 GWh came from combined cycle generators and 440 GWh from simple cycle generators, which are 0.25 and 2.84 percent of their observed output. Each is significant at the 10 percent level, and the total is significant at the 5 percent level. The steam turbine estimate of 38 GWh is not distinguishable from zero, and the bound adds 27 GWh of it, so I do not interpret it. Figure 6 shows that the displaced generation is concentrated in the afternoon and evening. Average installed storage in counties with gas generation was 1,170 MWh, so the estimates imply about 73 GWh of displaced gas generation per 100 MWh of storage per year.

The point estimate is large relative to what storage itself supplies. Storage in counties with gas generation discharged 196 GWh per year, about 0.46 full cycles per day, so the estimated displacement is more than four times its discharge. Even the lower bound of the 95 percent confidence interval, about 250 GWh per year, exceeds that discharge. Two mechanisms can explain this. First, storage might relieve local congestion and allow imported energy to replace local gas generation. Gas generation might also shift to counties without storage, which the county-level design cannot separate from a true reduction. I therefore treat the aggregate as an upper bound on the storage-induced gas generation displacement. The welfare analysis in Section 7 values storage by its observed operations rather than by this displacement for the same reason.

**Summary of displacement results** Overall, both empirical strategies provide consistent evidence that battery storage displaces natural gas generation during peak hours. The structural estimates confirm that generator output is responsive to marginal costs, consistent with standard optimal bidding behavior, and that storage disproportionately displaces higher-cost units. The two-stage estimates confirm the displacement and the cost gradient, and they remain robust even when excluding observations near generators' operational limits. Furthermore, estimates of the residual demand slope show that storage flattens local residual demand, consistent with the slope channel. For the subsequent welfare analysis, I rely on the technology-specific, capacity-weighted two-stage estimates of the capacity factor. Because this specification holds the generator-hour population fixed and retains observations at zero output or physical limits, it correctly aggregates to the total change in generation (MWh) that determines avoided emissions and the comparison measure of production cost.

**Figure 6:** Gas generation displaced by local storage, by hour




*Notes.* Annual displaced generation, in GWh, from the technology-specific two-stage estimates by hour, on all hours and weighted by capacity. Counterfactual output is observed output net of the estimated storage contribution, bounded between zero and capacity.

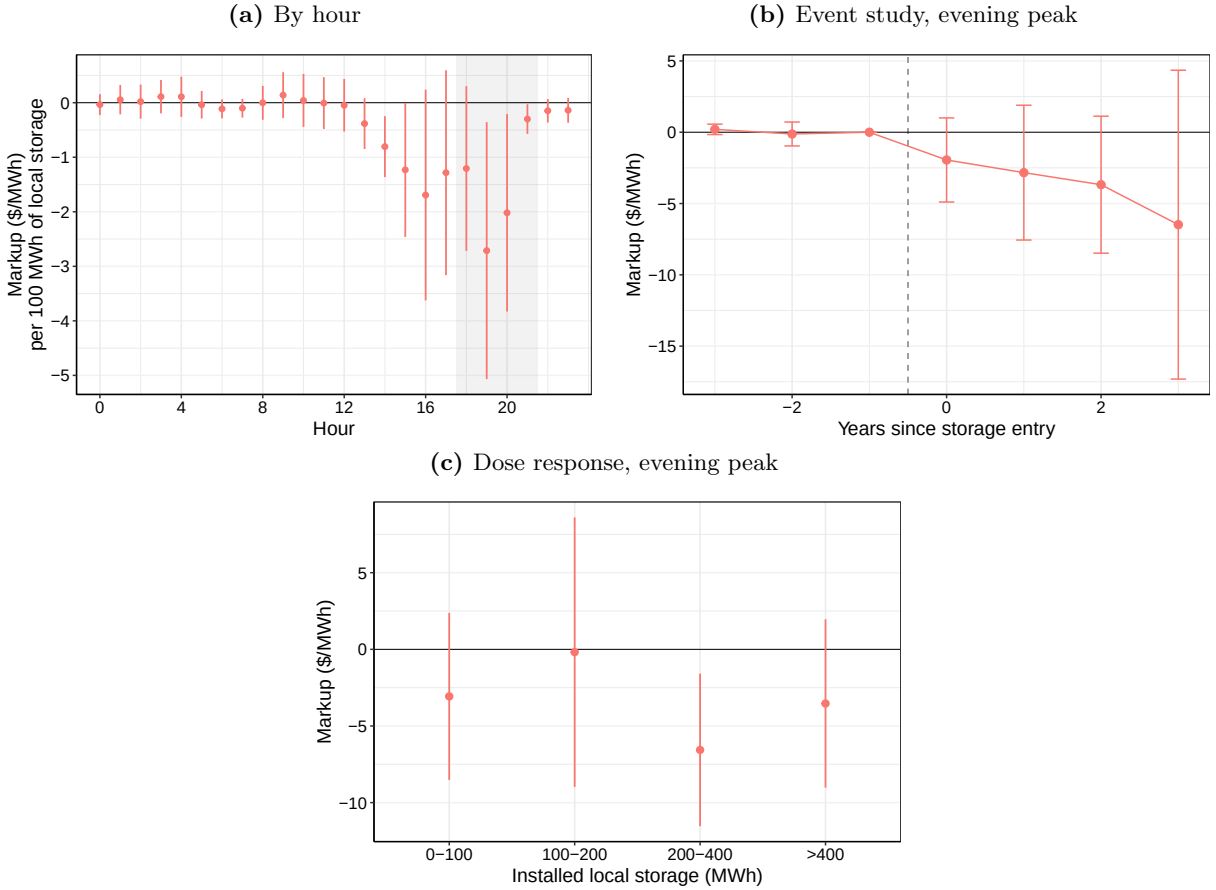
## 6.2 Market Power Results

This section evaluates how local battery storage affects the markups and operating profits of incumbent natural gas generators, decomposing this profit impact into competitive and market power rents. I first present the treatment effects on generator markups, followed by the hourly and daily impacts on operating profits. After decomposing these profit changes, I conclude by estimating the aggregate rent transfer away from natural gas generators. Throughout this section, all treatment effects are scaled per 100 MWh of installed local storage capacity, and all values are reported in 2020 dollars.

**Markups** Storage lowers the markups of gas generators in the late afternoon and evening. Figure 7 presents the two-stage estimates on the markup. Panel (a) shows that the markup does not change overnight or in the morning. Markup begins to fall at 1 PM, and the effect is largest at 7 PM, when an additional 100 MWh of local storage lowers the markup by \$2.71 per MWh (s.e. 1.17). The effect is also significant at 2 PM, 3 PM, 8 PM, and 9 PM. Over the evening peak from 5 PM to 9 PM, the markup falls by \$1.50 per MWh on average, about 5 percent of the pre-entry evening markup of \$31 per MWh. Panel (b) presents the corresponding event study estimates for the evening peak. The pre-treatment coefficients are near zero and statistically insignificant, supporting the identifying assumption. The post-entry coefficients grow more negative over time, although they are imprecisely estimated. Finally, Panel (c) demonstrates that this markup compression occurs across all dose bins, yielding an average marginal evening effect of $-1.31$ (s.e. 0.67)

per 100 MWh. The pattern is consistent across technologies (Appendix Figure C.13), with evening effects of −0.92 for combined cycle, −1.57 for simple cycle, and −1.70 for steam turbine generators.

**Figure 7:** Effect of local storage on markups

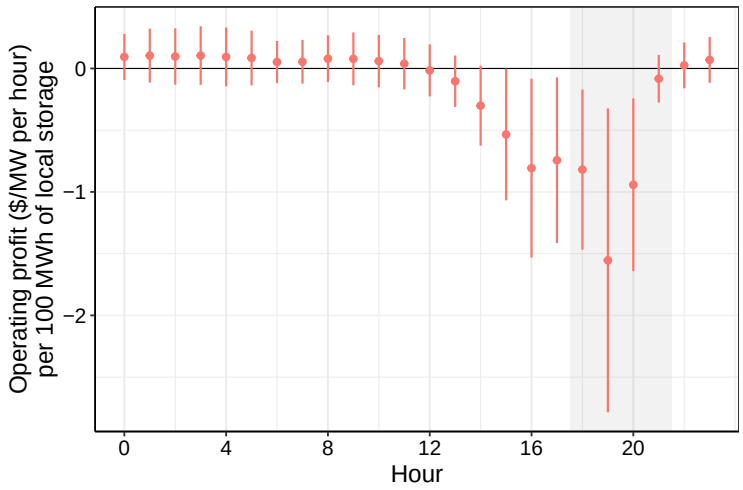



*Notes.* Two-stage difference-in-differences estimates. The markup is the nodal price minus marginal cost, set to zero when the generator does not run. Panel (a) reports the effect of 100 MWh of local storage by hour. Panel (b) reports a binary event study for 5 PM to 9 PM, with the year before entry as the reference. Panel (c) reports the effect of each dose bin relative to no storage for 5 PM to 9 PM. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Operating profit** The markup results translate into lost operating profit in the same hours. Figure 8 presents the effect on hourly operating profit per MW of capacity, weighted by capacity. Gas generators lose operating profit from 3 PM to 8 PM, with the largest loss of $1.55 per MW at 7 PM. Overnight, operating profit remains the same. Summed over the day, the hourly effects imply a loss of about $4.9 per MW-day.

The daily estimates confirm this loss. I aggregate the panel to the generator-day and estimate the effect on daily operating profit per MW, weighted by capacity. An additional 100 MWh of local storage lowers daily operating profit by $5.45 per MW-day (s.e. 1.85), or about 6 percent of the pre-entry average of $84 per MW-day. This is close to the sum of the hourly effects. Figure 9 presents

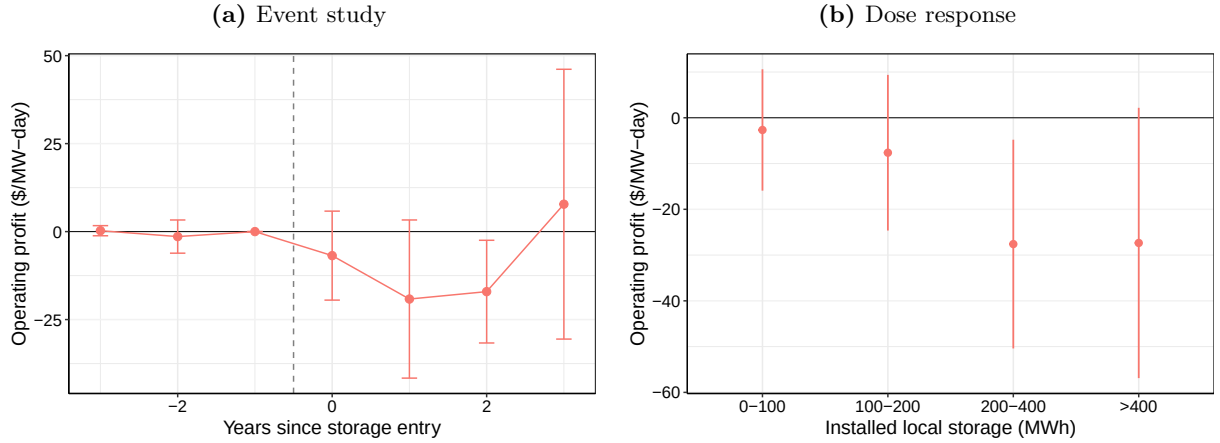
**Figure 8:** Effect of local storage on hourly operating profit




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on hourly operating profit per MW of nameplate capacity, weighted by capacity. The 95% confidence intervals use $t_{G-1}$ critical values and county clustering.

the event study and the dose response. The coefficients before entry are small and insignificant. After entry, operating profit falls by $19 per MW-day in the first year and $17 per MW-day in the second year, and the second-year effect is significant. The dose response shows that the loss grows with the installed dose. Counties with 200 to 400 MWh of storage lose $28 per MW-day relative to counties without storage, and counties with more than 400 MWh lose the same amount.

**Figure 9:** Effect of local storage on daily operating profit




*Notes.* Two-stage difference-in-differences estimates on the generator-day panel. The outcome is daily operating profit per MW of nameplate capacity, weighted by capacity, with generator and load-zone-by-date fixed effects. Panel (a) reports a binary event study with the year before entry as the reference. Panel (b) reports the effect of each dose bin relative to no storage. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Why distinguish competitive rent and market power rent** A fall in gas operating profit does not necessarily mean that storage reduces market power. A gas generator earns operating profit for two reasons. It earns a competitive rent when a more expensive unit sets the price, which it would earn even in a competitive market. Storage can lower competitive rent if its discharge lowers the load gas must serve, so a cheaper unit becomes marginal. It earns a market power rent when the price exceeds the competitive level. Storage can lower market power rents if it also flattens residual demand, which narrows the gap between the price and the competitive level. Only the second is a reduction in market power. Separating the two shows whether storage disciplines the conduct of gas generators or simply adds supply.

I decompose each gas generator's operating profit into a competitive rent and a market power rent, following Borenstein, Bushnell and Wolak (2002). I index generators by $i$, days by $t$, and hours by $h$, as in Section 5. Each generator belongs to one local market $n(i)$, and $\mathcal{G}_n$ denotes the set of gas generators in market $n$. For generator $i$ in hour $h$ of day $t$, $q_{iht}$ is its output in MWh, $p_{iht}$ is the real-time nodal price at its settlement point in \$/MWh, and $\bar{k}_{iht}$ is its available capacity in MWh. Its marginal cost $c_{it}$ in \$/MWh varies only by day, because it combines a weekly heat rate from CEMS with a daily fuel price.

I construct the competitive price from a cost-based supply stack. For each market $n$ and 15-minute interval, I rank the generators in $\mathcal{G}_n$ from lowest to highest marginal cost. I then fill the realized gas output $Q_{nht} = \sum_{j \in \mathcal{G}_n} q_{jht}$ from the bottom of this stack. The competitive price is the marginal cost of the last generator needed to serve that output,

$$p^c_{nht} = \min \left\{ c_{jt} \ : \ j \in \mathcal{G}_n, \sum_{l \in \mathcal{G}_n : \, c_{lt} \le c_{jt}} \bar{k}_{lht} \ge Q_{nht} \right\}. \tag{16}$$

This is the price that a cost-based dispatch of the same gas output would produce. It holds fixed the output of non-gas generators, storage, and imports into the market. I average the competitive price over the intervals in each hour.

Operating profit then splits into two parts by adding and subtracting $p^c_{n(i)ht}$,

$$\pi_{iht} \equiv (p_{iht} - c_{it})\, q_{iht} = \underbrace{(p^c_{n(i)ht} - c_{it})\, q_{iht}}_{\text{competitive rent}} + \underbrace{(p_{iht} - p^c_{n(i)ht})\, q_{iht}}_{\text{market power rent}}. \tag{17}$$

The competitive rent is what generator $i$ would earn on its realized output if the market cleared at cost. It is positive for units below the competitive margin. It can be negative for a unit that runs while its cost exceeds $p^c_{n(i)ht}$, for example because of minimum-run or commitment constraints. The market power rent is the price premium over the competitive benchmark, earned on each unit of output.

This central benchmark likely overstates the market power rent for two reasons. First, measured marginal cost omits startup, maintenance, and ramping costs, so the stack understates the competitive price. Second, $p_{iht}$ is nodal while $p^c_{n(i)ht}$ is market-level, so the premium also absorbs local congestion and losses. I therefore treat the central estimate as an upper bound.

I construct a lower bound from the generator that actually sets the price. ERCOT identifies the marginal generator in each interval. When a gas generator sets the nodal price, I use its marginal cost as the competitive price. Because withholding pushes more expensive generators to the margin, this cost is at or above the true competitive price, so the benchmark understates the market power rent. When no gas generator sets the price, I keep the central benchmark, so the sample composition stays the same. A gas generator sets the nodal price in 58 percent of intervals.

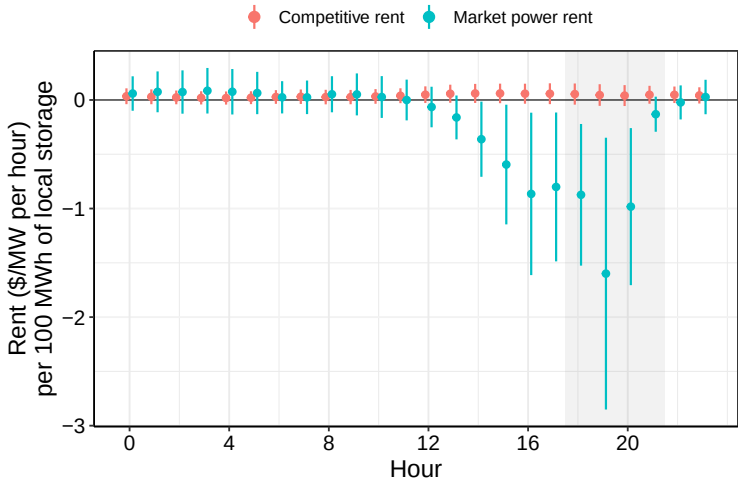
I estimate the effect of storage on each rent component under both benchmarks, using the same specification, sample, and weights. Because the estimator is linear in the outcome, the two effects sum exactly to the effect on operating profit.

**Rent decomposition** The loss in operating profit falls mainly on the market power rent. Figure 10 presents the hourly effects on the two rents under the central benchmark. The market power rent falls from 2 PM to 8 PM, by up to $1.60 per MW at 7 PM, and accounts for nearly all of the hourly profit loss. The competitive rent changes little in any hour. Table 6 presents the daily effects. Under the central benchmark, the market power rent falls by $6.27 per MW-day (s.e. 1.74), while the competitive rent rises by an insignificant $0.82. Under the price-setter benchmark, the market power rent falls by $3.96 per MW-day (s.e. 1.80), and the competitive rent falls by an insignificant $1.50. The market power rent therefore accounts for between 73 percent and all of the loss in operating profit, and it is significant under both benchmarks. The estimates by technology tell the same story. Under the price-setter benchmark, the market power rent accounts for 72 percent of the loss for combined cycle generators and 87 percent for simple cycle generators (Appendix Table C.8).

Storage lowers markups without displacing gas from being the price-setter. The probability that a gas generator sets the nodal price does not change significantly with local storage in any hour (Appendix Figure C.18). If storage reduced gas profits only by adding supply, it would push cheaper units or storage itself to the margin. I find no evidence of this in the data and gas generators continue to set the price, but at a smaller margin above cost. This pattern supports the interpretation that storage disciplines the market power of gas generators.

**Aggregate transfer** Scaling the daily effects by installed storage and generator capacity gives the aggregate transfer from gas generators. Local storage lowered the operating profit of gas generators by $47.4 million per year (s.e. 16.1), or about $4.1 million per year per 100 MWh of storage. Of this, the market power rent accounts for between $34.4 million per year (s.e. 15.6) under the price-setter benchmark and $54.5 million per year (s.e. 15.2) under the central benchmark. Aggregating over the technology-specific estimates yields a similar total loss of $53.2 million, so the aggregate number is robust to pooling across technologies. These amounts are transfers. The rents gas generators lose accrue to storage owners through arbitrage revenue and to consumers and other generators through lower prices, and I account for them in the welfare analysis of Section 7.

**Figure 10:** Effect of local storage on competitive and market power rents by hour




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the competitive rent and the market power rent in Equation (17) under the central benchmark, per MW of nameplate capacity per hour, weighted by capacity. The two effects sum to the effect on hourly operating profit. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Table 6:** Effect of local storage on daily operating profit and rents

| | Operating profit (1) | Competitive rent (2) | Market power rent (3) | Competitive rent (price-setter) (4) | Market power rent (price-setter) (5) |
|---|---|---|---|---|---|
| $/MW-day per 100 MWh | -5.45*** | 0.82 | -6.27*** | -1.50 | -3.96** |
| | (1.85) | (0.79) | (1.74) | (1.24) | (1.80) |
| Share of profit effect | 1.00 | -0.15 | 1.15 | 0.27 | 0.73 |
| Annual total ($ million) | -47.4 | 7.1 | -54.5 | -13.0 | -34.4 |
| | (16.1) | (6.8) | (15.2) | (10.8) | (15.6) |
| Per 100 MWh of storage | -4,057 | 609 | -4,666 | -1,114 | -2,943 |
| ($ thousand/year) | (1,378) | (585) | (1,297) | (921) | (1,337) |
| Observations (generator-days) | | | 671,972 | | |
| Counties | | | 49 | | |

*Notes.* Two-stage difference-in-differences estimates on the generator-day panel, weighted by capacity, with generator and load-zone-by-date fixed effects and the local renewable and temperature controls. Columns (2) and (3) use the central competitive benchmark. Columns (4) and (5) use the price-setter benchmark, which sets the competitive price to the marginal cost of the gas generator setting the nodal price and keeps the central benchmark when no gas generator sets it. The annual total multiplies the daily effect by installed storage and capacity on treated generator-days and divides by the 6.9 sample years. Dollar amounts are in 2020 dollars, deflated by year with the CPI-U before estimation. The last rows divide the annual total by the average installed storage in gas counties, 1,170 MWh. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Summary of market power results** Local storage reduces the markups and operating profits of gas generators in the late afternoon and evening, when storage discharges and gas generators would otherwise exercise market power. The daily profit loss of $5.45 per MW-day per 100 MWh of storage grows with the installed dose. Between 73 percent and all of this loss falls on the market power rent, and gas generators continue to set the price after storage enters. Storage therefore disciplines the market power of gas generators rather than only adding supply.

# 7 Welfare and Distributional Effects

This section asks who gains and loses from the competitive effect of storage, and whether storage raises total surplus. I first set out an accounting identity that separates transfers from changes in total surplus. I then measure each component for the storage installed in counties with gas generation, the storage behind the estimates in Sections 6.1 and 6.2. All amounts are annual averages over the sample, in 2020 dollars, the dollar year of the AP4 damages and the social cost of carbon. I deflate prices, costs, and profits by year with the annual average CPI-U (U.S. Bureau of Labor Statistics, 2025) before estimation, so the estimated effects are also in 2020 dollars.

## 7.1 Welfare Accounting

Load is inelastic, so storage changes total surplus only by changing the cost of serving load and the damages from emissions. I write the change in total surplus as

$$\Delta W = V^S + \Delta\text{Damage}^{\text{local}} + \Delta\text{Damage}^{\text{CO}_2} - K, \tag{18}$$

where $V^S$ is the value of the energy storage moves across hours, $\Delta$Damage are avoided damages, and $K$ is the annualized capital cost of storage.

The same inelastic load implies that any change in prices is a transfer. When storage lowers the price, consumers pay less, and generators and storage owners receive less, by the same amount. The value of energy shifting is therefore split among four groups,

$$V^S = \Delta CS + \Delta\pi^{\text{gas}} + \pi^S + \Delta\pi^{\text{other}}, \tag{19}$$

where $\Delta CS$ is the gain to consumers, $\Delta\pi^{\text{gas}}$ is the change in gas operating profit, $\pi^S$ is the arbitrage revenue of storage owners, and $\Delta\pi^{\text{other}}$ is the change in the profit of the remaining generators. Equation (19) is the key to the precision of the transfer results. I measure $V^S$ and $\pi^S$ from observed storage operations, so they carry no sampling error. The sum $\Delta CS + \Delta\pi^{\text{other}} = V^S - \Delta\pi^{\text{gas}} - \pi^S$ is therefore as precise as the estimated loss of gas producers, even though its split between consumers and other generators is not.

### 7.2 Measurement

**Value of energy shifting** I value storage operations at the social marginal cost of energy rather than at market prices. Let $s_{mht}$ be the net discharge of storage in county $m$ at hour $h$ of date $t$, positive when storage discharges. I compute

$$V^S = \sum_{m,h,t} \tilde{p}_{mht}\, s_{mht}, \tag{20}$$

where $\tilde{p}_{mht}$ is the marginal cost of the energy that storage displaces when it discharges and draws on when it charges. $V^S$ therefore measures the avoided production cost, net of round-trip losses. I use observed storage operations rather than the estimated displacement. Local storage displaced about 849 GWh of gas generation per year, but storage discharged only 196 GWh, and part of the difference may be gas generation that moved to counties without storage (Section 6.1).

The marginal source of energy differs across hours. When a gas generator sets the nodal price, I use the competitive price $p^c_{n(m)ht}$ of Section 6.2, the marginal cost of the last gas generator in a cost-based dispatch of realized gas output. When gas does not set the price, the marginal source is usually renewable, with a marginal cost near zero. In these hours I use the nodal price, floored at zero and capped at the gas benchmark:

$$\tilde{p}_{mht} = g_{mht}\, p^c_{n(m)ht} + (1 - g_{mht})\, \min\{p^c_{n(m)ht},\, \max\{\lambda_{mht}, 0\}\}, \tag{21}$$

where $g_{mht}$ is the share of the county's gas generators that ERCOT identifies as setting the price, $\lambda_{mht}$ is the average nodal price at the county's gas nodes, and $n(m)$ is the local market that contains county $m$. The floor removes negative prices, which reflect renewable production credits, a transfer rather than a cost. The cap prevents the markups of other price setters from counting as costs. As a comparison, I also report $V^S$ with the gas benchmark in every hour, and the avoided production cost implied by the local displacement, with replacement generation valued at the competitive price.

**Transfers** Gas producers lose the operating profit $\Delta\pi^{\text{gas}}$ estimated in Section 6.2. Storage owners earn arbitrage revenue $\pi^S = \sum_{m,h,t} \lambda^S_{mht} s_{mht}$, where $\lambda^S_{mht}$ is the nodal price at the storage node. Consumers gain from lower prices. I estimate the effect of storage on each county's daily nodal price with the two-stage design and apply it to local load,

$$\Delta CS = -\hat{\beta}^p \sum_{m,t} B_{mt}\, L_{mt}, \tag{22}$$

where $\hat{\beta}^p$ is the effect of 100 MWh of local storage on the county's daily price, $B_{mt}$ is installed storage capacity in county $m$ on date $t$, and $L_{mt}$ is the county's load on that date. The daily price averages the nodal prices at all non-storage nodes in the county and weights the hours of the day by the county's own hourly load. I observe hourly county load through 2023 and project 2024 load from each county's 2023 load profile, scaled by ERCOT load. Counties without gas generation

join the control group until they receive storage, which raises the number of counties to 133. I weight counties equally, because the load-weighted estimate shows a price trend before entry in the largest counties (Appendix C.4). The remaining generators take the residual of Equation (19), $\Delta\pi^{\text{other}} = V^S - \Delta CS - \Delta\pi^{\text{gas}} - \pi^S$.

**Avoided damages** I compute avoided emissions by multiplying the displaced generation of each generator-hour by that generator's weekly emission rates of $NO_x$, $SO_2$, and $CO_2$ from CEMS, and a $PM_{2.5}$ factor from U.S. Environmental Protection Agency (2020). I value local pollutants with the AP4 source-receptor matrices, which give the damage in each receptor county per ton emitted at each source (Tschofen, Azevedo and Muller, 2019). I match each generator to its AP4 source by plant code. Fifteen plants, holding 5 percent of gas capacity, have no AP4 source and take the source of the nearest plant of the same technology, in the same county when possible. I value $CO_2$ at a social cost of carbon of \$190 per metric ton (EPA, 2023). Because the damages follow the estimated change in gas output in every hour, any gas generation that storage draws on when charging is netted out. They use the local displacement, so they share its caveat: if gas generation moved to counties without storage, they overstate the system-wide reduction. I therefore treat them as upper bounds. The displacement is bounded so that output without storage lies between zero and capacity, and its standard errors come from the simulation in Section 6.1.

**Storage cost** The investment cost comprises a power cost of 1,440 \$/kW and an energy cost of 461.08 \$/kWh (EIA, 2024). I apply these unit costs to the average installed capacity of storage in counties with gas generation over the sample, 759 MW and 1,170 MWh, an average duration of 1.54 hours, so costs and benefits refer to the same capacity. Applying a capital recovery factor for a 15-year asset life, following the NREL Annual Technology Baseline assumption for utility-scale lithium-ion storage, at a 7% real discount rate gives an annualized cost of about \$179 million per year.[16]

### 7.3 Welfare Results

Table 7 presents the welfare accounting. I discuss the transfers first, because they are the competitive effect this paper measures, and then the change in total surplus.

The competitive effect of storage moves surplus away from gas producers. Gas producers lose \$47.4 million per year (s.e. 16.1), of which the market power rent accounts for \$34.4 million to \$54.5 million. Storage owners keep \$5.8 million per year from energy arbitrage, or \$8 per kW-year. By Equation (19), consumers and the other generators together gain \$41.1 million per year, with the same standard error as the gas producers' loss. This is the precise statement of who gains: almost all of the surplus storage takes from gas producers goes to the buyers and other sellers of energy, not to storage owners.

[16] I annualize using the capital recovery factor $\text{CRF} = r(1+r)^N/[(1+r)^N - 1]$, with asset life $N = 15$ and real discount rate $r = 0.07$, so that $\text{CRF} \approx 0.110$. The annualized cost excludes fixed operating costs.

**Table 7:** Welfare and distributional effects of local storage

| | Annual (2020 $ million) | Per 100 MWh of storage (2020 $ million/year) |
|---|---|---|
| *Transfers among market participants* | | |
| Gas producers | -47.4 | -4.05 |
| | (16.1) | (1.38) |
| Competitive rent, central benchmark | 7.1 | 0.61 |
| | (6.8) | (0.59) |
| Market power rent, central benchmark | -54.5 | -4.66 |
| | (15.2) | (1.30) |
| Market power rent, price-setter benchmark | -34.4 | -2.94 |
| | (15.6) | (1.34) |
| Storage owners | 5.8 | 0.50 |
| Consumers and other generators | 41.2 | 3.52 |
| | (16.1) | (1.38) |
| Consumers | 71.6 | 6.12 |
| | (55.5) | (4.75) |
| Other generators (residual) | -30.5 | -2.61 |
| *Change in total surplus* | | |
| Value of storage energy shifting | -0.5 | -0.04 |
| Avoided local damages | 13.2 | 1.13 |
| | (5.0) | (0.43) |
| Avoided $CO_2$ damages | 76.8 | 6.57 |
| | (28.7) | (2.45) |
| Total benefit, local damages only | 12.7 | 1.09 |
| | (5.0) | (0.43) |
| Total benefit, including $CO_2$ | 89.6 | 7.66 |
| | (33.2) | (2.84) |
| Annualized storage cost | -107.8 | -9.21 |
| Net benefit, including $CO_2$ | -18.2 | -1.56 |
| | (33.2) | (2.84) |
| Net benefit, lower capital cost | 13.0 | 1.11 |
| | (33.2) | (2.84) |
| *Comparison measures of production cost* | | |
| Value of storage energy shifting, gas benchmark | -0.5 | -0.05 |
| Avoided production cost, local displacement | 5.7 | 0.49 |
| | (2.2) | (0.18) |
| *Storage owner revenue and cost recovery* | | |
| Storage revenue, energy arbitrage | 5.8 | 0.50 |
| Arbitrage revenue minus value of energy shifting | 6.3 | 0.54 |
| Storage revenue, ancillary services | 35.6 | 3.04 |
| Storage revenue, total | 41.4 | 3.54 |
| Share of annualized cost recovered | 38% | |
| Share recovered, net of investment tax credit | 42% | |
| Share recovered, lower capital cost | 54% | |
| Net benefit, ancillary services at their price | 17.4 | 1.48 |

*Notes.* Annual averages over 2018 to 2024 for storage in counties with gas generation, in 2020 dollars. Energy shifting is valued at the social price of energy (Equations (20) and (21)). Avoided damages apply CEMS emission rates and AP4 damages to the bounded local displacement, with $CO_2$ at $190 per metric ton. Storage cost annualizes the EIA investment cost at 7 percent over 15 years. Other generators are the residual of the surplus identity in Equation (19). Ancillary service revenue does not enter total surplus. Per 100 MWh values divide by 1,170 MWh of installed storage. Standard errors in parentheses come from 500 draws of the estimated displacement coefficients for the damages and production cost rows, and from the clustered regressions for gas producers and consumers.

The split between consumers and other generators is less precise. My estimate for consumers is \$71.6 million per year (s.e. 55.5), which leaves the other generators with a loss of \$30.5 million. A loss for other generators is plausible, because lower prices also reduce the revenue of generators that do not lose market power, such as wind and solar units selling at the same nodes. The consumer estimate implies about \$94,000 per MW of storage per year, close to the \$124,000 per MW per year that Kirkpatrick (2026) estimates for the cost of serving load in California. Because the consumer estimate is not statistically significant, I interpret only the combined transfer.

**The private return to storage** Storage earns its arbitrage revenue from the markups it erodes. The difference between what storage earns and what its operations are worth is

$$\pi^S - V^S = \sum_{m,h,t} \left(\lambda^S_{mht} - \tilde{p}_{mht}\right) s_{mht}, \tag{23}$$

the gap between the market price and the social marginal cost of energy, weighted by storage operations. Storage discharges in the evening, when gas generators set the price above their cost, and charges when that gap is small. The same operations that earn \$5.8 million at market prices are worth −\$0.5 million at marginal cost, so the private return of storage arbitrage is entirely the markup of the marginal gas generators. As storage erodes those markups, it also erodes the price spread it earns. The private return to arbitrage therefore declines as storage disciplines gas generators.

Storage creates little surplus by shifting energy. Valued at the social price of energy, storage operations slightly raise production costs, by \$0.5 million per year. Storage charges overnight and in the morning and discharges in the evening peak (Figure 1). The cost spread between these hours is small, and storage loses about 19 percent of its energy to round-trip efficiency losses, so the two roughly cancel. About half of the charging occurs in hours when gas does not set the nodal price, typically when renewable output is on the margin. Valuing that energy at the renewable margin changes the result little, because the nodal price in these hours is close to the cost of the marginal gas generator. With the gas benchmark in every hour, the value is also −\$0.5 million. The comparison based on the local displacement, which also counts any reallocation of output from higher-cost to lower-cost gas generators, gives a gain of \$5.7 million per year (s.e. 2.2), still small.

Avoided damages are the largest measured benefit, but they rest on the local displacement. Avoided local damages total \$13.2 million per year (s.e. 5.0), and avoided $CO_2$ damages \$76.8 million per year (s.e. 28.7), for a total of \$89.6 million, or \$12.7 million without $CO_2$. These are upper bounds. The displacement is more than four times what storage discharged, and the damages assume that none of the displaced gas generation moved elsewhere.

The measured benefits do not cover the capital cost of storage. The annualized capital cost is \$179 million per year, or \$15.3 million per 100 MWh of storage. Even with avoided damages at their upper bound, the net benefit is −\$90 million per year. This comparison omits the main service that ERCOT storage provides. Storage in counties with gas generation cycled only 0.46 times per

day and earned $35.6 million per year from ancillary services, 86 percent of its $41.4 million total revenue. Ancillary services hold capacity in reserve for grid reliability, and their social value does not appear in energy-market accounts. Valuing them at their market prices, which likely overstates their value when reserve prices exceed the cost of the next available provider, raises the net benefit to −$54 million per year. These estimates imply that energy arbitrage and its competitive effect alone do not justify the investment, not that the investment lowers welfare.

Ancillary services provide most of storage revenue. Together with arbitrage, they cover about 23 percent of the annualized capital cost of storage in counties with gas generation, or $54 per kW-year.[17] The gap is partly a feature of the sample, which excludes Winter Storms Uri and Elliott, when reserve and energy prices reached their caps. Storage in these counties earned $82 million from ancillary services in these 12 days, about 22 percent of all its revenue over the sample period.

The avoided local damages accrue mostly outside the counties that host storage. Storage counties receive 29 percent of the avoided damages, the rest of Texas 19 percent, and other states 52 percent, because fine particles formed from Texas emissions travel downwind. Per resident, the benefit is concentrated in Texas: $0.22 per person per year in counties with storage and $0.19 in the rest of Texas, compared with $0.02 outside the state (Appendix Figure C.20).

**Summary of welfare results** The competitive effect of storage is a transfer. Storage moved about $47 million per year away from incumbent gas generators, most of it market power rent. Storage owners kept about $6 million, and consumers and the other generators gained the remaining $41 million, with most of it going to consumers by my estimates, although that split is imprecise. Storage created little value by shifting energy at marginal cost, and its measured benefits, even with avoided damages at their upper bound, fall short of its capital cost. Its private return from arbitrage comes from the same markups it disciplines.

# 8 Conclusion

This paper shows that battery storage disciplines the market power of natural gas generators. Using the staggered rollout of county-level storage capacity in ERCOT, I find that local storage makes the residual demand of nearby gas generators more elastic and displaces their output in the afternoon and evening. It lowers their evening markups and their daily operating profits by about 6 percent per 100 MWh of storage. Between 73 percent and all of the profit loss is market power rent, and gas generators continue to set the price at a smaller margin. These results support a simple mechanism: storage flattens the residual demand of a strategic generator and raises the cost of withholding output. The competitive effect of storage moved about $47 million per year away from gas producers, most of it market power rent. Storage owners kept about $6 million, and

[17] The capital cost is the full resource cost of storage. The investment tax credit introduced by the Inflation Reduction Act, available to standalone storage from 2023, shifts part of this cost from storage owners to taxpayers and does not change total surplus. Net of the credit, storage revenue covers about 25 percent of the annualized cost to owners, compared with 23 percent before the credit, because most storage in the sample entered before standalone storage became eligible.

consumers and the other generators gained the rest. By contrast, storage created little value by shifting energy at marginal cost. Its arbitrage revenue comes from the markups it erodes, and it earns most of its revenue from ancillary services rather than energy. Even with ancillary services, storage revenue covers about a quarter of its annualized capital cost over the sample.

These results carry three policy implications. First, evaluations of storage should count its competitive effect. Standard cost-benefit analyses value storage by the production cost and emissions it avoids. In my setting, the production cost savings are small relative to the transfer away from gas producers. The avoided emissions are larger, but they rest on a local displacement that I treat as an upper bound. Omitting competition therefore understates what consumers gain from storage. Second, storage complements market power mitigation. ERCOT combines permissive offer caps with energy-only pricing, and storage lowers markups in the congested local markets where offer mitigation is hardest to target. Interconnection and siting policies that speed storage into these markets would deliver competitive benefits beyond the energy it supplies. Third, consistent with results from prior works (Karaduman, 2023; Butters, Dorsey and Gowrisankaran, 2025), energy arbitrage alone cannot justify storage investment, because its revenue shrinks as storage erodes markups. The design and pricing of ancillary service and reliability products will therefore shape how much storage enters and, through entry, how much competitive discipline it provides.

My analysis has limitations that point to future work. First, the county-level design measures local effects. If gas generation shifts to counties without storage, the local displacement and the avoided emissions overstate the system-wide effects. The welfare accounts also omit the reliability value of ancillary services. Second, I abstract away from storage ownership, and my estimates average across owners. My model implies that an owner of both storage and generation discharges less, because discharging lowers the price its other generation receives. Storage owned by incumbent gas generators or by large renewable portfolios may therefore discipline markups less than storage owned by independent firms. Third, I do not model curtailment directly. When gas does not set the nodal price, I value the energy storage shifts at the nodal price floored at zero, which approximates the near-zero cost of renewable energy that would otherwise be curtailed. This approximation is coarse in congested hours, when the nodal price can exceed the cost of curtailed energy, so my estimates may understate the value of storage in those hours. Curtailment is concentrated in the West and South, where storage is common but gas generators are few, so this value likely falls largely outside my sample. Fourth, the estimates reflect storage at its current penetration. As storage grows, it may set prices more often and exercise market power of its own. Finally, the estimates come from an energy-only market with high offer caps and narrow local mitigation. Markets with capacity payments and tighter mitigation leave less market power for storage to discipline, so the mechanism should carry over, but the effect is likely smaller. Where storage also earns capacity payments, its revenue mix and the share of its capital cost it recovers would also differ.

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# Supplementary Appendix

## A Detailed Theoretical Derivations

This appendix extends the model of Section 4 in three directions. It shows how transmission congestion determines how much of a battery's discipline reaches a gas generator, generalizes storage from a price taker to a strategic actor, and derives how ownership changes the discipline storage provides. Throughout, I keep the notation of the main text.

### A.1 Congestion and the Local Market

Congestion determines how much of a remote battery's discipline reaches the gas firm, which is why I define the market locally. Consider two regions, $W$ and $E$, joined by a line of capacity $\bar{F}$ that carries flow $F$ from $W$ to $E$. Region $E$ holds the strategic gas firm $f$, a fringe $S^c_E(p_E) = \alpha_E + \gamma_E p_E$, and load $D_E$. Region $W$ holds a fringe $S^c_W(p_W) = \alpha_W + \gamma_W p_W$, load $D_W$, and a storage fleet with level $\bar{s}_t$ and slope $\sigma_t$. Nodal balance and the line constraint are

$$\text{(E)} \quad q_{ft} + \alpha_E + \gamma_E p_E + F = D_E, \tag{A.1}$$

$$\text{(W)} \quad \alpha_W + (\gamma_W + \sigma_t) p_W + \bar{s}_t = D_W + F, \tag{A.2}$$

$$F \le \bar{F}, \qquad p_E - p_W = \lambda \ge 0, \qquad \lambda(\bar{F} - F) = 0, \tag{A.3}$$

so the price gap equals the congestion shadow price $\lambda$.

When the line is slack, remote storage reaches the gas firm in full. With $\lambda = 0$, both regions clear at one price $p$, and adding (A.1) and (A.2) eliminates the flow,

$$q_{ft} = (D_E + D_W) - (\alpha_E + \alpha_W + \bar{s}_t) - (\gamma_E + \gamma_W + \sigma_t)\, p. \tag{A.4}$$

The firm faces the slope $\gamma_E + \gamma_W + \sigma_t$, because the flow adjusts to carry storage output into region $E$.

When the line binds, remote storage does not reach the gas firm at all. The import into $E$ is fixed at $F = \bar{F}$, so (A.1) gives $q_{ft} = D_E - \alpha_E - \gamma_E p_E - \bar{F}$, and the firm faces the slope $\gamma_E$ alone. Storage clears within $W$, lowering $p_W$ and widening $\lambda$, but it leaves the firm's output and markup unchanged. Storage located in $E$, by contrast, enters (A.1) directly and gives the firm the slope $\gamma_E + \sigma_t$ regardless of the line.

These three cases order storage by its share of the firm's competitive set. The share is $\sigma_t/(\gamma_E + \sigma_t)$ for local storage, the smaller $\sigma_t/(\gamma_E + \gamma_W + \sigma_t)$ for remote storage on a slack line, and zero for remote storage behind a binding constraint. Local storage therefore delivers the full effect, remote storage an attenuated one, and constrained remote storage none. Storage entry can itself change congestion, because storage in $W$ lowers $p_W$ and raises desired exports. I abstract from this by assuming no constraint is marginally binding, following Woerman (2023). The same logic implies that the effect of storage should scale with its size relative to the local market, which motivates

defining treatment at the county level.

## A.2 Strategic Storage

Strategic storage weakens the discipline that price-taking storage imposes, and the price-taking model is its limiting case. I follow Andres-Cerezo and Fabra (2025); Andrés-Cerezo and Fabra (2023), in which storage internalizes the price effect of its own dispatch, and focus on the discharge period. Netting out the fringe, the price is $p = (a - q_f - s)/\gamma$ with $a = D_H - \alpha_H$. The gas firm has marginal cost $c_f$, and storage has marginal cost $m_s = p_L/\eta$.

A conduct parameter $\theta \geq 0$ indexes how much storage withholds. Storage perceives marginal revenue $p - \theta s/\gamma$, so the two first-order conditions are

$$p - c_f = \frac{q_f}{\gamma} \quad \text{(gas firm)}, \qquad p - m_s = \theta\,\frac{s}{\gamma} \quad \text{(storage)}. \tag{A.5}$$

Storage is a price taker when $\theta = 0$ and a Cournot competitor when $\theta = 1$. Solving,

$$s^\star(\theta) = \frac{a + \gamma c_f - 2\gamma m_s}{1 + 2\theta}, \qquad q_f^\star(\theta) = \tfrac{1}{2}\big(a - s^\star - \gamma c_f\big), \qquad \mu_f^\star(\theta) = \frac{q_f^\star}{\gamma}. \tag{A.6}$$

More strategic storage discharges less and leaves a higher gas markup,

$$\frac{\partial s^\star}{\partial \theta} < 0, \qquad \frac{\partial \mu_f^\star}{\partial \theta} = \frac{a + \gamma c_f - 2\gamma m_s}{\gamma\,(1 + 2\theta)^2} > 0 \quad \text{whenever } s^\star > 0. \tag{A.7}$$

Competitive storage delivers the largest markup reduction. As $\theta$ grows, storage withholds more, and the gas markup returns toward its level without storage. Storage that is present but strategic can therefore provide little discipline. The main text studies the case $\theta = 0$, in which the slope $\sigma_t$ comes from aggregating heterogeneous price-taking batteries.

## A.3 Ownership

Portfolio ownership reduces the discipline storage provides through two wedges. Suppose the storage owner also sells generation $G_H$ in the discharge period and $G_L$ in the charge period, and its storage is large enough to move prices, $\partial p_t/\partial s_t = -1/E_t$. Maximizing portfolio profit gives the first-order condition

$$\underbrace{\eta p_H - p_L}_{\text{arbitrage spread}} + \underbrace{\left(\frac{s_L}{E_L} - \eta\frac{s_H}{E_H}\right)}_{\text{storage-side market power}} + \underbrace{\left(\frac{G_L}{E_L} - \eta\frac{G_H}{E_H}\right)}_{\text{portfolio}} = 0. \tag{A.8}$$

A price-taking battery with no other generation keeps only the arbitrage spread and discharges fully, as in Section 4.1. The other two terms reduce discharge.

The storage-side term makes a large storage owner withhold. Because $s_L < 0 < s_H$, the term is negative: the owner recognizes that discharging lowers the peak price and charging raises the

off-peak price, so it discharges less and responds less to price.

The portfolio term makes an owner with peak generation withhold. Its peak component, $-\eta G_H/E_H$, is negative, because discharging lowers the price at which the owner's other generation sells. This applies to owners of gas generation and equally to owners of wind or solar, whose output also clears at the peak price. A battery co-located with solar faces a related limit, because its charging is tied to the solar profile, which lowers its responsiveness to price.

Both wedges point in the same direction. Storage owned by firms with large generation portfolios or large storage positions should discharge less and impose less discipline on gas markups than storage owned by independent, storage-only firms.

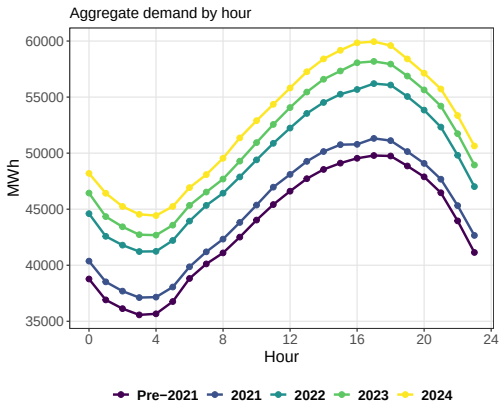
# B Additional Descriptive Statistics

## B.1 Trends in Demand and Supply Patterns

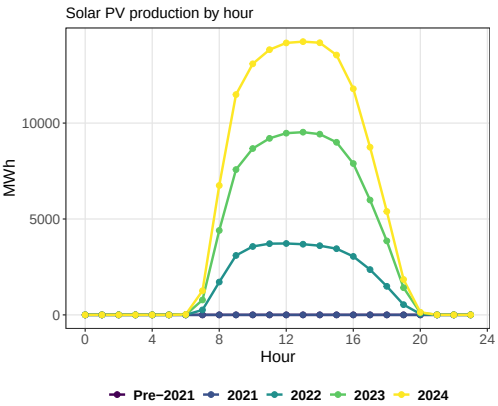
Figure B.1 documents the hourly feature of the Texas electricity market over my sample period. First, in Panel B and C, the daily renewable production pattern have changed significantly over the years. Before 2021, the average renewable generation is relatively flat throughout the day, with its highest production at night due wind domination in the renewable generation. As more solar PV capacity is installed, we see a significant increase in renewable generation during the daytime hours, peaking around 1-2 pm. Figure B.1a shows that the hourly demand pattern has also evolved over the years, generally increasing across all hour of day. The hourly pattern shows a typical daily cycle with peaks in the morning and evening hours. These two things combined has implications for the net demand pattern in Figure B.1d, which has evolved from a relatively flat pattern to a more pronounced "duck curve" pattern, with a significant dip during the daytime hours when solar PV production is high and a steep ramp in the morning and evening hours when demand is high and solar PV production is low.

**Figure B.1:** Hourly demand and supply patterns in ERCOT

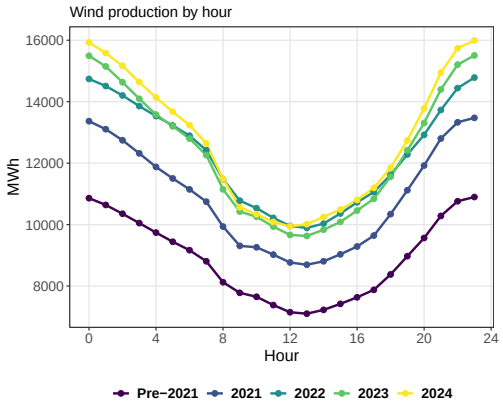
**(a)** Demand



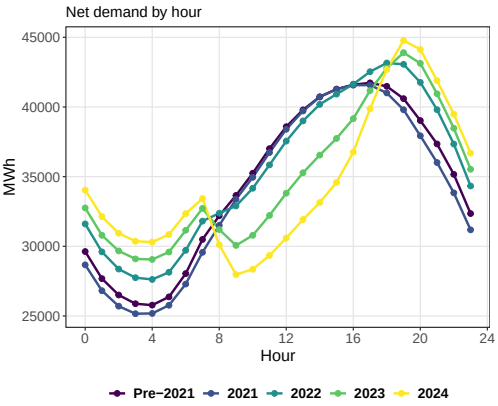

**(b)** Solar PV production



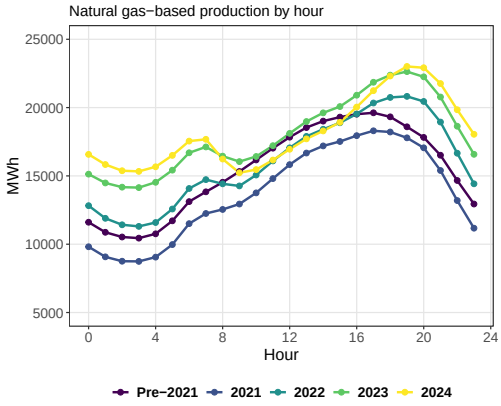

**(c)** Wind production




**(d)** Net demand




**(e)** Natural gas-based production

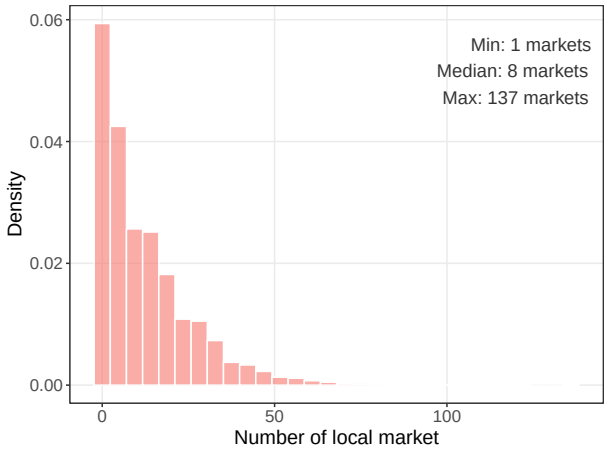



*Notes:* These figures show the average renewable generation, load, net load, and natural gas-based generation for each hour of the day. The net load is defined as the total load minus the total renewable generation.

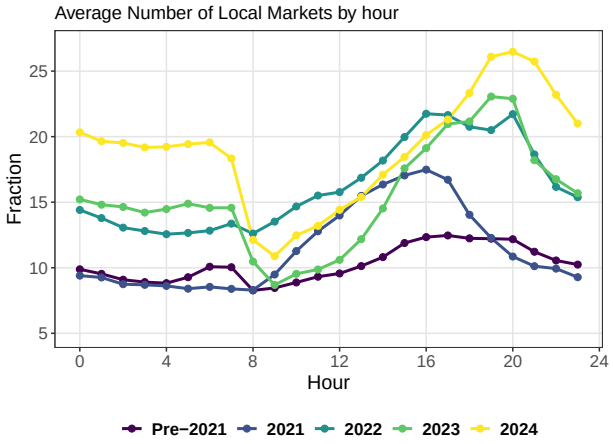
## B.2 Local Market Descriptive Statistics

**Figure B.2:** Local market statistics

**(a)** Number of local market distribution



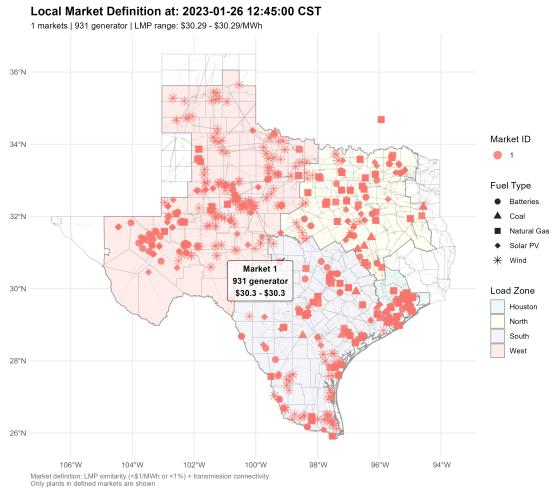

**(b)** Number of local market by year



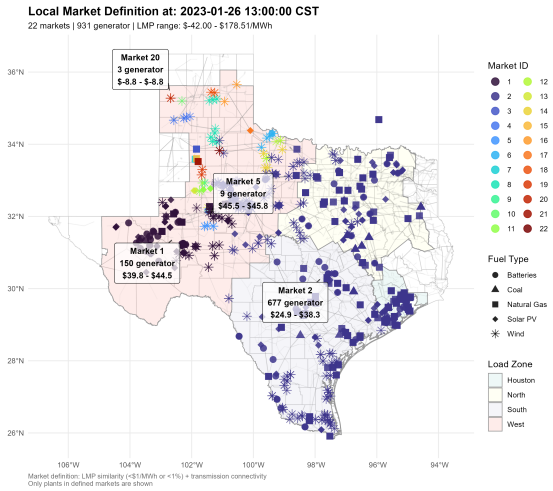

*Notes:* Panel (a) and (b) summarize the distribution of the number of local markets and the trends in hourly variation of the number of local markets by year.

**Figure B.3:** Example of local market definition

**(a)** Example of interval with no congestion




**(b)** Example of interval with congestion

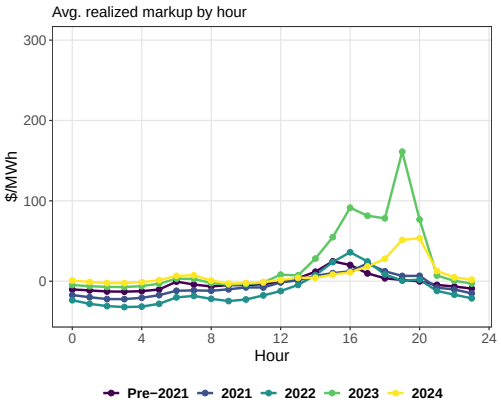



*Notes:* Panel (a) and (b) show an example of market definition for intervals 15 minutes apart. Each dot is a generator, with colors representing distinct local markets. In panel (a), there is no congestion and all generators are in the same local market, while in panel (b), congestion splits the generators into 22 local markets. Across the sample, the number of local markets varies from 1 to 137, with a median of 8. Figure B.2 presents the distribution of local markets statistics within my sample.

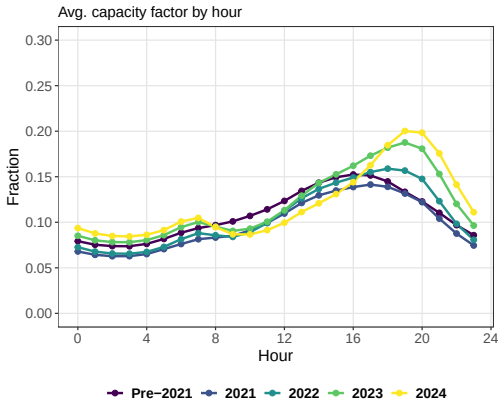
## B.3 Natural Gas Generators Descriptive Statistics

**Figure B.4:** Natural gas unit markup and capacity factor by hour of day

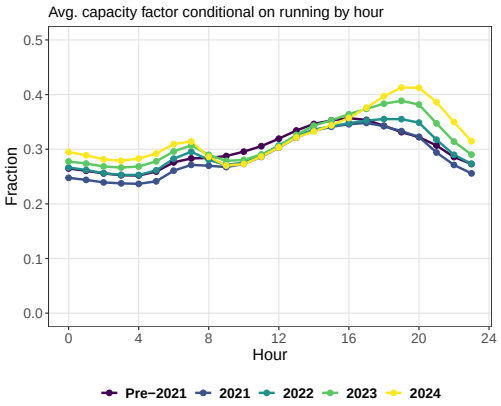
**(a)** Markup



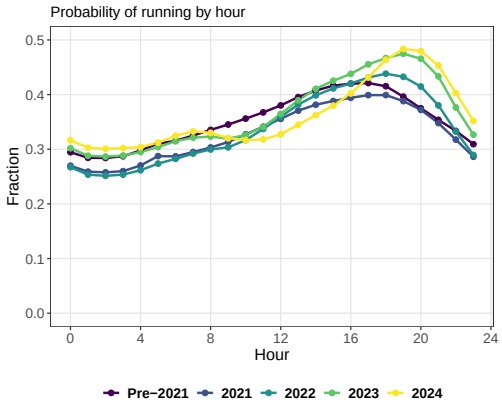

**(b)** Capacity Factor



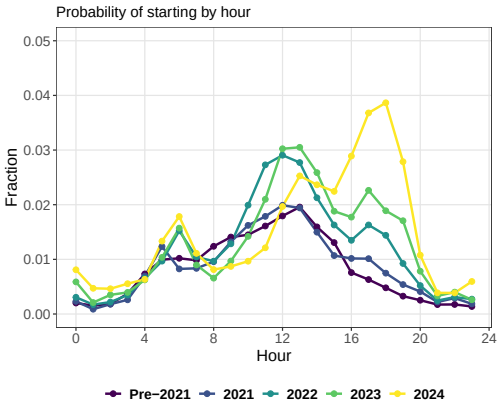

**(c)** Capacity Factor when Running



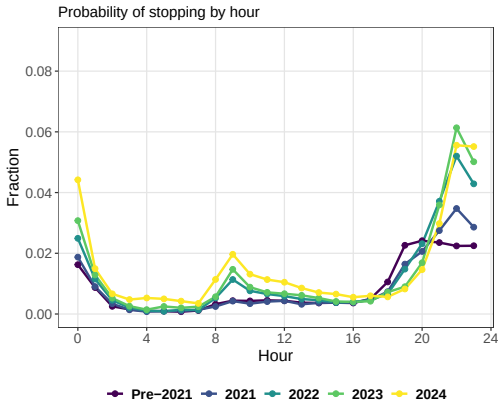

**(d)** Probability of Running




**(e)** Probability of Starting




**(f)** Probability of Stopping




*Notes:* Each panel shows the average of each variable for each hour of the day, plotted separately for the pre-2021 period and each year from 2021 through 2024. The data exclude period of extreme event.

# C Additional Results

## C.1 Selection in Battery Siting

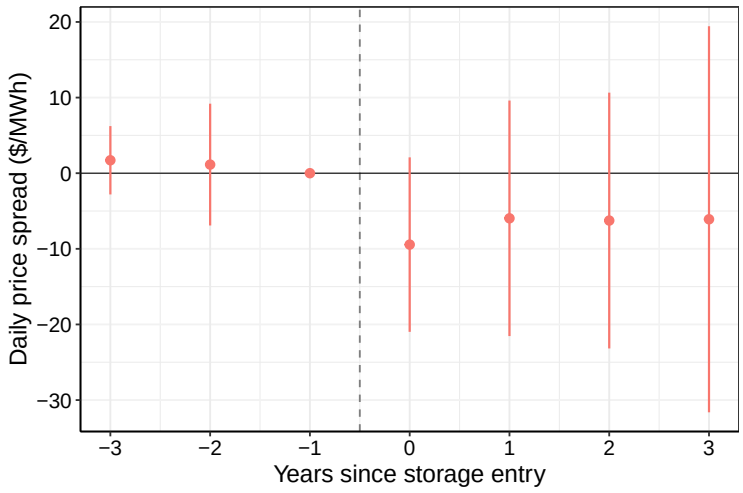
I test whether batteries were sited where prices or gas generator conduct made entry attractive. Table C.1 relates storage entry to pre-period county characteristics. Figure C.5 tests whether price spreads were rising before entry. Table C.2 tests whether batteries locate near gas units with high pre-period markups.

**Table C.1:** Pre-period county characteristics and storage entry

| | Has storage | | Log storage (MWh) | | Entry year | |
|---|---|---|---|---|---|---|
| Sample | All | Gas | All | Gas | All treated | Gas treated |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Mean price ($10/MWh) | 0.661 | 0.699 | 1.485 | -3.777 | -0.838 | -2.569 |
| | (0.500) | (1.388) | (2.272) | (6.980) | (3.109) | (4.567) |
| Mean daily spread ($10/MWh) | 0.044** | 0.250 | 0.158 | 0.898 | -0.163 | -0.805 |
| | (0.022) | (0.161) | (0.100) | (0.885) | (0.123) | (0.727) |
| Mean evening price ($10/MWh) | -0.355 | -0.840 | -0.688 | 0.279 | 0.395 | 1.582 |
| | (0.268) | (0.938) | (1.203) | (4.686) | (1.668) | (3.521) |
| SD of daily price ($10/MWh) | -0.123 | -0.181 | -0.436 | -0.196 | 0.648 | 2.003 |
| | (0.089) | (0.249) | (0.413) | (1.670) | (0.464) | (1.196) |
| Load growth, 2016–2019 (log) | 0.980 | 1.947 | 3.644 | 9.066 | 11.638 | 2.647 |
| | (0.660) | (1.320) | (3.508) | (6.647) | (7.740) | (11.552) |
| Solar capacity, 2019 (GW) | 0.472*** | 4.134** | 3.253*** | 9.581 | -1.104* | -29.754*** |
| | (0.164) | (1.982) | (0.744) | (12.409) | (0.618) | (5.059) |
| Wind capacity, 2019 (GW) | 0.008 | 0.001 | 0.097 | -0.213 | 0.342 | 0.443 |
| | (0.062) | (0.226) | (0.308) | (1.184) | (0.351) | (0.766) |
| Load zone FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Joint test, price variables | 2.22 | 1.28 | 4.62 | 2.97 | 3.90 | 3.20 |
| F (p-value) | (0.073) | (0.295) | (0.002) | (0.033) | (0.009) | (0.036) |
| Observations | 108 | 46 | 108 | 46 | 49 | 30 |
| $R^2$ | 0.141 | 0.221 | 0.191 | 0.208 | 0.218 | 0.412 |

*Notes.* One observation per county. Characteristics are measured over 2018–2019 untreated days. Prices are in 2020 dollars and scaled to $10/MWh. Gas columns restrict to counties with gas generation, the estimation sample. Entry year columns use treated counties only. Solar capacity in gas counties is small, so the solar coefficient in Column (6) reflects a narrow range of variation. All columns include load zone fixed effects. The joint test is a Wald test that the four price coefficients are zero. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Figure C.5:** County daily price spread before and after storage entry




*Notes.* Two-stage difference-in-differences estimates of the county daily price spread, the daily maximum minus minimum of the county mean nodal price, by year relative to first local storage entry. The first stage uses untreated county-days, with county and load-zone-by-date fixed effects and the same controls as the main estimates. The 95% confidence intervals use standard errors from a county block bootstrap of both stages. The year before entry is the reference.

**Table C.2:** Battery proximity and gas unit pre-period characteristics

| | Battery within 10 km (1) | Log distance to battery (2) |
|---|---|---|
| Evening markup ($10/MWh) | 0.002 | -0.010 |
| | (0.003) | (0.009) |
| Mean evening price ($10/MWh) | 0.007 | 0.005 |
| | (0.009) | (0.015) |
| Capacity factor | 0.036 | -0.326 |
| | (0.139) | (0.247) |
| Pre-period heat rate | 0.005 | -0.018 |
| | (0.004) | (0.019) |
| Capacity (100 MW) | -0.003 | 0.012* |
| | (0.002) | (0.007) |
| County FE | ✓ | ✓ |
| Mean of outcome | 0.284 | 2.928 |
| Joint test, markup and price | 1.13 | 1.06 |
| F (p-value) | (0.333) | (0.356) |
| Observations | 473 | 473 |
| Adjusted $R^2$ | 0.736 | 0.841 |

*Notes.* One observation per gas unit. Characteristics are measured over untreated hours. The evening markup is the nodal price minus marginal cost from 5 PM to 9 PM when the unit runs. Batteries are those operating by December 2024. All columns include county fixed effects, so the comparison is within county. The joint test is a Wald test that the markup and price coefficients are zero. Standard errors clustered by county in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

## C.2 Additional Displacement Results

This appendix reports the full structural estimates, the robustness of the two-stage estimates, and additional results on heterogeneity and the residual demand slope.

**Structural estimates** Table C.3 reports the structural estimates of Equation (10) across samples, weights, and fixed effects. The IV estimate is −0.79 on interior hours (Column 2) and −0.81 when I include hours at the operating limits (Column 3). Excluding hours within 2 percent of capacity of either limit shrinks it to −0.26 and makes it insignificant (Column 4). Capacity weights give −0.66 (Column 5), and load-zone-by-date-by-hour fixed effects, which absorb the common natural gas price shock, give −0.81 (Column 6). The cost coefficient and the interaction term are negative and significant in every column except the 2 percent band.

**Table C.3:** Structural estimates across samples, weights, and fixed effects

| | Capacity factor (pp) | | | | | |
|---|---|---|---|---|---|---|
| | OLS | | | IV | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Storage $B$ (100MWh), $\hat{\delta}^{(1)}$ | -0.785*** | -0.790*** | -0.811*** | -0.259 | -0.656** | -0.814** |
| | (0.291) | (0.292) | (0.261) | (0.175) | (0.263) | (0.343) |
| Cost $\tilde{c}$ (centered, \$/MWh), $\hat{\delta}^{(2)}$ | -0.070*** | -0.070*** | -0.070*** | -0.081*** | -0.100*** | -0.070*** |
| | (0.019) | (0.017) | (0.019) | (0.013) | (0.017) | (0.020) |
| Cost $\tilde{c} \times B$, $\hat{\delta}^{(3)}$ | -0.016*** | -0.016*** | -0.019*** | 0.000 | -0.011*** | -0.017*** |
| | (0.004) | (0.003) | (0.003) | (0.008) | (0.004) | (0.004) |
| Generator FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Load zone-Month-of-sample-Hour FE | ✓ | ✓ | ✓ | ✓ | ✓ | |
| Load zone-Date-Hour FE | | | | | | ✓ |
| Local renewable capacity and potential | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Local temperature | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Weights | None | None | None | None | Capacity | None |
| Sample | Interior | Interior | Running | Interior 2% band | Interior | Interior |
| Observations | 3,770,307 | 3,770,307 | 5,439,859 | 1,744,113 | 3,770,307 | 3,761,401 |
| First-stage $F$, $\tilde{c}$ | | 327.3 | 626.1 | 207.8 | 407.0 | 298.0 |
| First-stage $F$, $\tilde{c} \times B$ | | 2,447.3 | 2,351.4 | 1,762.3 | 534.2 | 9,073.3 |
| Partial $R^2$, $\tilde{c}$ | | 0.397 | 0.376 | 0.374 | 0.410 | 0.565 |
| Partial $R^2$, $\tilde{c} \times B$ | | 0.810 | 0.802 | 0.846 | 0.853 | 0.835 |

*Notes.* Estimates of Equation (10). The outcome is the hourly capacity factor in percentage points. Marginal cost is centered within technology and instrumented by the pre-entry heat rate times the Henry Hub price. Column (3) uses all hours with positive output, and Column (4) drops interior hours within 2 percent of capacity of either operating limit. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4 reports the structural estimates by technology. None of the technology-specific storage coefficients is statistically significant. The interaction term is negative and significant for combined cycle and simple cycle generators, so storage displaces the higher-cost units within these technologies more. It is positive for steam turbines. Figure C.6 presents the hourly structural estimates. The displacement at mean cost is between 0.2 and 0.4 percentage points for combined cycle generators and between 0.4 and 1.2 percentage points for simple cycle generators, and it is imprecise in most

hours. The cost gradient has the predicted sign in every hour for both technologies. Figure C.7 shows that the instruments explain 32 to 46 percent of the remaining variation in cost and 79 to 87 percent of the remaining variation in the interaction for these two technologies in every hour.

**Table C.4:** Structural estimates by technology

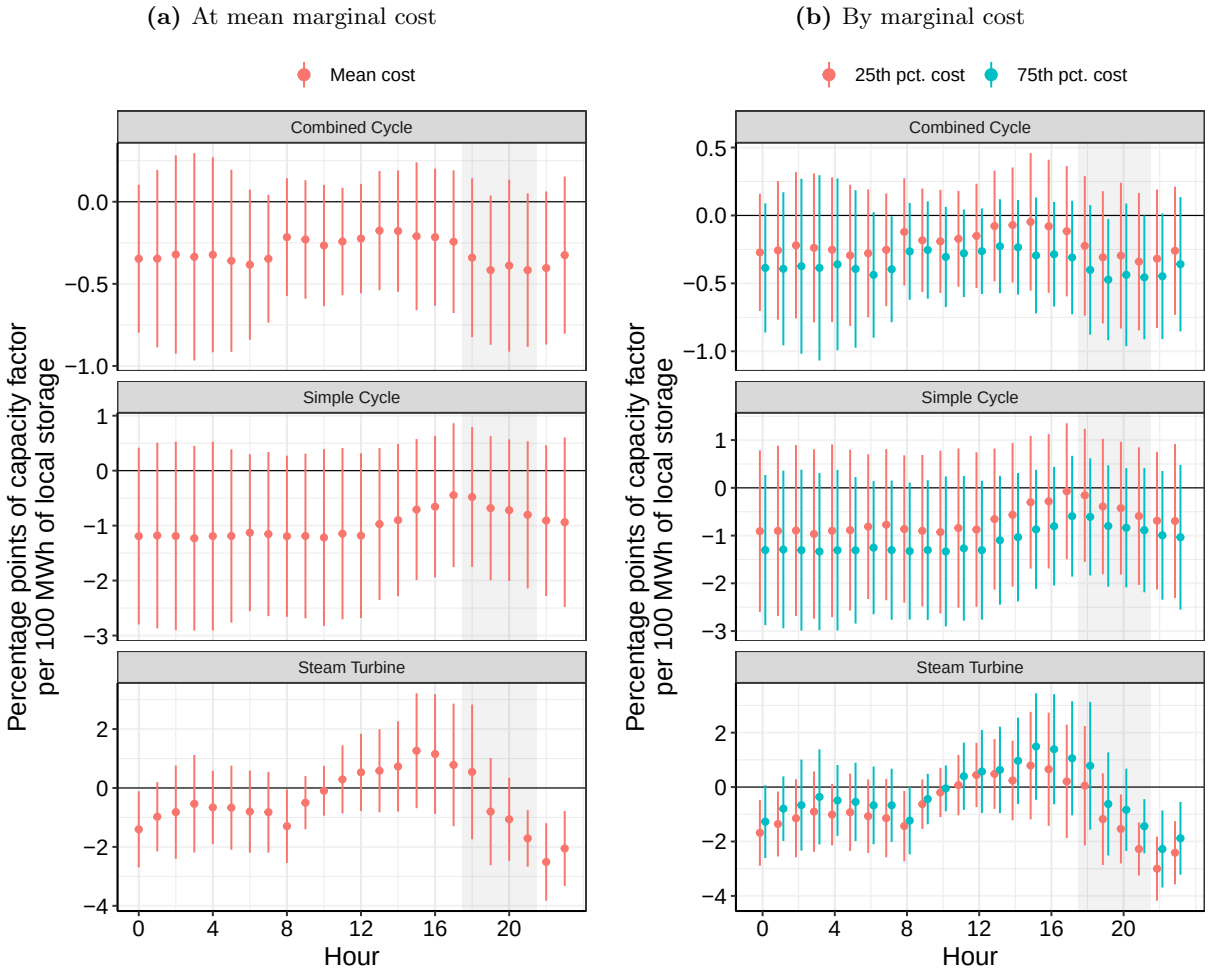
| | Capacity factor (pp) | | | |
|---|---|---|---|---|
| | All gas | Combined Cycle | Simple Cycle | Steam Turbine |
| | (1) | (2) | (3) | (4) |
| Storage $B$ (100MWh), $\hat{\delta}^{(1)}$ | -0.790*** | -0.298 | -1.022 | 0.169 |
| | (0.292) | (0.209) | (0.641) | (0.547) |
| Cost $\tilde{c}$ (centered, \$/MWh), $\hat{\delta}^{(2)}$ | -0.070*** | -0.092*** | -0.038 | -0.120*** |
| | (0.017) | (0.016) | (0.026) | (0.037) |
| Cost $\tilde{c} \times B$, $\hat{\delta}^{(3)}$ | -0.016*** | -0.009** | -0.024*** | 0.029*** |
| | (0.003) | (0.004) | (0.005) | (0.007) |
| Generator FE | ✓ | ✓ | ✓ | ✓ |
| Load zone-Month-of-sample-Hour FE | ✓ | ✓ | ✓ | ✓ |
| Local renewable capacity and potential | ✓ | ✓ | ✓ | ✓ |
| Local temperature | ✓ | ✓ | ✓ | ✓ |
| Sample | Interior | Interior | Interior | Interior |
| Observations | 3,770,307 | 1,821,957 | 1,608,347 | 339,526 |
| First-stage $F$, $\tilde{c}$ | 327.3 | 250.7 | 114.3 | 89.7 |
| First-stage $F$, $\tilde{c} \times B$ | 2,447.3 | 1,879.4 | 12,777.0 | 348.2 |
| Partial $R^2$, $\tilde{c}$ | 0.397 | 0.413 | 0.367 | 0.187 |
| Partial $R^2$, $\tilde{c} \times B$ | 0.810 | 0.845 | 0.818 | 0.647 |

*Notes.* Estimates of Equation (10) on interior hours, separately by technology. Cost and the instrument are centered within technology. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Two-stage estimates across samples** Table C.5 reports the pooled two-stage estimates across samples and weights. The storage coefficient is $-0.55$ on interior hours, $-0.34$ on hours with positive output, and $-0.44$ when I exclude hours within 2 percent of either operating limit. It is $-0.79$ on all hours and $-0.90$ on all hours with capacity weights. Unlike the IV estimate, the two-stage estimate remains significant when I exclude hours near the operating limits.

**Reconciling the structural and two-stage estimates** Table C.6 changes one feature at a time by technology and hour group. For combined cycle generators, the four estimates are close in every hour group, so the choice of specification matters little. For simple cycle generators, the two-stage estimate on the interior sample is larger and more precise than the IV estimate, at 0.8 to 1.6 percentage points, and changing the sample to all hours moves the estimate toward the welfare specification. For steam turbines, the estimates on the interior sample are unreliable because storage changes which steam turbines operate in the interior. Table 5 shows that storage moves steam turbines away from maximum output and into the interior, which changes the composition of the interior sample. The estimates on all hours do not have this problem.

**Figure C.6:** Hourly structural displacement by technology



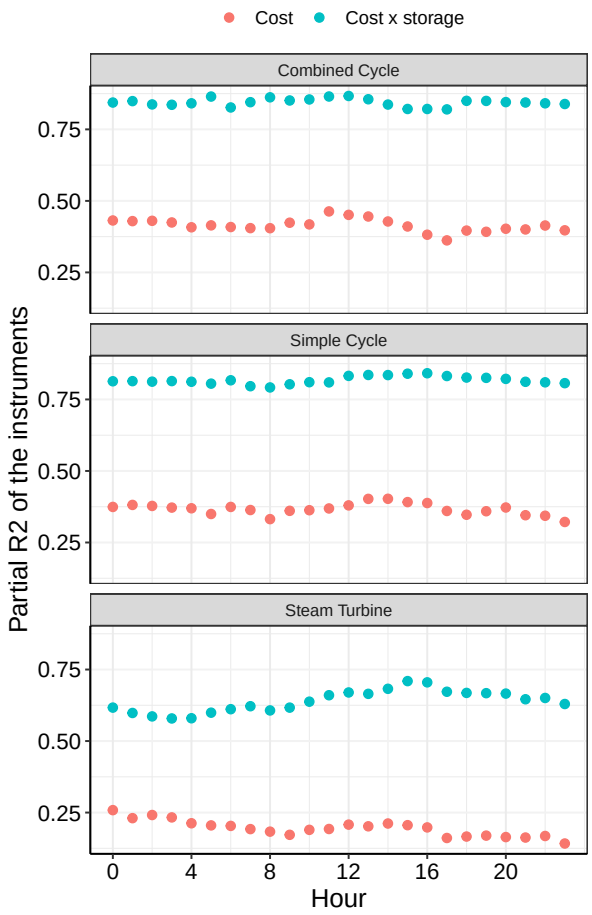

*Notes.* Estimates of Equation (10) by technology and hour on interior hours, in percentage points of capacity factor per 100 MWh of local storage. Panel (b) evaluates the displacement at the 25th and 75th percentiles of marginal cost within each technology. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Bounded counterfactual** Table C.7 compares displaced generation with and without the bounds on counterfactual output. Bounding raises the total from 811 GWh to 849 GWh per year, mostly among steam turbines, whose counterfactual output hits zero in 30 percent of generator-hours with local storage.

### Event study estimates by technology

**Local renewable controls** Figure C.9 compares the welfare specification with and without the local renewable controls. The controls lower the combined cycle estimates by about 0.1 percentage points in every hour. For simple cycle generators, they remove most of the overnight displacement but leave the afternoon displacement nearly unchanged. Solar capacity that enters alongside storage

**Figure C.7:** First-stage strength by hour




*Notes.* Partial $R^2$ of the excluded instruments for the cost term and for the cost-by-storage interaction in each hourly first stage, by technology.

therefore does not explain the afternoon displacement.

**Heterogeneity** Figure C.10 presents the displacement for all gas generators by region and by renewable conditions. In the Houston and North load zones, storage lowers the capacity factor by 1.5 percentage points in the evening and 1.2 percentage points at midday, and the evening effect is significant in every hour. In the West and South load zones, the evening effect is 1.1 percentage points and not significant. The displacement is larger on days with above-median solar output, by about 0.4 percentage points at midday and 0.3 percentage points in the evening. It is also larger on days with above-median wind output, especially overnight, when it is 0.8 percentage points against 0.3 percentage points on low-wind days.

**Residual demand slope** Figure C.11 presents the effect of storage on the measured residual demand slope by technology. The residual demand of combined cycle generators becomes significantly flatter between 6 PM and 11 PM and steeper at 3 PM. The effects for simple cycle and steam

**Table C.5:** Two-stage estimates across samples and weights

| | Capacity factor (pp) | | | | |
|---|---|---|---|---|---|
| | 2SDD (1) | 2SDD (2) | 2SDD (3) | 2SDD (4) | 2SDD (5) |
| Storage $B$ (100MWh) | -0.553*** | -0.342** | -0.437** | -0.794*** | -0.904*** |
| | (0.165) | (0.129) | (0.212) | (0.224) | (0.266) |
| Predetermined cost $\tilde{z} \times B$ | -0.013*** | -0.009** | -0.001 | -0.002 | -0.004 |
| | (0.003) | (0.004) | (0.010) | (0.004) | (0.008) |
| Generator FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Load zone-Date-Hour FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Local renewable capacity and potential | ✓ | ✓ | ✓ | ✓ | ✓ |
| Local temperature | ✓ | ✓ | ✓ | ✓ | ✓ |
| Sample | Interior | Running | Interior 2% band | All hours | All hours |
| Weights | None | None | None | None | Capacity |
| Observations | 3,705,777 | 5,362,817 | 1,696,824 | 13,472,250 | 13,472,250 |
| Counties | 49 | 49 | 49 | 49 | 49 |

*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, in percentage points, with storage interacted with predetermined cost. Standard errors clustered by county in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

turbine generators are imprecise. Figure C.12 presents the event study for the slope by hour group. The coefficients two and three years before entry are small and insignificant in every hour group, so treated and untreated counties followed parallel paths before entry. After entry, the midday slope rises by 12 and 28 MW per $/MWh in the second and third year. The post-entry effects in the other hour groups are imprecise.

**Table C.6:** Reconciling the structural and two-stage estimates

| | (1) IV Interior Unweighted | (2) 2SDD Interior Unweighted | (3) 2SDD All hours Unweighted | (4) 2SDD All hours Capacity-weighted |
|---|---|---|---|---|
| *Combined Cycle* | | | | |
| Overnight (0–6) | -0.344 | -0.289 | -0.216 | -0.203 |
| | (0.263) | (0.217) | (0.186) | (0.243) |
| Midday (10–16) | -0.216 | -0.170 | -0.203 | -0.225 |
| | (0.180) | (0.172) | (0.177) | (0.154) |
| Evening (17–21) | -0.356 | -0.341* | -0.451** | -0.438** |
| | (0.230) | (0.201) | (0.187) | (0.184) |
| *Simple Cycle* | | | | |
| Overnight (0–6) | -1.169 | -1.485*** | -0.789 | -1.202 |
| | (0.753) | (0.206) | (0.677) | (0.931) |
| Midday (10–16) | -0.988 | -1.596*** | -0.674** | -0.774 |
| | (0.667) | (0.213) | (0.317) | (0.542) |
| Evening (17–21) | -0.641 | -0.812*** | -0.813 | -0.802** |
| | (0.607) | (0.250) | (0.582) | (0.357) |
| *Steam Turbine* | | | | |
| Overnight (0–6) | -0.847 | -0.126 | -0.435*** | -0.320 |
| | (0.626) | (0.157) | (0.067) | (0.372) |
| Midday (10–16) | 0.736 | 4.382*** | 0.291 | 0.434 |
| | (0.652) | (1.150) | (0.225) | (0.706) |
| Evening (17–21) | -0.282 | 3.160* | -0.626*** | 0.170 |
| | (0.756) | (1.665) | (0.202) | (0.606) |

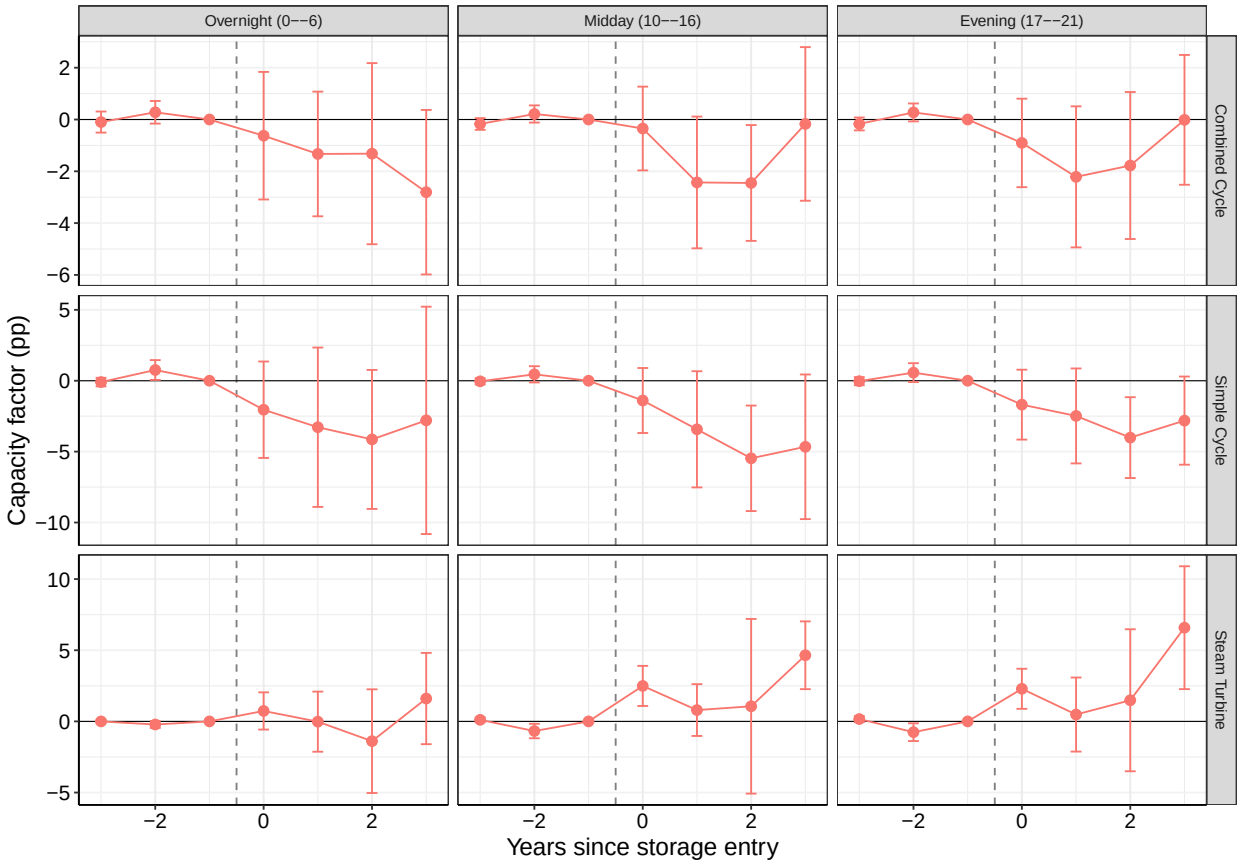
*Notes.* Effect of 100 MWh of local storage on the capacity factor, in percentage points, by technology and hour group. Each column changes one feature relative to the previous one: the estimator, then the sample, then the weights. Column (4) is the specification used in the welfare analysis. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Table C.7:** Displaced gas generation with bounded and unbounded counterfactual output

| | Displaced generation (GWh per year) | | |
|---|---|---|---|
| | Bounded | Unbounded | Generator-hours bounded (%) |
| Combined Cycle | 371.0 | 370.4 | 1.3 |
| | (191.4) | (203.3) | |
| Simple Cycle | 440.2 | 429.8 | 7.1 |
| | (217.4) | (258.1) | |
| Steam Turbine | 37.8 | 10.9 | 29.7 |
| | (66.4) | (93.5) | |
| All gas | 849.0 | 811.1 | 10.1 |
| | (304.8) | (341.6) | |

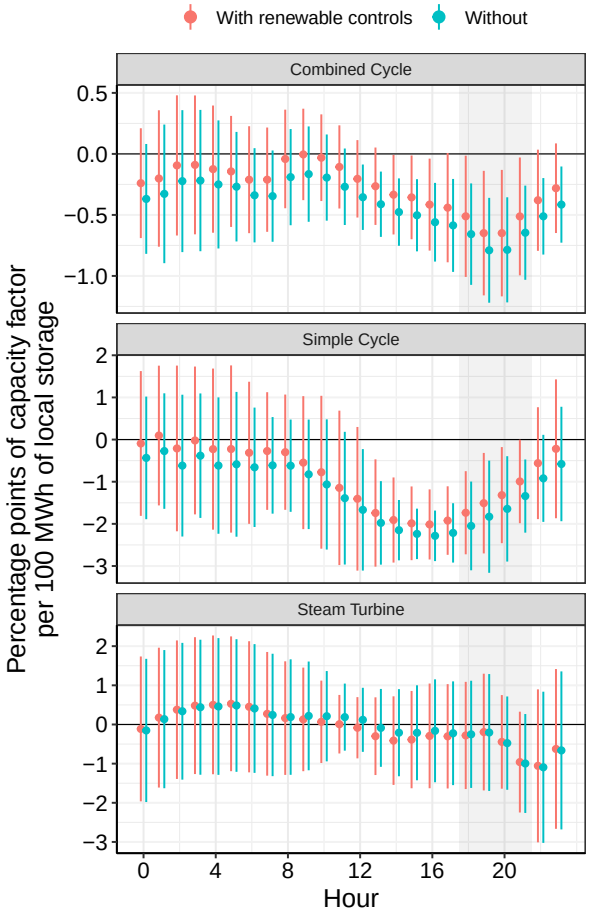
*Notes.* Annual gas generation displaced by local storage, from the technology-specific two-stage estimates with the cost interaction. The bounded counterfactual keeps output without storage between zero and capacity. Standard errors for bounded totals come from 500 draws of the hourly coefficients; standard errors for unbounded totals use the delta method. The last column is the share of generator-hours with local storage in which a bound binds.

**Figure C.8:** Event studies of the capacity factor, by technology and hour group




*Notes.* Two-stage difference-in-differences estimates of the effect of local storage entry on the hourly capacity factor, in percentage points, by years relative to first local entry. The year before entry is the reference. Rows are technologies; columns are hour groups. Estimates use all hours, weighted by capacity, with the predetermined cost shifter in the first stage. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

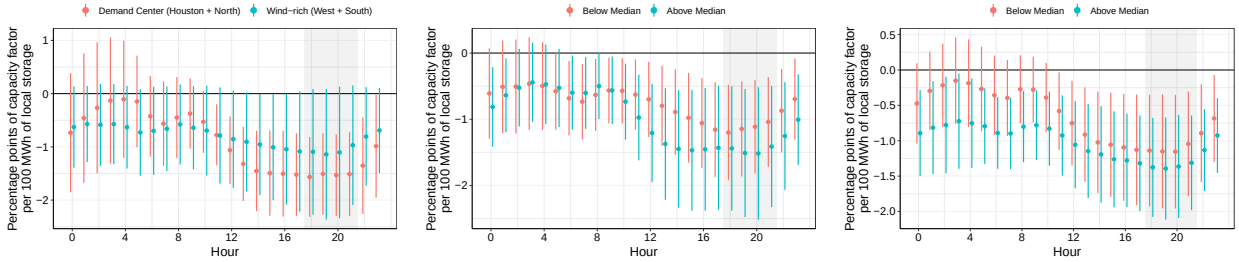
**Figure C.9:** Displacement with and without local renewable controls




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, by hour and technology, on all hours and weighted by capacity. The controls are county solar and wind capacity and their potential output, capacity times irradiance or a normalized wind power curve. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

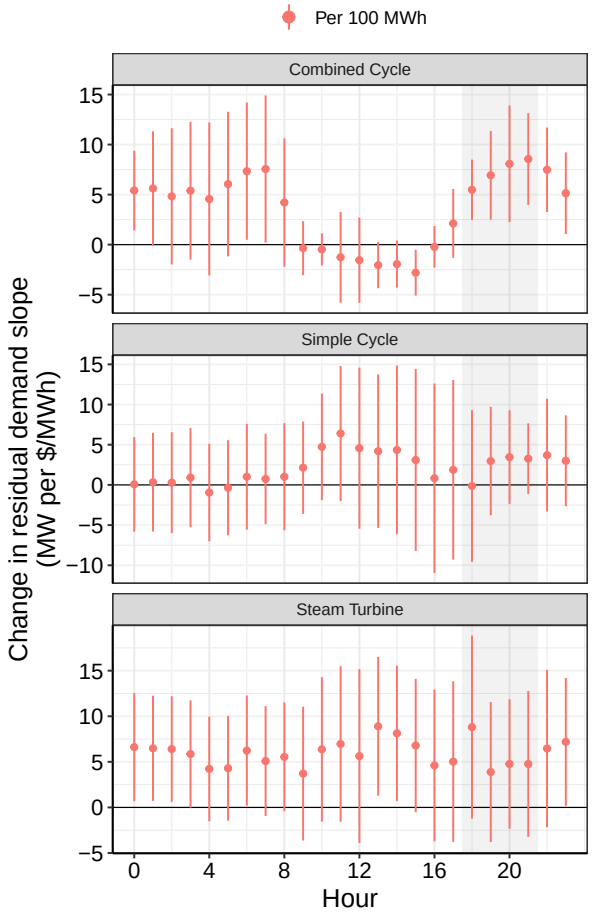
**Figure C.10:** Displacement by region and renewable conditions

**(a)** Load zone **(b)** Daily maximum solar irradiance **(c)** Daily maximum wind speed



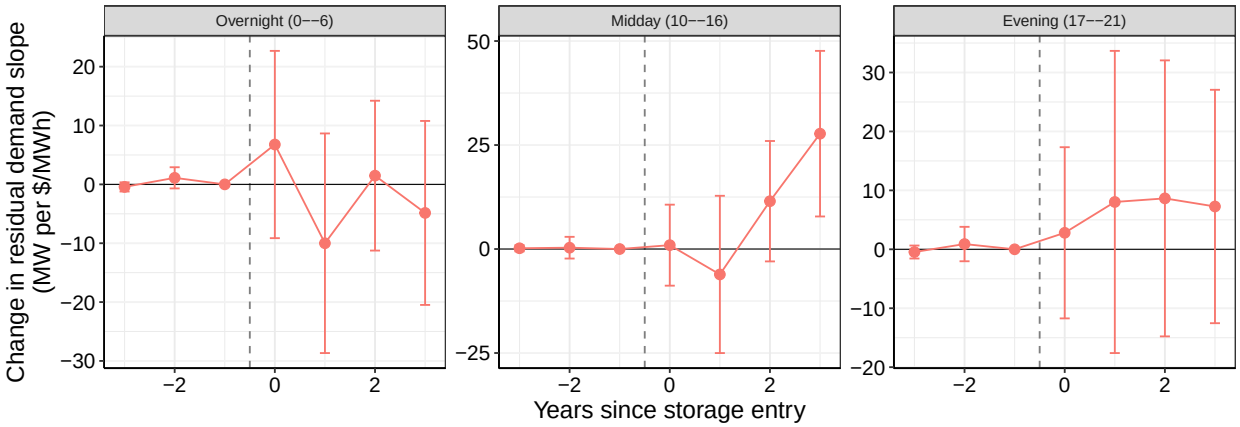

*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor of all gas generators, by hour, on all hours and weighted by capacity. Panel (a) splits the sample into the Houston and North load zones and the West and South load zones. Panels (b) and (c) split days at the sample median of statewide daily maximum solar irradiance and wind speed. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Figure C.11:** Residual demand slope by technology




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the local residual demand slope faced by each technology, in MW per $/MWh, by hour. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Figure C.12:** Event study: residual demand slope



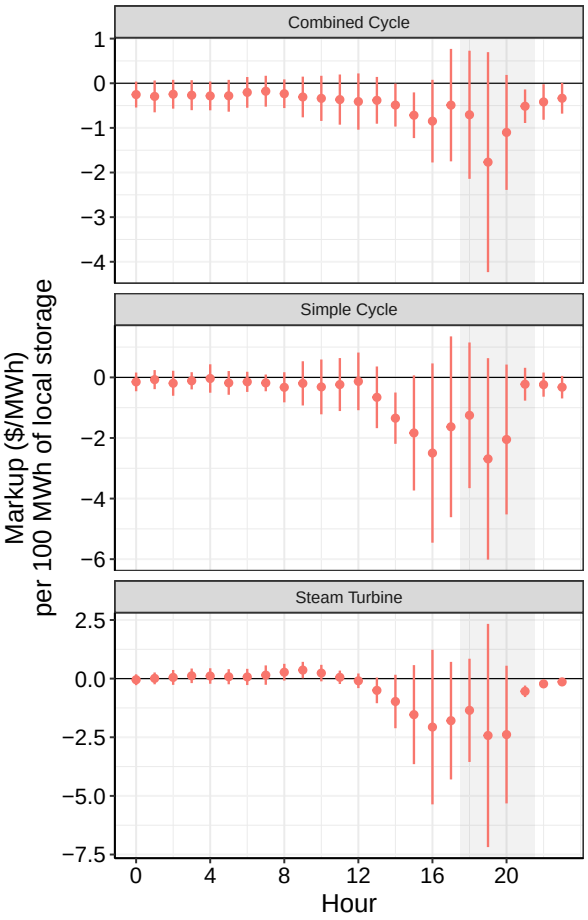

*Notes.* Binary event study from Equation (14) by hour group, with years relative to the first local storage entry binned at three years. The year before entry is the reference. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

## C.3 Additional Market Power Results

This appendix reports the market power results by technology and the event studies not shown in the main text.

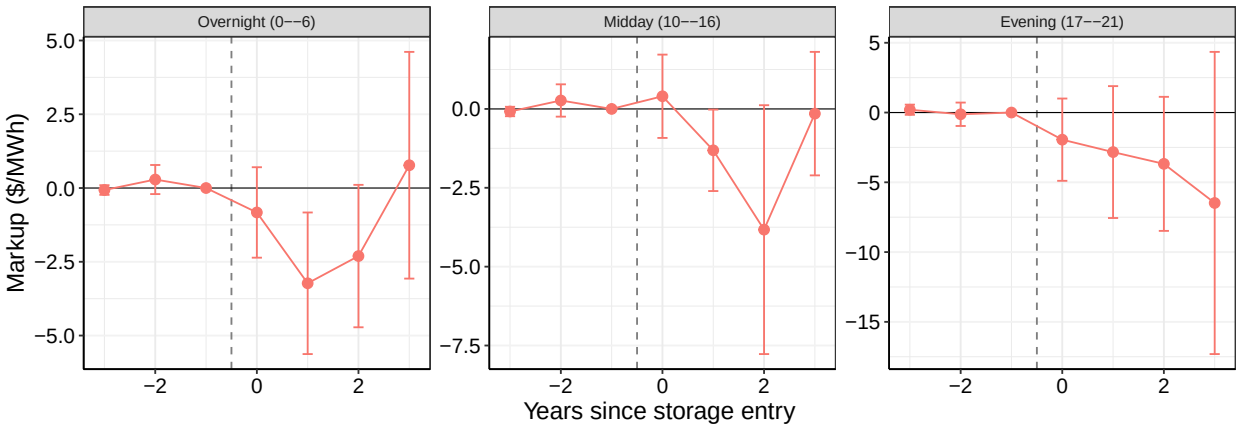
**Markups by technology and hour group** Figure C.13 presents the hourly markup effects by technology. All three technologies show the same pattern as the pooled estimate, with no change overnight and a fall in the late afternoon and evening. The evening effects average −0.92 for combined cycle, −1.57 for simple cycle, and −1.70 for steam turbine generators. Figure C.14 presents the markup event studies for the overnight, midday, and evening hour groups. The coefficients before entry are small and insignificant in every hour group. After entry, markups fall in all three hour groups. The overnight and midday declines in the first year after entry are individually significant, while the evening coefficients are larger but imprecise. The overnight decline does not appear in the dose estimates by hour, and it reverses by the third year after entry.

**Figure C.13:** Effect of local storage on markups by technology




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the markup, in $/MWh, by hour and technology. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Figure C.14:** Event study: markups by hour group



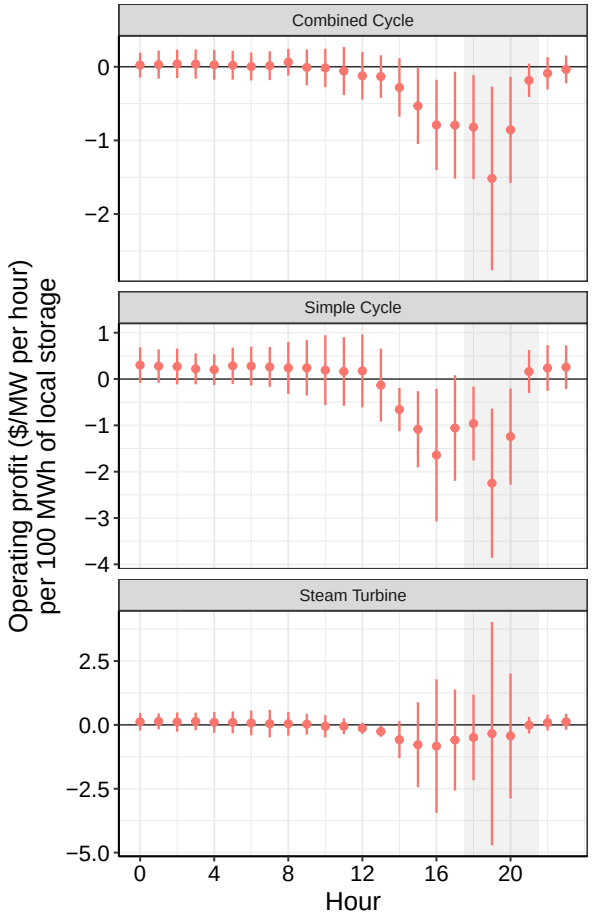

*Notes.* Binary event studies of the markup for midnight to 6 AM, 10 AM to 4 PM, and 5 PM to 9 PM, with years relative to the first local storage entry binned at three years and the year before entry as the reference. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Operating profit and rents by technology** Figure C.15 presents the hourly operating profit effects by technology. Combined cycle and simple cycle generators lose operating profit in the evening, by \$0.83 and \$1.07 per MW per hour on average from 5 PM to 9 PM. Simple cycle generators also gain a small amount overnight. The steam turbine estimates are imprecise. Table C.8 presents the daily effects by technology. Under the central benchmark, the market power rent falls by \$6.24 per MW-day for combined cycle and \$6.69 per MW-day for simple cycle generators, and the competitive rent does not change significantly. For steam turbines, the operating profit effect is not significant.

**Market power rent event study** Figure C.16 presents the event study for the daily market power rent. The coefficients before entry are small and insignificant. After entry, the market power rent falls by \$7 to \$12 per MW-day in the year of entry and the two years after it. The coefficient three or more years after entry is positive, and all post-entry coefficients are imprecise.

**Alternative competitive benchmark** Table C.9 reports the decomposition under a third benchmark, which sets the competitive price to the highest marginal cost among gas generators actually running. This benchmark attributes only 18 percent of the profit loss to market power, and the market power rent is not significant. It is, however, mechanically biased by the treatment. Storage discharge displaces exactly the most expensive running generators, so the benchmark falls when storage enters, and the decomposition records that fall as a loss of competitive rent. I therefore use the price-setter benchmark as the lower bound in the main text. Figure C.17 presents the hourly rents under the price-setter benchmark, which follow the same pattern as under the central benchmark.

**Figure C.15:** Effect of local storage on hourly operating profit by technology




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on hourly operating profit per MW of nameplate capacity, by technology, weighted by capacity. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Table C.8:** Effect of local storage on daily operating profit and rents by technology

| | Operating profit (1) | Competitive rent (2) | Market power rent (3) | Competitive rent (price-setter) (4) | Market power rent (price-setter) (5) |
|---|---|---|---|---|---|
| *Combined Cycle* | | | | | |
| $/MW-day per 100 MWh | -6.38** | -0.14 | -6.24*** | -1.76 | -4.62** |
| | (2.46) | (0.51) | (2.24) | (2.49) | (2.19) |
| Share of profit effect | 1.00 | 0.02 | 0.98 | 0.28 | 0.72 |
| Annual total ($ million) | -36.5 | -0.8 | -35.7 | -10.1 | -26.4 |
| | (14.1) | (2.9) | (12.8) | (14.2) | (12.5) |
| Per 100 MWh of storage | -3,122 | -68 | -3,054 | -861 | -2,261 |
| ($ thousand/year) | (1,207) | (250) | (1,099) | (1,219) | (1,071) |
| *Simple Cycle* | | | | | |
| $/MW-day per 100 MWh | -6.53** | 0.15 | -6.69*** | -0.82 | -5.71** |
| | (2.72) | (1.52) | (2.16) | (1.93) | (2.71) |
| Share of profit effect | 1.00 | -0.02 | 1.02 | 0.13 | 0.87 |
| Annual total ($ million) | -13.9 | 0.3 | -14.2 | -1.7 | -12.1 |
| | (5.8) | (3.2) | (4.6) | (4.1) | (5.7) |
| Per 100 MWh of storage | -1,185 | 28 | -1,213 | -149 | -1,036 |
| ($ thousand/year) | (494) | (275) | (392) | (350) | (492) |
| *Steam Turbine* | | | | | |
| $/MW-day per 100 MWh | -3.31 | -2.99*** | -0.32 | -1.63 | -1.68 |
| | (5.75) | (0.92) | (5.54) | (1.00) | (4.95) |
| Share of profit effect | 1.00 | 0.90 | 0.10 | 0.49 | 0.51 |
| Annual total ($ million) | -2.8 | -2.6 | -0.3 | -1.4 | -1.4 |
| | (4.9) | (0.8) | (4.7) | (0.9) | (4.2) |
| Per 100 MWh of storage | -242 | -219 | -24 | -120 | -123 |
| ($ thousand/year) | (421) | (67) | (405) | (73) | (362) |
| Observations (generator-days) | | | 203,395 | | |
| Counties | | | 32 | | |

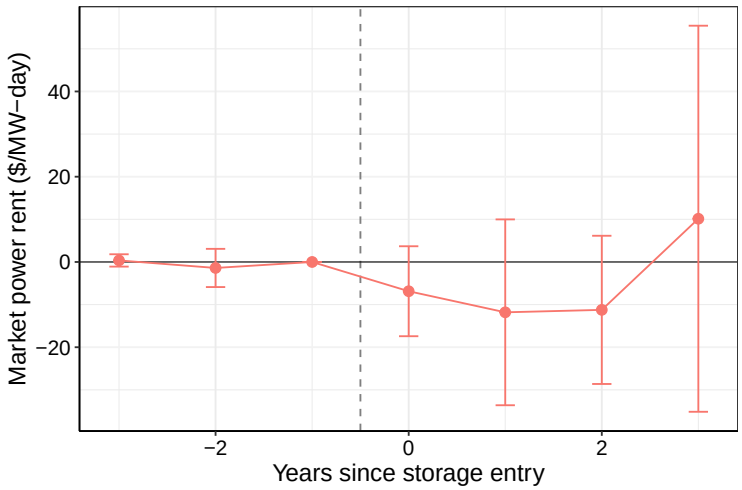
*Notes.* Two-stage difference-in-differences estimates on the generator-day panel by technology, weighted by capacity. Columns are defined as in Table 6. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Table C.9:** Rent decomposition under the highest-running-cost benchmark

| | Operating profit (1) | Competitive rent (highest running cost) (2) | Market power rent (highest running cost) (3) |
|---|---|---|---|
| $/MW-day per 100 MWh | -5.45*** | -4.47** | -0.98 |
| | (1.85) | (2.11) | (2.53) |
| Share of profit effect | 1.00 | 0.82 | 0.18 |
| Annual total ($ million) | -47.4 | -38.9 | -8.6 |
| | (16.1) | (18.3) | (22.0) |
| Per 100 MWh of storage | -4,057 | -3,324 | -733 |
| ($ thousand/year) | (1,378) | (1,567) | (1,885) |
| Observations (generator-days) | | 671,972 | |
| Counties | | 49 | |

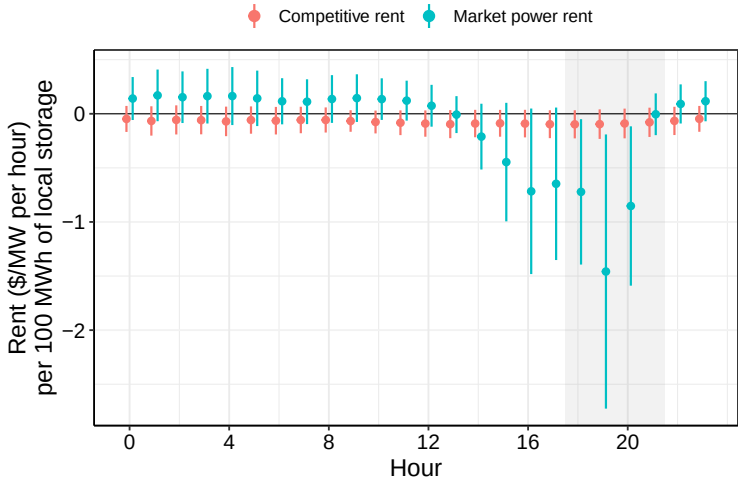
*Notes.* Two-stage difference-in-differences estimates on the generator-day panel, weighted by capacity. The competitive price is the highest marginal cost among gas generators running in the local market and interval. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Figure C.16:** Event study: daily market power rent




*Notes.* Binary event study of the daily market power rent per MW of capacity under the central benchmark, weighted by capacity, with the year before entry as the reference. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

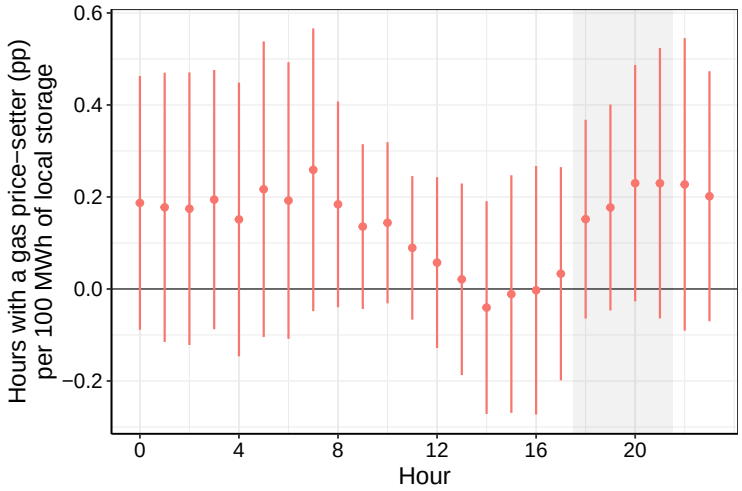
**Figure C.17:** Hourly rents under the price-setter benchmark




*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the competitive and market power rents under the price-setter benchmark, per MW of capacity per hour, weighted by capacity. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

**Gas price-setting** Figure C.18 presents the effect of local storage on the probability that a gas generator sets the nodal price. Before storage entry, a gas generator sets the price in 53 to 69 percent of intervals, depending on the hour. The effect of storage is small and not significant in any hour, ranging from −0.04 to 0.26 percentage points per 100 MWh.

**Figure C.18:** Effect of local storage on gas price-setting



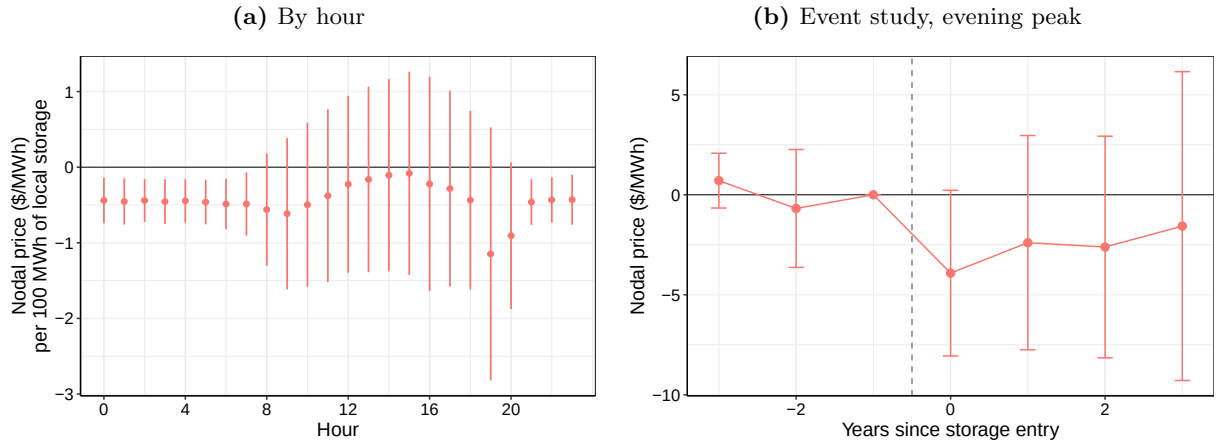

*Notes.* Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the share of intervals in the hour in which a gas generator sets the nodal price, in percentage points. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

## C.4 Additional Welfare Results

**Effect on nodal prices** Figure C.19 presents the effect of local storage on the county nodal price, which enters the consumer surplus calculation in Equation (22). I average the nodal prices at all non-storage nodes in each county and hour. The sample includes counties with gas generation and, until they receive storage, counties without gas generation. Panel (a) shows that the price falls in every hour, by \$0.08 to \$1.15 per MWh per 100 MWh of storage. The largest decline is at 7 PM, when storage discharges. The declines overnight and late in the evening are individually significant, while those from 8 AM to 8 PM are not. Panel (b) presents the event study for the evening peak. The coefficients before entry are small and insignificant. After entry, the evening price falls by \$1.6 to \$3.9 per MWh, although the individual coefficients are imprecise.

**Consumer surplus across specifications** The consumer surplus estimate in Table 7 weights counties equally. Weighting by county load instead gives a consumer loss of \$15.3 million per year (s.e. 34.4). The load-weighted estimate shows a significant price decline two years before entry, because prices in the largest counties were already rising relative to their load zones before storage entered. Controlling for county load does not remove this trend, while the unweighted estimate shows no trend before entry. Other choices change the estimate little. Adjusting the county price

**Figure C.19:** Effect of local storage on county nodal prices



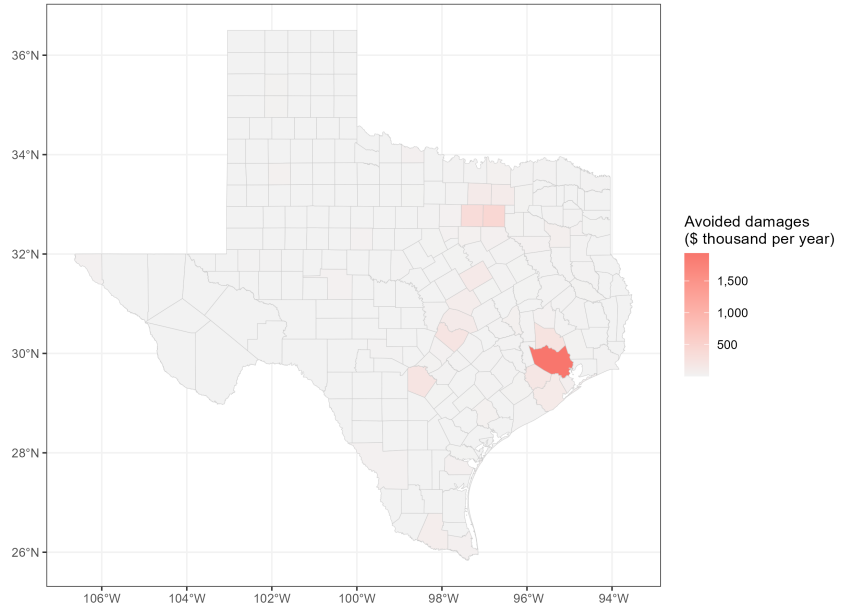

*Notes.* Two-stage difference-in-differences estimates on the county-by-hour panel of nodal prices, in 2020 dollars per MWh, averaging all non-storage settlement points in each county, with county and load-zone-by-date-by-hour fixed effects and the local renewable and temperature controls. The sample includes counties with gas generation and, until they receive storage, counties without gas generation. Counties are weighted equally. Panel (a) reports the effect of 100 MWh of local storage by hour. Panel (b) reports a binary event study for 5 PM to 9 PM, with the year before entry as the reference. The 95% confidence intervals use $t_{G-1}$ critical values and standard errors clustered at the county level.

for changes in which nodes report gives $68.8 million (s.e. 55.9), and weighting the hours of the day equally gives $61.4 million (s.e. 49.2). Following the design of Kirkpatrick (2026), with fixed effects for county, hour, quarter, year, and weekend and county-by-hour slopes on the load zone price, gives −$26.1 million (s.e. 103.6), which is consistent with the main estimate given its precision. None of these estimates is statistically significant, which is why I interpret only the combined transfer to consumers and other generators.

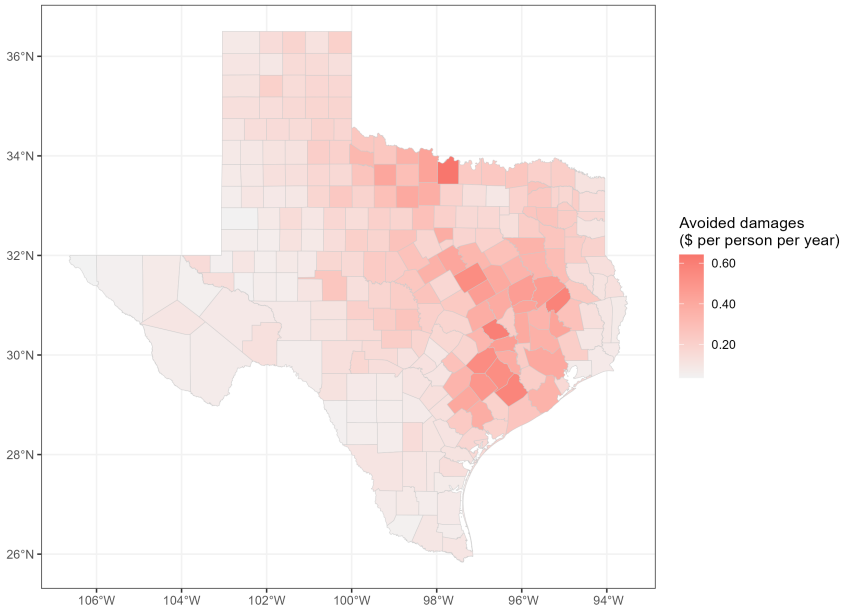
**Distribution of avoided damages across Texas counties** Figure C.20 maps the avoided local damages across Texas counties. Panel (a) shows the total avoided damages in each receptor county, which are largest in the most populated counties of Harris, Dallas, Tarrant, and Bexar. Panel (b) divides the damages by county population. Per resident, the avoided damages are more evenly spread and remain highest in and around the counties that host gas generation and storage.

**Figure C.20:** Avoided local damages by Texas county

**(a)** Total




**(b)** Per capita




*Notes.* Annual avoided damages from $NO_x$, $SO_2$, and $PM_{2.5}$ in each receptor county, in 2020 dollars, from the avoided emissions of gas generators displaced by local storage and the AP4 source-receptor matrices. Panel (b) divides by county population. The estimates use the local displacement and are an upper bound if displaced gas generation moves to counties without storage.

# D Additional Tables

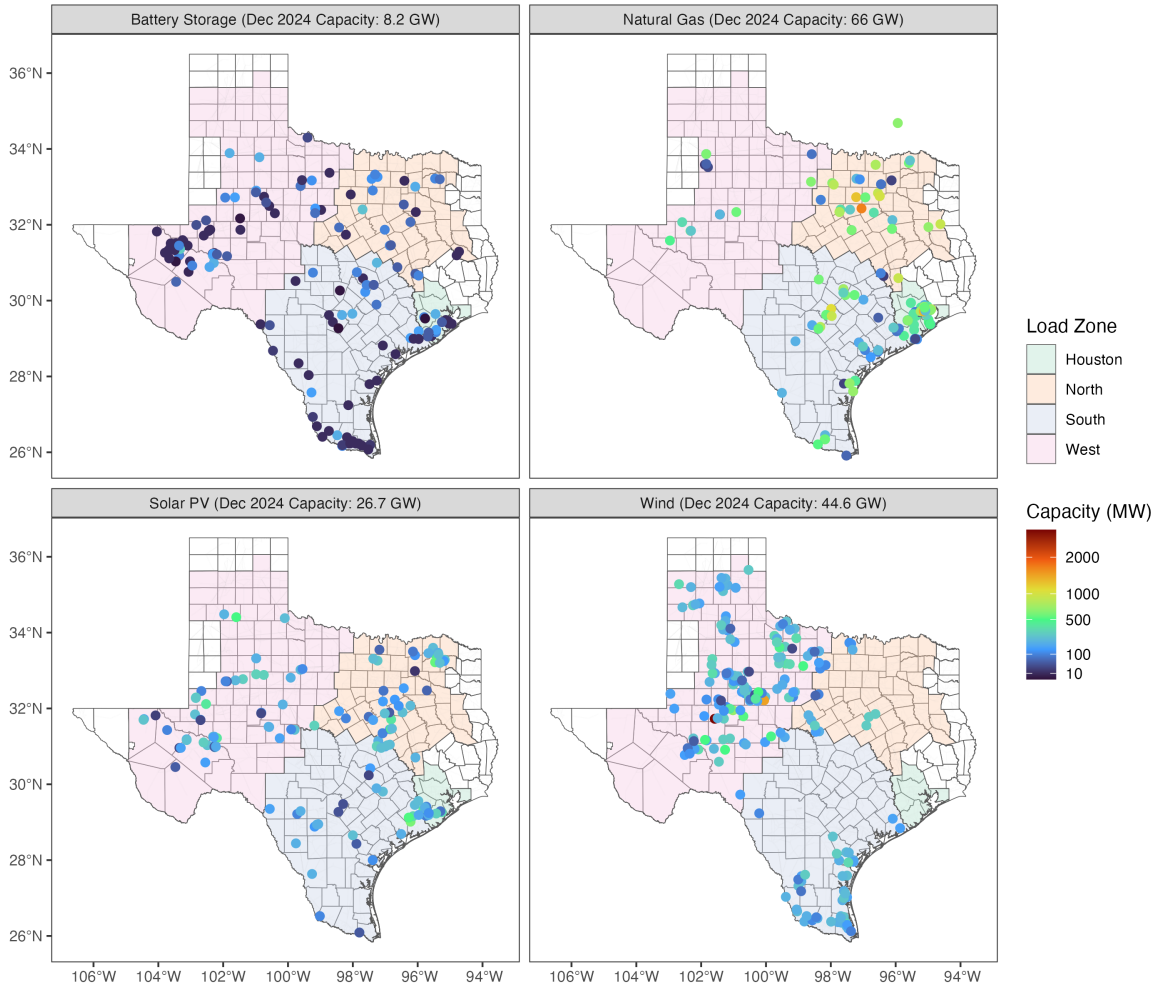
**Table D.10:** Generation Mix Summary

| | Capacity | | | Mean Generation | | Units | |
|---|---|---|---|---|---|---|---|
| | GW | Duration (hr) | % | GWh | % | Count | % |
| *Panel A: Fuel Type* | | | | | | | |
| Natural Gas | 65.23 | | 38.9 | 20.482 | 44.5 | 159 | 24.2 |
| Coal | 20.19 | | 12.1 | 8.252 | 17.9 | 17 | 2.6 |
| Wind | 41.57 | | 24.8 | 10.453 | 22.7 | 192 | 29.2 |
| Solar PV | 26.63 | | 15.9 | 2.062 | 4.5 | 133 | 20.2 |
| Batteries | 8.04 | 1.33 | 4.8 | 0.043 | 0.1 | 142 | 21.6 |
| Nuclear | 5.23 | | 3.1 | 4.666 | 10.1 | 2 | 0.3 |
| Other | 0.64 | | 0.4 | 0.079 | 0.2 | 12 | 1.8 |
| Total | 167.54 | | 100.0 | 46.036 | 100.0 | 657 | 100.0 |
| *Panel B: Top 5 Natural Gas Firms* | | | | | | | |
| Luminant | 12.91 | | 19.8 | 4.767 | 21.7 | 18 | 11.3 |
| Calpine Corporation | 10.04 | | 15.4 | 6.877 | 31.3 | 17 | 10.7 |
| Constellation Energy | 6.09 | | 9.3 | 2.007 | 9.1 | 10 | 6.3 |
| NRG Energy | 5.64 | | 8.6 | 0.943 | 4.3 | 12 | 7.6 |
| City of San Antonio | 5.53 | | 8.5 | 1.449 | 6.6 | 10 | 6.3 |
| Others | 25.03 | | 38.4 | 5.943 | 27.0 | 92 | 57.9 |
| Total | 65.23 | | 100.0 | 21.985 | 100.0 | 159 | 100.0 |
| *Panel C: Top 5 Storage Firms* | | | | | | | |
| Engie S.A. | 1.83 | 1.02 | 22.8 | 0.017 | 27.8 | 24 | 16.9 |
| Enel S.p.A. | 1.19 | 1.69 | 14.8 | 0.014 | 22.6 | 15 | 10.6 |
| Plus Power | 0.79 | 1.92 | 9.8 | 0.009 | 14.8 | 4 | 2.8 |
| Key Capture Energy | 0.63 | 1.83 | 7.8 | 0.002 | 3.5 | 11 | 7.8 |
| Luminant | 0.44 | 2.05 | 5.5 | 0.003 | 5.4 | 4 | 2.8 |
| Others | 3.16 | 1.22 | 39.3 | 0.016 | 25.8 | 84 | 59.1 |
| Total | 8.04 | | 100.0 | 0.061 | 100.0 | 142 | 100.0 |

*Notes.* Panel A shows capacity, generation, and number of units by fuel type. Panel B shows capacity, generation, and number of units by firms for natural gas unit, while Panel C shows specifically for storage firms. Discrepancies in generation between each panel arise from differences in operating intervals when aggregating generation over multiple units. I define capacity using maximum observed generation from my data, this creates discrepancies in generation capacity compared to (ERCOT, 2025). The largest firm in ERCOT owns 15 percent of total capacity and contributes, on average, about 20 percent of electricity generation in each interval. The second to fifth largest firms own between 4 to 6 percent of total capacity, contributing on average between 5 to 14.6 percent of electricity generation. For storage firms, the largest owns 21.9 percent of storage capacity, while the second to fifth largest own between 3.6 to 10 percent of storage capacity. Based on just the top five storage firms, there is significant heterogeneity in storage characteristics in terms of unit capacity and duration.

# E Additional Figures

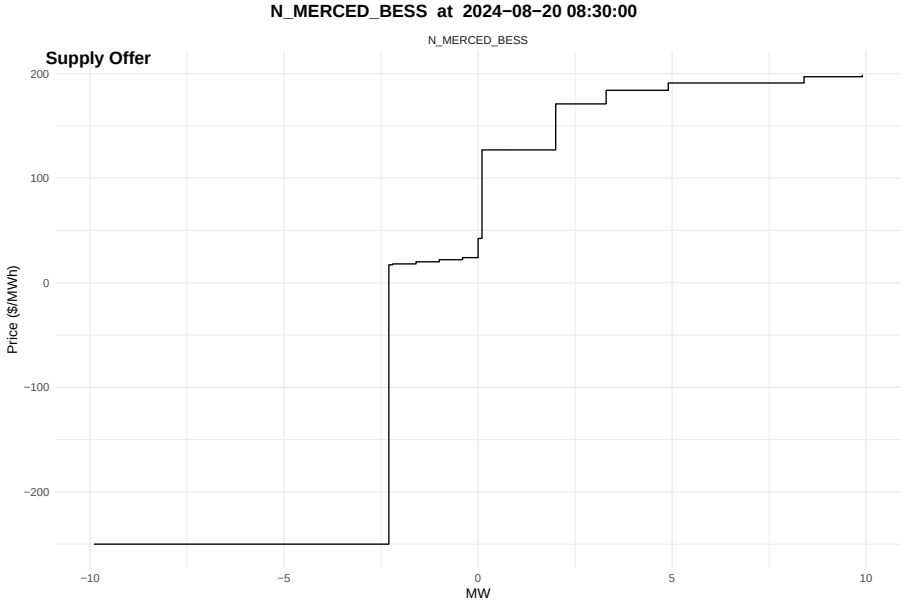
**Figure E.21:** Location of Generator in ERCOT




*Notes:* This figure shows the spatial distribution of each generator unit based on their fuel type in ERCOT. Natural gas and coal units are concentrated in the populated area of Texas, where the demand center is, such as the cities of Austin, Dallas, Houston, and San Antonio. Solar PV and storage are relatively spread evenly throughout the state, while wind is concentrated in the West and South load zone, the area with the highest wind potential.

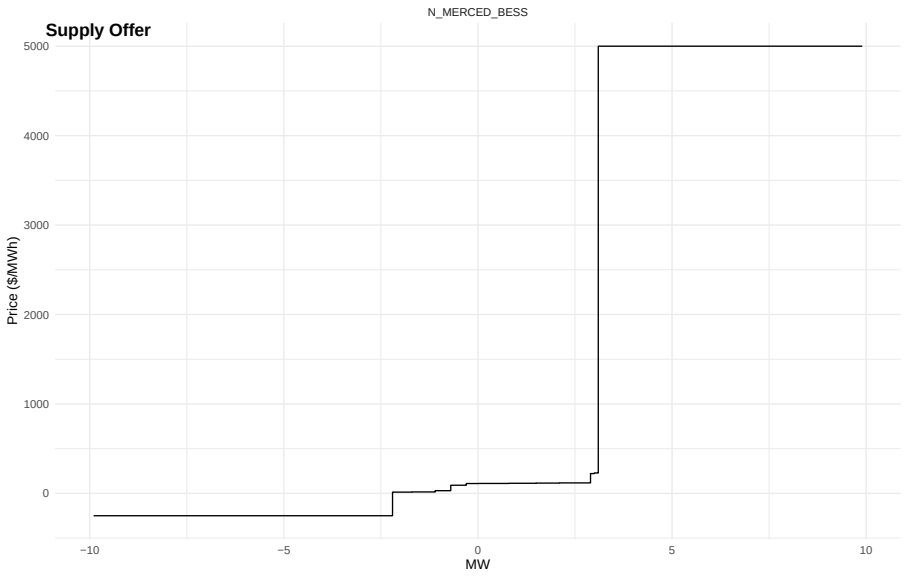
**Figure E.22:** Example Storage Offer

**(a)** Example of unconstrained storage offer




N_MERCED_BESS details:
Capacity: 9.9 MW, Storage duration: 1 hour,
Owner: SMT TX MANAGEMENT LLC
SCED Outcome: Charge 2.3 MW
Generation Resource Status: ON
AS Responsibility: 0 MW of RRSFFR, 0 MW of ECRS, 0 MW of REGUP, 0 MW of REGDN, 0 MW of RRS, 0 MW of NSRS
Load Resource Status: ONCLR
AS Responsibility: 0 MW of RRSFFR, 0 MW of ECRS, 0 MW of REGUP, 0 MW of REGDN, 0 MW of RRS, 0 MW of NSRS

**(b)** Example of energy constrained storage offer




N_MERCED_BESS details:
Capacity: 9.9 MW, Storage duration: 1 hour,
Owner: SMT TX MANAGEMENT LLC
SCED Outcome: Discharge 3.09 MW
Generation Resource Status: ONREG
AS Responsibility: 0 MW of RRSFFR, 3 MW of ECRS, 0.9 MW of REGUP, 0 MW of REGDN, 3 MW of RRS, 0 MW of NSRS
Load Resource Status: ONRGL
AS Responsibility: 0 MW of RRSFFR, 0 MW of ECRS, 0 MW of REGUP, 0 MW of REGDN, 0 MW of RRS, 0 MW of NSRS

**Figure E.23:** Installed capacity by Fuel Type from 2018-2024

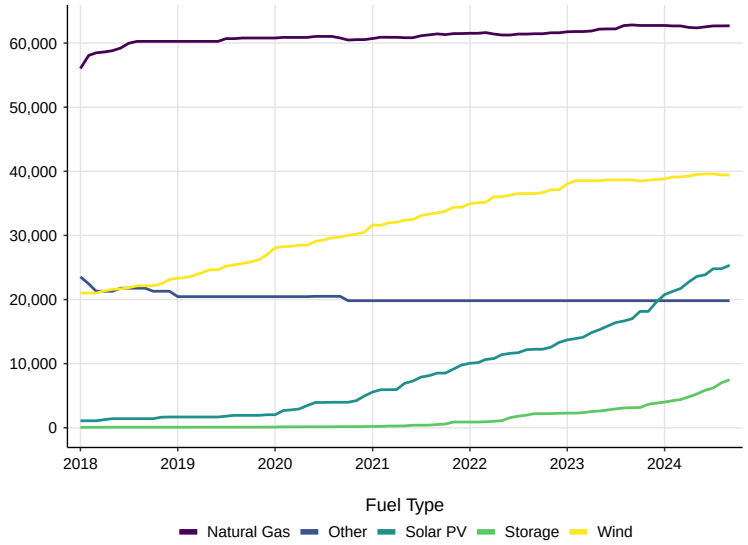
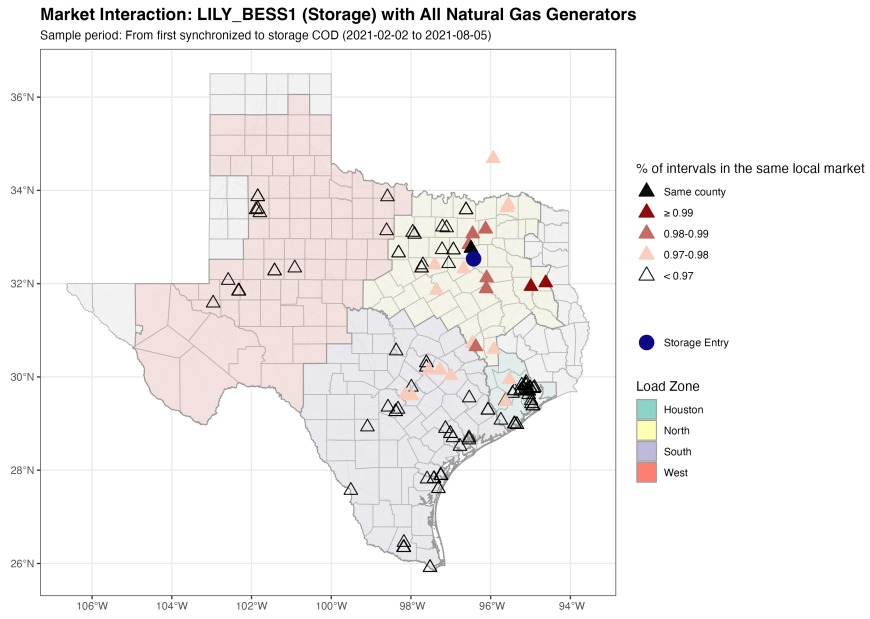
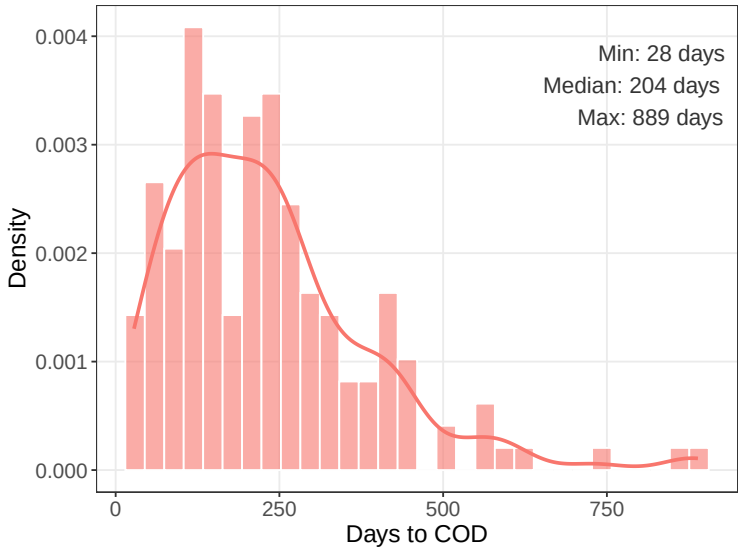



**Figure E.24:** Treatment assignment and market interaction frequency




*Notes:* This figure shows which natural gas generators are treated by the LILY_BESS1 storage unit. The blue dot shows the location of the storage unit. The solid black triangle represents generators located in the same county as the storage unit. The red triangle represents generators sharing the same local market with the storage unit more than 97% of the intervals prior to storage COD. The void triangle represents generators that do not share the same local market with the storage unit more than 97% of the intervals prior to storage COD.

**Figure E.25:** Distribution of storage first observation to COD




*Notes:* This figure shows the distribution of the time gap between the first date a storage unit shows up in the SCED data (synchronized) and the COD date.